



High-Performance Liquid Chromatography (HPLC): From Theory to Practice – A Comprehensive Guide to Analytical Method Development & Validation

HPLC From Zero to Hero

كروماتوغرافيا السائل عالية الأداء HPLC من النظرية إلى التطبيق – دليل شامل لتطوير والتحقق من صحة طرق التحليل.

HPLC من الصفر إلى الاحتراف

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Theoretical framework,الشق النظري

تحتوي هذه المحاضرة على شقين الجزء الاول نظري مقسم بتدريج شيق لفهم هذا الجهاز المهم في المختبرات العلمية والدوائية والصناعية وقد تم تبسيط المحاضرة من المبادئ الى العمق وعرضها باللغة العربية والتى تفتقر بمحاضرات متدرجة من السهل الى الصعب مع ترك المصطلحات باللغة الانجليزية وتفسيرها بالعربية والانجليزية

This lecture is structured in **two complementary parts**. **Part One** presents the **theoretical framework**, systematically organized in a progressive and engaging manner to facilitate comprehensive understanding of this pivotal analytical instrument widely utilized in scientific, pharmaceutical, and industrial laboratories.

The lecture content has been systematically simplified and scaffolded, progressing from foundational principles to advanced conceptual depth. It is delivered in Arabic—a language currently underserved by pedagogically structured educational resources that follow a graduated learning trajectory from introductory to complex levels. To bridge this gap and enhance international scientific literacy, all technical terminology and nomenclature are preserved in their original English form, accompanied by bilingual explanations in both Arabic and English to reinforce conceptual understanding and facilitate seamless engagement with global scientific literature

الشق العملي *practical part* الجزء الثاني

اما الجزء الثاني فكيف تطور **Method Validation وهي بعد الجزء النظري وهي باللغة الانجليزية لغة النشر العلمي** لتطوير طريقة فحص على جهاز HPLC حسب بحثي المنشور 2026في مجلة Wiley

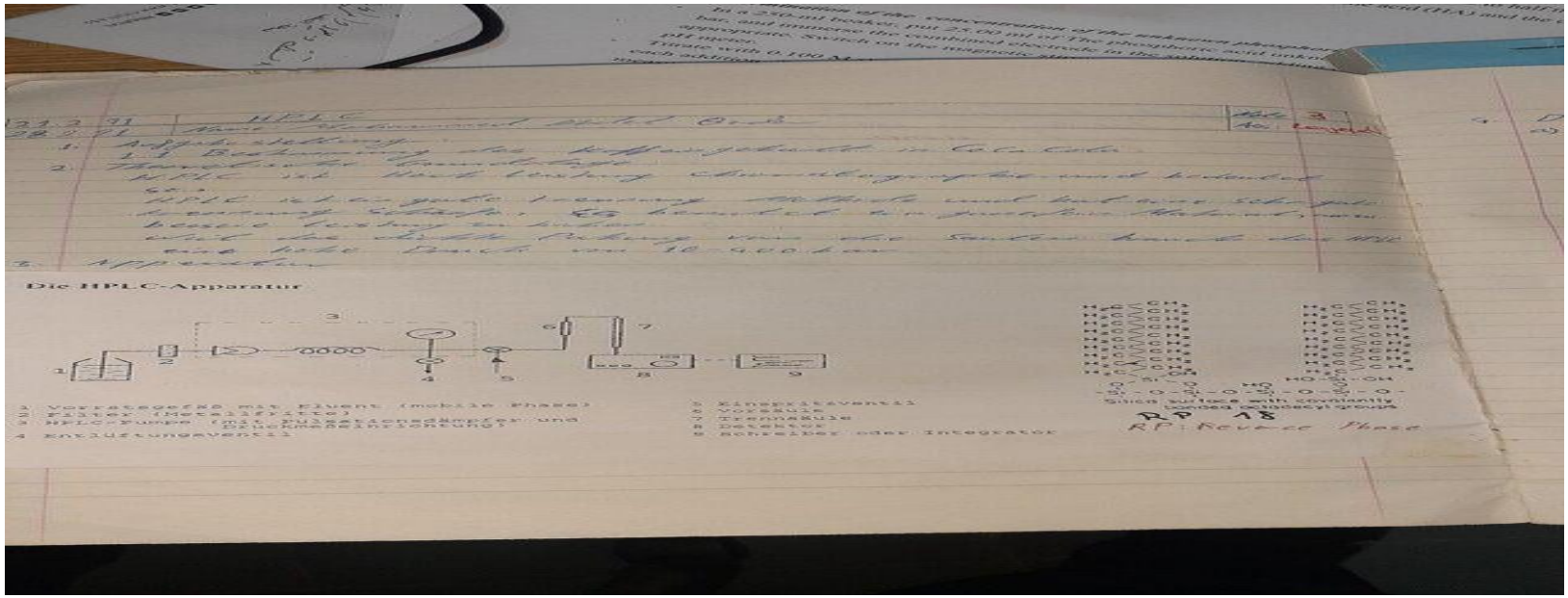
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With regard to the practical component, the accompanying lecture material has been delivered exclusively in English, as this constitutes the internationally recognized language of scientific publication, technical documentation, and professional communication within the fields of analytical chemistry, pharmaceutical sciences, and laboratory quality management.

Following the theoretical lecture

https://www.researchgate.net/publication/400769184_A_High-Performance_Liquid_Chromatography_Method_for_Simultaneous_Analysis_of_Naringin_Ascorbic_Acid_and_Caffeine_Design_Development_and_Validation

اول تطبيق عملي خلال دراستي في المانيا 1991
My first hands-on practical application during my studies in Germany, 1991



My first hands-on practical application during my studies in Germany in 1991 involved the analysis of caffeine in cola beverages using High-Performance Liquid Chromatography (HPLC).


Mobiles are silent
Ask, ask and ask
Step by step

Chromatography Word *meaning* ما معنى كلمة الكروماتوجرافي

تقسم كلمة Chromatographyالى مقطعين

الاول chromaبمعنى اللون "Color"والمقطع الثاني graphyبمعنى كتابة "Writing"

وقت اكتشفت على يد العالم Michael Tswettعام 1906

*The term **Chromatography** is etymologically derived from two Greek roots: **chroma** (χρῶμα), meaning 'color' or 'pigment', and **graphy** (γραφία), meaning 'writing' or 'recording'—literally translating to 'color writing'. This nomenclature reflects the technique's historical origins in separating colored plant pigments. The method was first developed and demonstrated by the Russian-Italian botanist **Mikhail Semyonovich Tsvet** (commonly cited as Michael Tswett) in **1906**, during his pioneering work on plant pigment separation using calcium carbonate columns."*

طرق الفصل الكروماتوجرافي
Chromatographic Separation Methods

يُعرف الفصل الكروماتوجرافي بأنه عبارة عن عملية ديناميكية طبيعية Dynamic Physical Process يتم من خلالها فصل مخلوط من مكونات جزيئية بسبب الاختلافات في درجة ميلها Affinity لأن تتوزع بين وجهين Two Phases أحدهما يطلق عليه اسم "الوجه الثابت Immobile " (Stationary) Phase والآخر هو الوجه المتحرك . Mobile Phase

*"Chromatographic separation is defined as a dynamic physical process through which a mixture of molecular components is resolved based on differential affinities for partitioning between two distinct phases: the **stationary phase** (or immobile phase) and the **mobile phase**."*

ما هي أنواع الوجه المتحرك ؟
?What is Mobile Phase *physical states*

الوجه المتحرك Mobile Phase يمكن أن يكون
1- سائل Liquid فيضم عددا من الطرق أو التكنيكات تسمى "التحليل الكروماتوجرافي في الوجه السائل " Liquid Chromatography .
2- غاز Gas يسمى "التحليل الكروماتوجرافي في الوجه الغازي " Gas Chromatography

*The **mobile phase** in chromatographic systems may exist in one of two physical states:*

1. ***Liquid***: *When the mobile phase is a liquid solvent or mixture of solvents, the technique is classified as **Liquid Chromatography (LC)**, encompassing various methodologies including High-Performance Liquid Chromatography (HPLC), Ultra-High-Performance Liquid Chromatography (UHPLC), and Size-Exclusion Chromatography (SEC).*
2. ***Gas***: *When the mobile phase is an inert carrier gas (e.g., helium, nitrogen, or hydrogen), the technique is designated as **Gas Chromatography (GC)**, primarily applied to the separation of volatile and thermally stable compounds."*

ماذا يمثل الوجه الثابت والوجه المتحرك في نظام الفصل الكروي ؟

*What is the functional role and physicochemical representation of the **stationary phase** and the **mobile phase** in chromatographic separation systems?"*

الوجه الثابت Immobile (Stationary) Phase فيمثل سطح الإدمصاص Adsorption Surface بما حوله من وسط مائي أو خلافه) .الادمصاص هو تراكم ذرات أو جزيئات على سطح المادة .وتخلق هذه العملية طبقة من الجزيئات أو الذرات التي تراكمت بكثافة على سطح الممتصات .وهو يختلف عن الامتصاص Absorption حيث تنتشر المادة في السائل أو الصلب لتشكل محلولا(

*The **stationary phase** (also termed the immobile phase) functions as an **adsorption surface**, surrounded by an aqueous or alternative medium. **Adsorption** is defined as the accumulation of atoms, ions, or molecules at the interface of a solid or liquid surface, forming a concentrated layer of adsorbed species on the adsorbent material. This phenomenon is fundamentally distinct from **absorption**, wherein a substance permeates and disperses uniformly throughout the bulk of a iquid or solid phase to form a*

homogeneous solution."

ماذا يمثل الوجه الثابت والوجه المتحرك؟

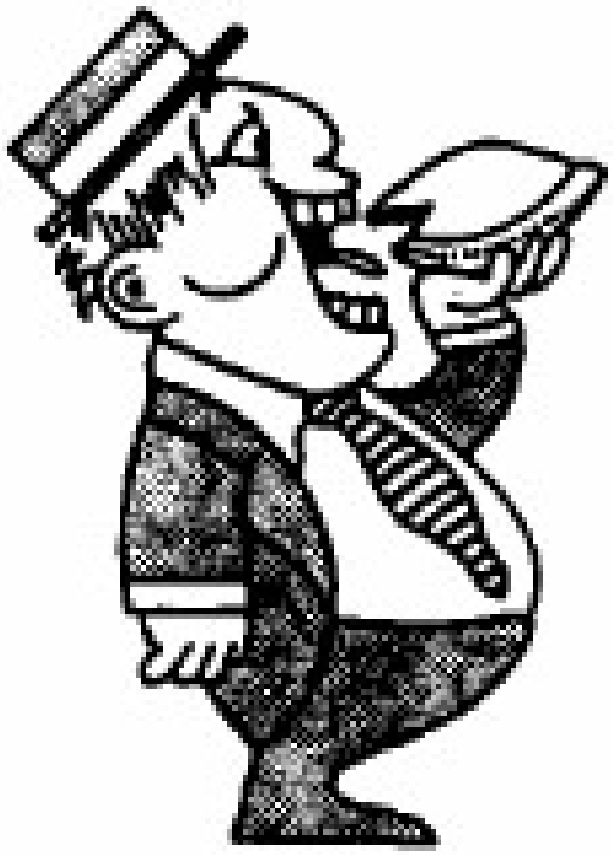
*What is the functional role and physicochemical representation of the **stationary phase** and the **mobile phase** in chromatographic separation systems?"*

أما الوجه المتحرك Mobile Phase يمثله المذيب أو المذيبات Solvents المستخدمة في عملية الفصل والذي ينتشر أو يحتك خلاله مخلوط المكونات المراد فصلها .

***Mobile Phase**: The liquid or gaseous solvent system that carries the sample through the chromatographic column, enabling separation via competitive interactions with the stationary phase*

الفرق بين الادمصاصِ Adsorption والامتصاص Absorption؟

Adsorption vs. Absorption: What is the key difference?





ما هي العوامل التي تؤثر على توزيع المكونات المخلوطة بين الوجه المتحرك والوجه الثابت ؟

What physicochemical and operational factors govern the distribution equilibrium of analytes between the mobile phase and the stationary phase in chromatographic systems?"

1

تتوزع مكونات المخلوط نفسها بين الوجه المتحرك والوجه الثابت بدرجات متفاوتة تبعا لعدة عوامل منها:

-1الوزن الجزيئي للمكون أو المكونات المراد فصلها.

-2التركيب الكيماوي للمكونات وما تحتويه من مجاميع وظيفية)نشطة (
Functional (Active) Groups.

The components of a mixture partition between the mobile and stationary phases to varying degrees, governed by several intrinsic factors, including:

1. *__Molecular weight__ of the target analyte(s) to be separated.
2. ***Chemical structure*** *of the components and the* ***functional (active) groups*** *they possess."*

ما هي العوامل التي تؤثر على توزيع المكونات المخلوطة بين الوجه المتحرك والوجه الثابت ؟

What physicochemical and operational factors govern the distribution equilibrium of analytes between the mobile phase and the stationary phase in chromatographic systems?"

2

-3طبيعة سطح الإدمصاص.

-4نوع المذيبات المستخدمة في عملية الفصل.

-5درجة حرارة الوسط.

-6درجة تشبع الوسط.

3-Nature of the adsorption surface.

4-Type of solvents employed in the separation process.

5-Temperature of the system/medium.

6-Degree of saturation of the medium.

Introduction to Liquid Chromatography

التحليل الكروماتوجرافي في الوجه السائل Liquid Chromatographyعبارة عن تكنيك للفصل يتم خلاله فصل المكونات بين وجهين الوجه الثابت Phase Stationary ذو المساحة السطح الكبير والوجه المتحرك السائل Mobile Liqid

*Liquid Chromatography (LC) is a separation technique in which mixture components are resolved through differential partitioning between two phases: the **stationary phase**, characterized by a high specific surface area, and the **liquid mobile phase**.*

Introduction to Liquid Chromatography

والوجه المتحرك عادة ما ينساب Flows فوق الوجه الثابت والوجه الثابت إما أن يكون :

1- مادة صلبة نشطة Active Solid

2- سائل غير متحرك Immobile Liquid

"The mobile phase flows through or over the stationary phase, which may be:

*1. An **active solid** (adsorbent material)*

*2. An **immobilized liquid** (liquid phase physically or chemically bound to a solid support)"*

Introduction to Liquid Chromatography

وعلى حسب عملية اختيار الوجه المتحرك ونوع جزيئات Particle Type الوجه الثابت وكذا ظروف التشغيل تتحرك المكونات المراد فصلها بمعدلات مختلفة مما ينتج عنه عملية الفصل.

Depending on the selection of the mobile phase, the particle type of the stationary phase, and the operating conditions, the target analytes migrate at different rates, thereby achieving chromatographic separation.

أقسام التحليل الكرماتوجرافي في الوجه السائل:
Classifications of Liquid Chromatography:

1

1-كروماتوجرافي الادمصاص

Adsorption Chromatography

(Liquid-Solid Chromatography)

وهذا التكتيك يعتمد مبدئيا على الاختلاف في درجة ميل المكونات سطح الإدمصاص الصلب والمستخدم كوجه ثابت.وغالبا ما تتم عملية الفصل على أساس التفاعلات القطبية المتبادلة Polar Interactions

1) Adsorption Chromatography *(Liquid-Solid Chromatography, LSC)*

This technique fundamentally relies on the differential affinity of analyte components for the solid adsorbent surface, which functions as the stationary phase. Separation is primarily governed by variations in polar interactions between the analytes and the active sites of the solid phase.

أقسام التحليل الكرماتوجرافي في الوجه السائل:
Classifications of Liquid Chromatography:
2

2- كرماتوجرفي الفصل السائل -سائل

Partition Chromatogaphy

(Liqid-Liquid Chromatography)

يعتمد هذا التكتيك على أساس خصائص الذائبية النسبية Relative Solubilityللمواد المذابة Solutesفي كل من الوجه السائل المتحرك Mobile Liquidوالوجه السائل الثابت Immobile Liquidهذا ويتم تشبع أو تشرب Impregnationللوجه السائل الساكن على مادة خاملة مسامية حاملة Inter Support.

2) Partition Chromatography *(Liquid-Liquid Chromatography, LLC)*

This technique is fundamentally based on the differential solubility (partitioning) of solutes between the mobile liquid phase and the stationary (immobile) liquid phase. The stationary liquid phase is typically impregnated or coated onto a porous, inert solid support material.

أقسام التحليل الكرماتوجرافي في الوجه السائل:
Classifications of Liquid Chromatography:

3-كرماتوجرافي التبادل الأيوني

Ion Exchange Chromatography

وهذا التكنيك يعتمد على ميل الايونات المتواجدة في المحلول والمحملة بشحنات كهربائية للاتجاه نحو ألأماكن المحملة بشحنة كهربائية معاكسة Opposite Polarity بالوجه الثابت .

Ion Exchange Chromatography

This technique relies on the tendency of ions present in a solution—carrying electrical charges—to migrate toward sites bearing opposite electrical charges (opposite polarity) on the stationary phase.

أقسام التحليل الكرماتوجرافي في الوجه السائل:
Classifications of Liquid Chromatography:

4-كروماتوجرفي الترشيح خلال الجيل

Gel Filtration Chromatography

يعتمد هذا التكنيك على مقدرة مواد مسامية معينة Porous Materialsمتواجدة في صورة جيل مثل مادة السفادكس Sephadexوالمعباة في عمود أن تقوم بفصل المكونات الموجودة في المخلوط وذلك حسب حجم وشكل الجزيئات

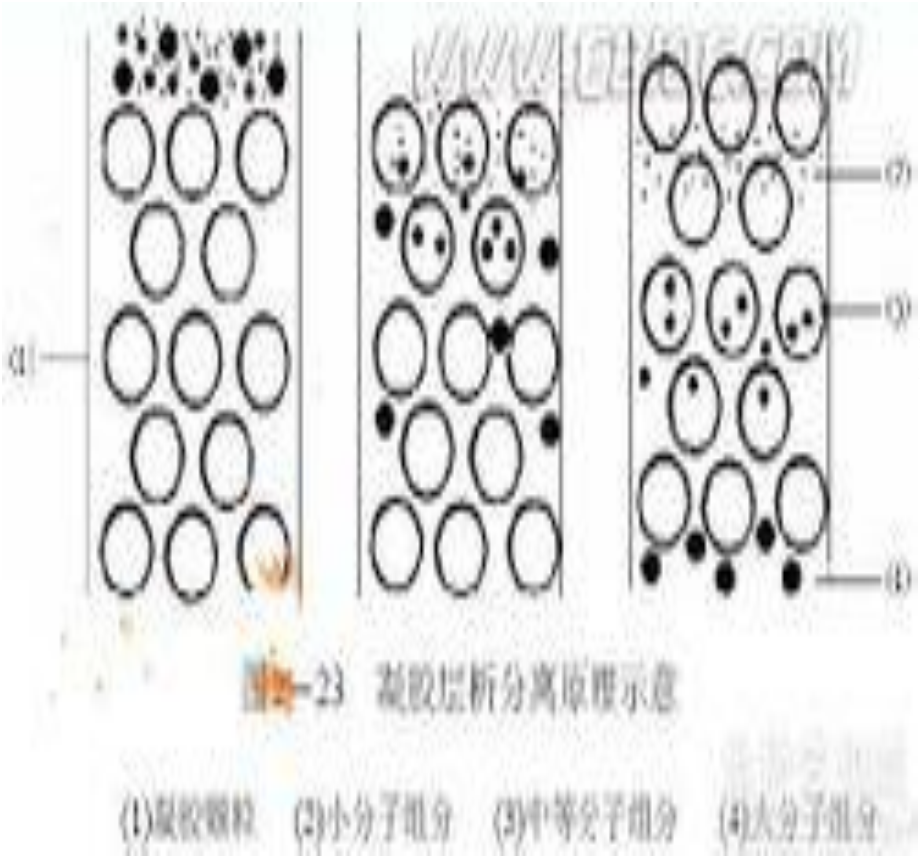
Size and Shape of the Molecules

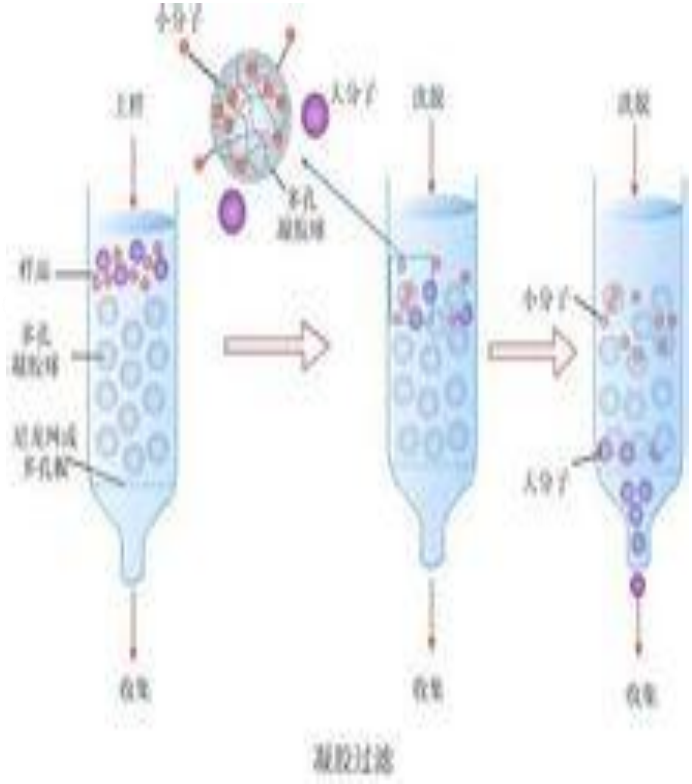
4. Gel Filtration Chromatography

This technique relies on the ability of certain porous materials—present in gel form, such as Sephadex—and packed into a column to separate the components of a mixture based on the size and shape of the molecules.

كروماتوجرفي الترشيح خلال الجيل
Gel Filtration Chromatography

- Sephadex





Gel Filtration Chromatography

Gel Type	Fractionation Range (molecular weight)
Sephadex G-10	<700
Sephadex G-25	1,000-5,000
Bio-Gel P-60	3,000-60,000
Sephadex G-75	3,000-70,000
Sephadex G-100	4,000-100,000
Bio-Gel P-100	5,000-100,000
Sephadex G-200	5,000-250,000
Bio-Gel P-200	30,000-200,000
Sephacryl S-300	10,000-1,500,000
Sepharose 2B	70,000-40,000,000

التكنيكيات التحليلية للتحليل الكرماتوجرافي في الوجه السائل Liquid Chromatography

Analytical Techniques for Chromatographic Analysis in the Liquid Phase (Liquid Chromatography)

يمكن تطبيق أساسيات التحليل الكرماتوجرافي في الوجه السائل بصفة أساسية في صورة أربعة أنواع من التكنيكيات التحليلية وهي:

أ-التحليل الكروماتوجرافي على الورق Paper Chromatography

ب -التحليل الكرمااتوجرافي على الطبقة الرقيقة Thin-Layer Chromatography TLC

ج -كرماتوجرافي العمود Column Chromatography

د -التحليل الكروماتوجرافي في الوجه السائل عالي الكفاءة High Performance Liquid Chromatography (HPLC))

The fundamentals of chromatographic analysis in the liquid phase can be primarily applied in the form of four main analytical techniques, namely:

a. Paper Chromatography

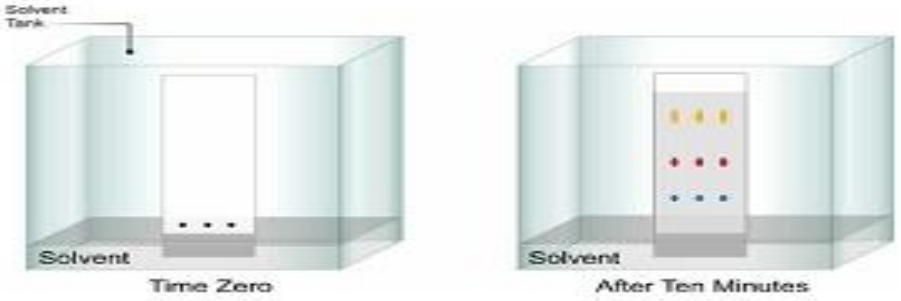
b. Thin-Layer Chromatography (TLC)

c. Column Chromatography

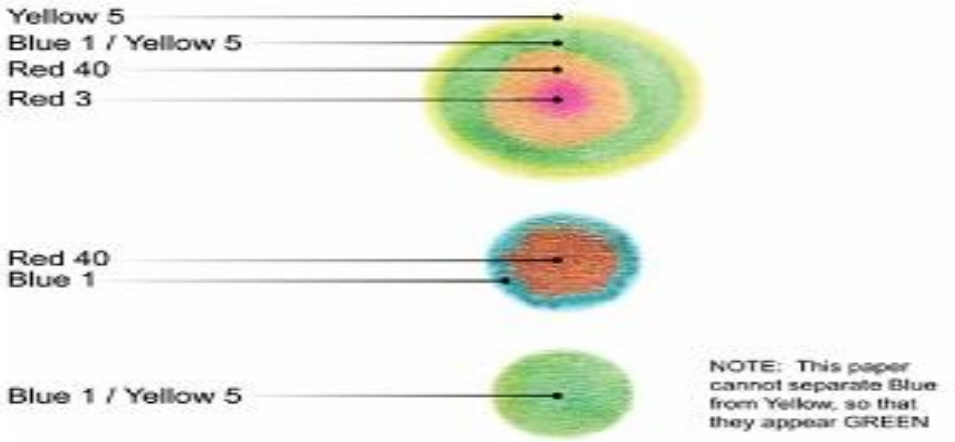
d. High Performance Liquid Chromatography (HPLC)

Chromatography)

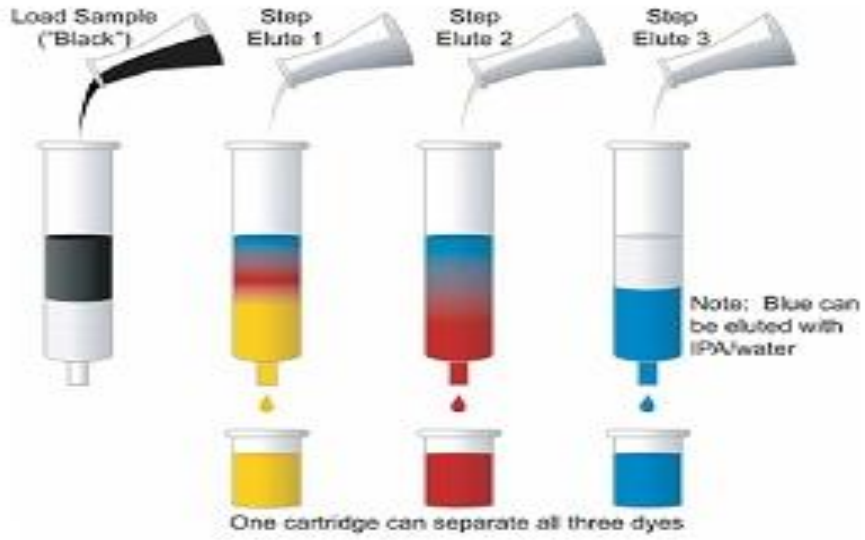
التكنيكيات التحليلية للتحليل الكرماتوجرافي في الوجه السائل Liquid Chromatography



Thin-layer Chromatography (TLC)
التحليل الكروماتوجرافي على الطبقة الرقيقة



Paper Chromatography التحليل الكروماتوجرافي على الورق



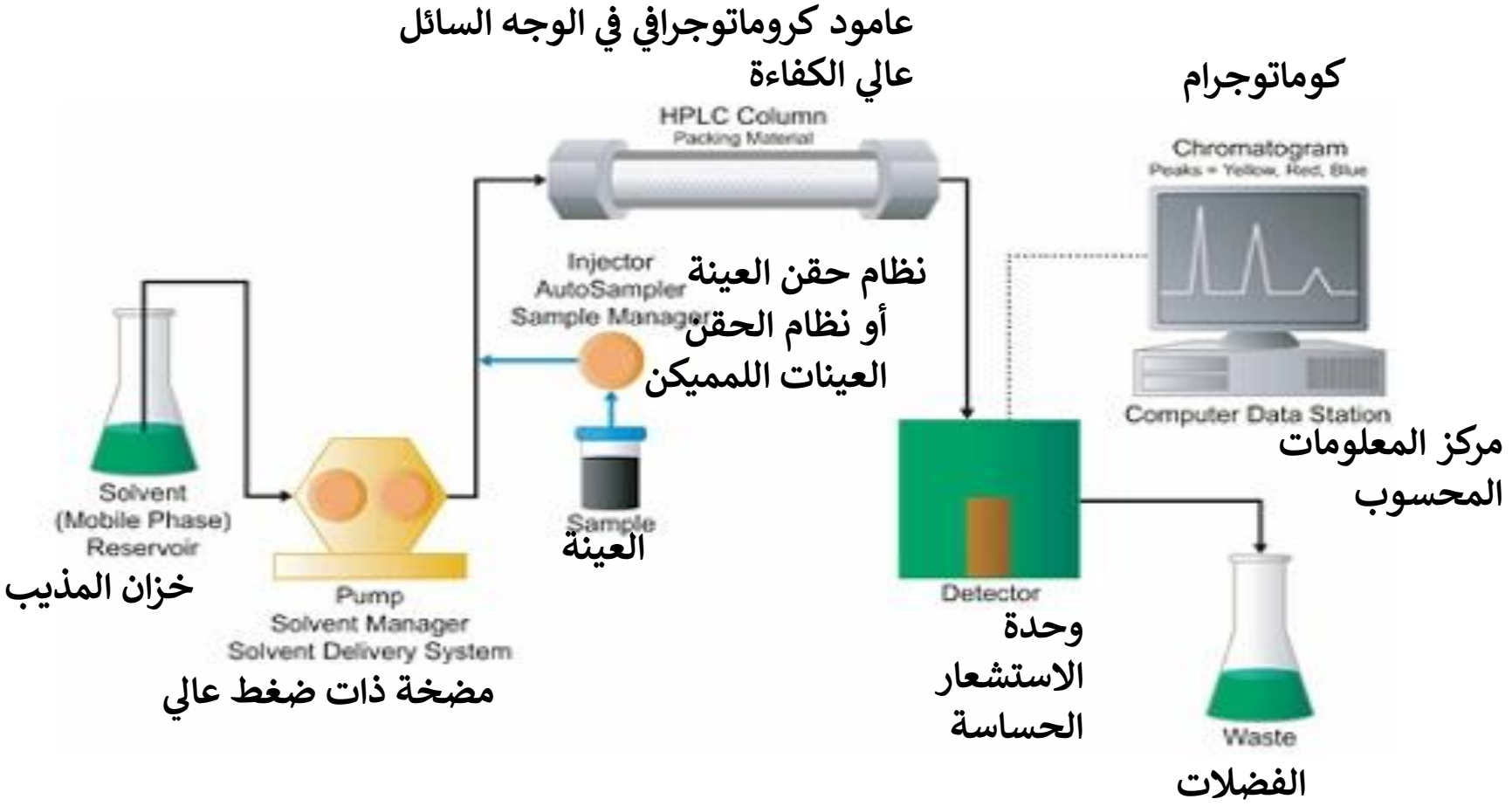
Column Chromatography
كرماتوجرافيا العمود

التكنيكيات التحليلية للتحليل الكرماتوجرافي في الوجه السائل
Liquid Chromatography

Analytical Techniques for Chromatographic Analysis in the Liquid Phase (Liquid Chromatography)

HPLC Column Chromatography
عامود كروماتوجرافي في الوجه السائل عالي الكفاءة





Schematic Diagram of the High Performance Liquid Chromatography (HPLC) Instrument

مخطط جهاز التحليل الكروماتوجرافي في الوجه السائل عالي الكفاءة (HPLC)

مقدمة في التحليل عن طريق الكرماتوجرافي عالي الكفاءة)عالي السرعة (للسوائل
Introduction to Analysis by High Performance Liquid Chromatography (HPLC) (High-Speed Liquid Chromatography)

تعتبر هذه الطريقة أهم طرق الفصل الكرماتوجرافي للسوائل وفيها يكون الوسط الساكن Immobile (Stationary) Phase على هيئة جسيمات دقيقة الحجم ويدفع بالوسط المتحرك)السائل Mobile Phase (مضخة عند ضغوط تصل الى 8000رطل على البوصة المربعة (Psi)وتستخدم المضخة لتسير سريان الوسط المتحرك خلال عامود بسرعة تتراوح بين 0.05-5مل /دقيقة بالرغم بأن جسيمات الوسط الساكن يصل قطرها إلى عدة ميكرمترات فقط وهذا يعني مساحات كبيرة .

This method is considered the most important chromatographic separation technique for liquids. In this technique, the stationary phase consists of fine particles, and the mobile phase (liquid) is propelled by a pump at pressures reaching up to 8,000 pounds per square inch (psi). The pump is used to maintain the flow of the mobile phase through the column at a flow rate ranging from 0.05 to 5 mL/min, despite the fact that the particles of the stationary phase have diameters of only a few micrometers—which corresponds to very large surface areas.

وقد أمكن بالفعل الحصول على سرعات سريان يتراوح بين 4-1مل /دقيقة باستخدام أعمدة مملوءة بجسيمات يصل نصف قطرها الى 5مايكرومترات حيث ان المكرومتر (الميكرومتر)إنجليزية إنجلترا) (Micrometre: إنجليزية أمريكا (micrometer: هو وحدة طول في النظام الدولي للوحدات تعادل جزء من مليون من المتر يرمز إليه بـ (.µm 1ميكرومتر 10^{-6} =متر.

It has indeed been possible to achieve flow rates ranging between 1–4 mL/min using columns packed with particles with radii up to 5 micrometers. The micrometer (British English: *micrometre*; American English: *micrometer*) is a unit of length in the International System of Units (SI), equivalent to one millionth of a meter, and is denoted by the symbol µm.

1 micrometer = 10^{-6} meter.

وتستعمل هذه الطريقة للمركبات

- صعبة التطاير
- التي تتأثر بالحرارة
- المواد غالية الثمن والمواد الصعب حصول عليها وسبب ذلك انها تستعمل كميات قليلة جدا من المادة المراد فحصها تبدأ بأقل من 10ميكرولتر ($\mu l = 10^{-6}$ liter)الى 100ميكرولتر

This method is employed for compounds that are:

- Non-volatile (difficult to vaporize)
- Heat-sensitive (thermolabile)
- Expensive or difficult to obtain

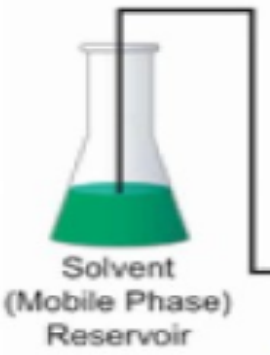
The reason for this suitability is that the technique requires only very small quantities of the sample to be analyzed—ranging from less than 10 microliters ($\mu L = 10^{-6}$ liter) up to 100 microliters.

اقسام جهاز HPLC
Parts of HPLC System

1-خزان المذيبات A reservoir Hold The Solvent

خزان المذيبات والذي يحتوي على الوجه المتحرك mobile phase لأنه يتحرك وهو اما ان يكون مذيب أو حامض أو بفر أو خليط منها أو من بعضها وذلك حسب العينة المراد فصلها

Solvent Reservoir, which contains the mobile phase—so named because it moves through the system. The mobile phase may consist of a single solvent, an acid, a buffer, or a mixture of these components, depending on the nature of the sample to be separated



2-مضخة ذات الضغط العالي A High-Pressure Pump

هي عبارة عن مدير نظام الإيصال)المزود Solvent delivery System (أو مدير المذيبات Solvent manager يقوم بضخ الوجه المتحرك خلال العامود عند ضغوط عالية ويحدد معدل التدفق عادتا بوحدة مل /دقيقة (milliliters /minute)

This is the Solvent Delivery System (or Solvent Manager), which pumps the mobile phase through the column at high pressures and controls the flow rate, typically expressed in milliliters per minute (mL/min).



Parts of HPLC System

أقسام جهاز HPLC

3- المُحّقن)نظام حقن العينة An Injector (



مدير العينة أو نظام حقن العينة هو نظام يحقن ويوصل العينة المراد فصلها وتحليلها في تيار الوجه المتحرك mobile phaseالذي ينقلها إلى عامود الفصل وهذا النظام إما أن يكون يدوي ويسمى ب manual Injectorاو آلي ويسمى ب Auto Samplerوسوف يتم شرح هذه الأنواع فيما بعد

The sample manager or sample injection system is a system that injects and introduces the sample to be separated and analyzed into the mobile phase stream, which then transports it to the separation column. This system is either manual, referred to as a Manual Injector, or automated, referred to as an Autosampler. These types will be explained in detail later.



4- عامود الفصل The Column

عامود الفصل الخاص بأجهزة HPLCتحتوي على مواد تعبئة صلبة كروماتوجرفي تساعد على الفصل Separation ومواد التعبئة ويسمى Immobile (Stationary) Phaseالوجه الثابت يحتوي على مواد فصل تختلف من عامود إلى أخر وسوف نفصل أنواع مواد التعبئة واستعمالاتها فيما بعد

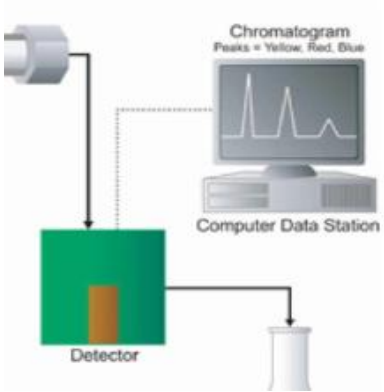
The separation column in HPLC instruments contains solid chromatographic packing materials that facilitate the separation process. These packing materials are referred to as the Immobile (Stationary) Phase. The stationary phase comprises separation materials that vary from one column to another. We will elaborate on the different types of packing materials and their applications in detail later.

5-وحدات الاستشعار الحساسة Detectors

6-ووحدة تسجيل المعلومات)المحوسب Computer Data Station)

5. Sensitive Detection Units (Detectors)

6. Computer Data Station (Data Acquisition and Recording Unit)



وحدة الاستشعار الحساسة تعطي إشارات كهربائية حيث تولد Chromatogramوالتي يمكن منها تحديد كمية المادة ونوعها إذا قورنت مع مادة مرجعية معروفة Standard Reference

The sensitive detection unit generates electrical signals that produce a chromatogram, from which the quantity and identity of the analyte can be determined by comparison with a known Standard Reference material.

ويوجد منها أنواع عدة على حسب مبدأ التعرف وقراءة المادة وسوف نشرح *أنواع وحدات الاستشعار الحساسة واستعمالاتها فيما بعد .*

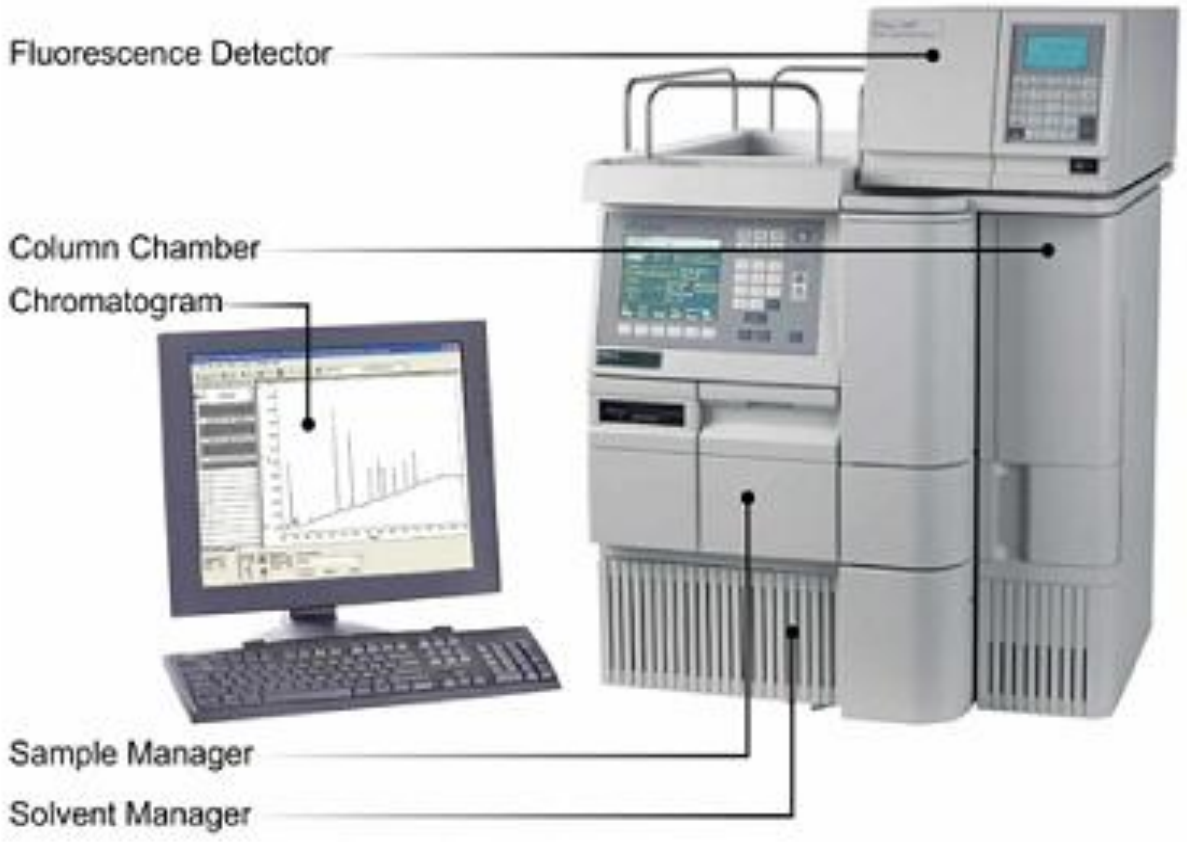
There are several types of these detectors, depending on the principle of analyte recognition and signal generation. The various types of sensitive detection units and their applications will be explained in detail later.

أما وحدة تسجيل المعلومات المحسوبة فتحول الإشارات الكهربائية إلى معلومات Chromatogramيمكن قراءته وتحليله وتخزين المعلومات في هذه الوحدة

As for the Computer Data Station (Data Acquisition Unit), it converts the electrical signals into a chromatogram—a readable and analyzable output—and stores the acquired data within this unit for further processing, reporting, and archival purposes.

شكل توزيع وحدات جهاز حديث من HPLC

Schematic Layout of the Components of a Modern HPLC Instrument



أنماط وأسس التحليل المستعملة في أجهزة HPLC
Analytical Modes and Fundamental Principles Employed in HPLC Instruments

1. الموديل الأول يسمى بنظام Isocratic Elution
2. الموديل الثاني يسمى بنظام Gradient Elution

1. The first model is called the Isocratic Elution System.
2. The second model is called the Gradient Elution System

أنماط وأسس التحليل المستعملة في أجهزة HPLC

Analytical Modes and Fundamental Principles Employed in HPLC Instruments

الموديل الاول يسمى بنظام Isocratic Elution

حيث جميع ظروف الفحص ثابتة لا تتغير خلال عملية الفحص

الموديل الثاني يسمى بنظام Gradient Elution

حيث ان هناك تغيير في احد ظروف الفحص مثل سرعة سريان المحلول ,نسبة خلط المحاليل التغير في درجة حرارة عامود الفصل والاغلب هو خلط المحاليل

Isocratic Elution System

In this mode, all analytical conditions remain constant and unchanged throughout the entire chromatographic run (e.g., mobile phase composition, flow rate, and column temperature are kept fixed).

Gradient Elution System

In this mode, one or more analytical conditions are deliberately varied during the run—such as the mobile phase flow rate, the solvent mixing ratio, or the column temperature. The most commonly applied variation is the change in solvent composition (i.e., gradually altering the proportion of solvents in the mobile phase) to optimize separation efficiency, reduce analysis time, and improve resolution of complex samples.

أنماط وأسس التحليل المستعملة في أجهزة HPLC
Analytical Modes and Fundamental Principles Employed in HPLC Instruments

ويقسم نظام الشطف بتدرج الطور المتحرك Gradient Elution إلى:

1.High- Pressure (Gradient System)نظام التدرج ذو الضغط العالي
2. Low- Pressure(Gradient System)نظام التدرج ذو الضغط المنخفض

The Gradient Elution System is further divided into two types:

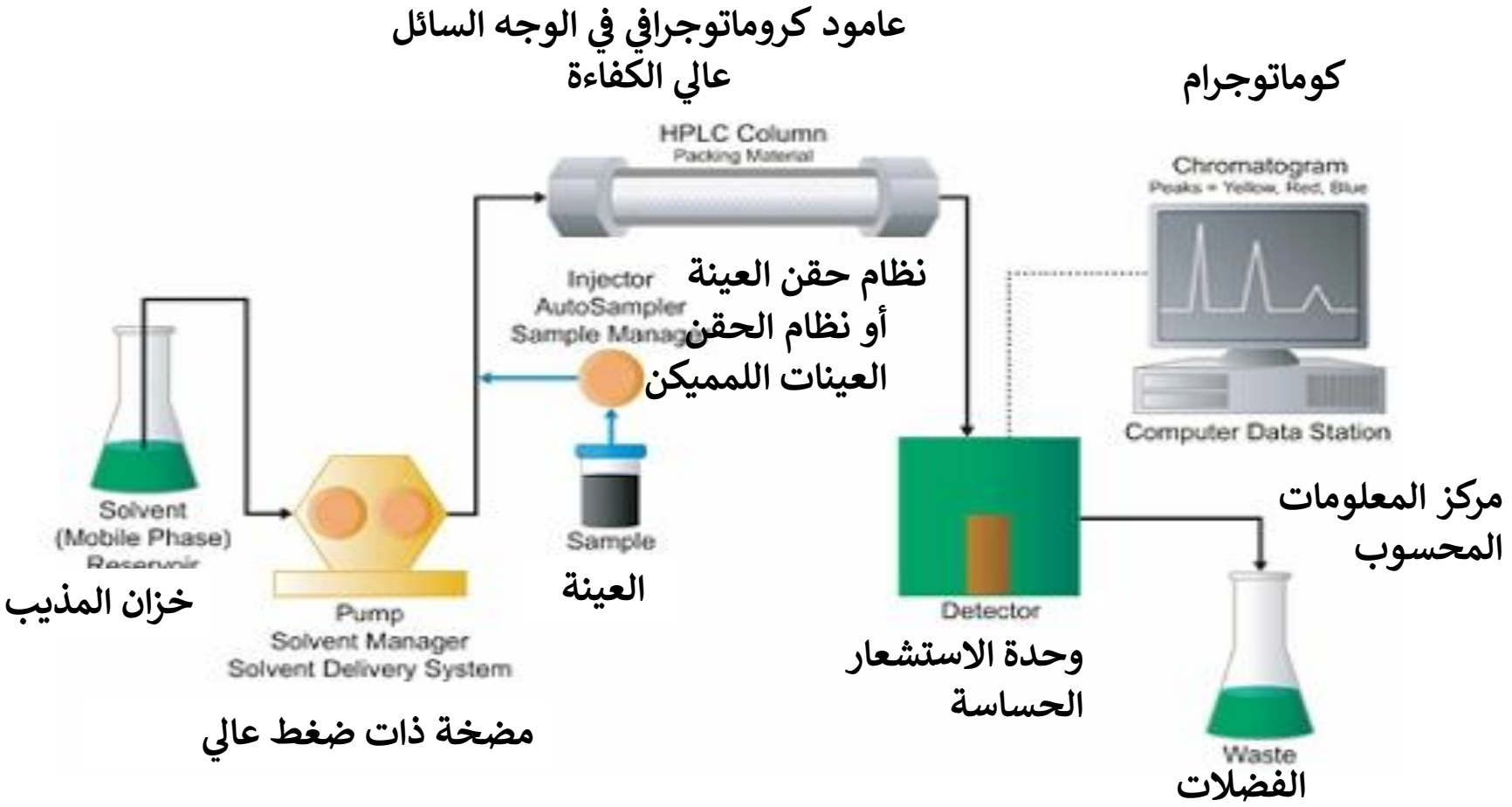
1. High-Pressure Gradient System
2. Low-Pressure Gradient System

نمط أو طراز نظام الشطف المتساوي
Isocratic Elution System (Mode/Model)

يعمل هذا النظام ويقوم على أساس خلط الوجه المتحرك (Mobile Phase)خارج نظام جهاز HPLCحيث أنها تحتوي على وعاء الوجه المتحرك ومضخة واحدة تزود عامود الفصل وهو ابسط أنظمة HPLCوالأرخص ثمنا كجهاز .

This system operates on the principle of mixing the mobile phase externally, outside the HPLC instrument itself. It consists of a mobile phase reservoir and a single pump that delivers the mobile phase to the separation column. This configuration represents the simplest and most cost-effective design among HPLC systems in terms of instrument cost.

Isocratic LC System(ISOCRATIC ELUTION) mode



جهاز التحليل الكروماتوجرافي في الوجه السائل عالي الكفاءة (HPLC)

Isocratic LC System(ISOCRATIC ELUTION)

Gradient Elution System نمط أو موديل

Gradient Elution System (Mode/Model)

نظام تطور عن نظام Isocratic Elution System وهو يقوم على خلط الوجه المتحرك Mobile Phase داخل جهاز HPLC حسب متطلبات الفحص والتحليل المدخلة

This system represents an advancement over the Isocratic Elution System. It operates by mixing the mobile phase internally within the HPLC instrument, according to the analytical requirements and parameters programmed into the system.

Type of Gradient HPLC System أنواع

وهو يقسم الى نوعين

1-(High –Pressure (Gradient System

2-(low –Pressure(Grandient System

This system is divided into two types:

1. High-Pressure Gradient System
2. Low-Pressure Gradient System

Feature	High-Pressure Gradient System	Low-Pressure Gradient System
Pumps	Two or more high-pressure pumps	Single high-pressure pump
Mixing Point	Post-pump (near column inlet)	Pre-pump (via proportioning valve)
Gradient Precision	Very high; rapid changes possible	Good; slightly slower response
Cost	Higher	More economical
Typical Use	Advanced research, complex gradients	Routine analysis, quality control

الخاصية	نظام التدرج عالي الضغط (High-Pressure Gradient)	نظام التدرج منخفض الضغط (Low-Pressure Gradient)
المضخات	مضختان أو أكثر عالية الضغط	مضخة واحدة عالية الضغط
نقطة الخلط	بعد المضخة)قرب مدخل العمود(	قبل المضخة عبر صمام تناسبي/نسبي
دقة التدرج	عالية جداً؛ إمكانية إجراء تغييرات سريعة	جيدة؛ استجابة أبطأ قليلاً
التكلفة	أعلى	أكثر اقتصادية
الاستخدام النموذجي	الأبحاث المتقدمة، التدرجات المعقدة	التحليلات الروتينية، مراقبة الجودة

High –Pressure Gradient System

وهو نظام ان وجد على ابسط صوره يحتوي على وعائيين محاليل ومضختان ويسطر عليهما من خلال الضابط المتدرج Grand Controllerحيث انه يسحب اقل او اكثر من المحاليل على حسب متطلبات الفحص وفصل المادة المضختان ذات الضغط العالي يرسلان المحاليل المسحوبة الى نظام خلط)خلاط Mixer (والتي تنقلها بعد ذلك الى عامود الفصل

فائدة ذلك توفير لمحاليل المخلوطة حيث انها تخلط حسب الحاجة والسيئة في هذا النظام انها تحتوي على مضختان وهذه المضخات مرتفعة السعر وتحتاجان الى صيانة دائمة .

This system, in its simplest configuration, consists of two solvent reservoirs and two high-pressure pumps, which are controlled by a Gradient Controller. The controller regulates the withdrawal rate from each solvent reservoir—drawing more or less from either solution—according to the analytical requirements and the separation needs of the target compound.

The two high-pressure pumps then deliver the withdrawn solvents to a mixing system (Mixer), which subsequently transfers the homogeneously mixed mobile phase to the separation column.

Advantages:

- **Efficient solvent utilization: Solvents are mixed on-demand, precisely according to the programmed gradient, minimizing waste and ensuring fresh preparation of the mobile phase.**

Disadvantages:

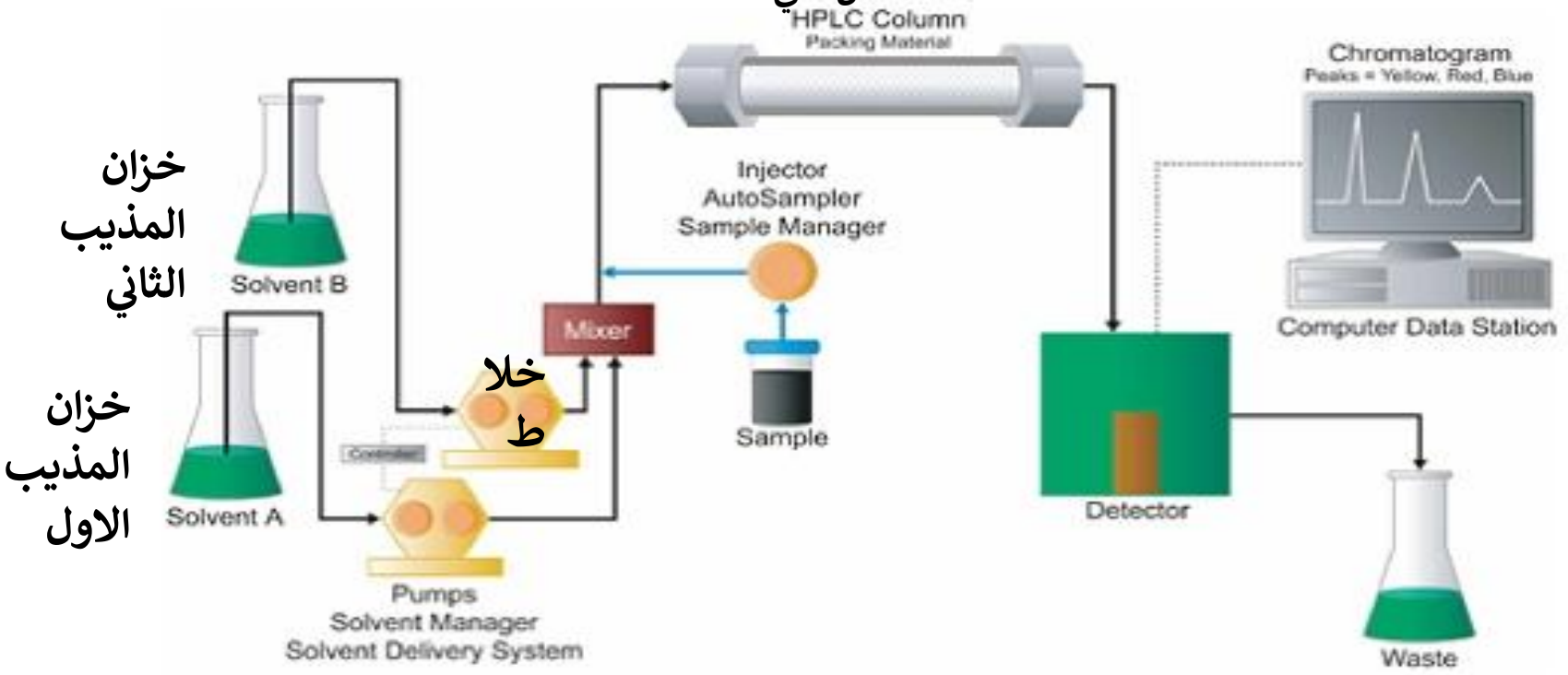
- **Higher instrument cost: The system requires two (or more) high-pressure pumps, which are significantly more expensive than a single-pump configuration.**
- **Increased maintenance requirements: Multiple high-pressure pumps entail more frequent calibration, servicing, and potential for mechanical wear, leading to higher operational costs and downtime.**

أنماط وأسس التحليل والفصل والاستخراج جهاز التحليل الكروماتوجرافي في الوجه السائل عالي الكفاءة (HPLC)

(

Gradient ELUTION) mode High-Pressure (Gradient System)

عامود كروماتوجرافي في الوجه السائل عالي الكفاءة



مضخات ذات ضغط عالي مع نظام تحكم متدرج

low –Pressure(Grandient System)

يعمل هذا النظام على خلط المحاليل لتكوين الوجه المتحرك Mobile Phaseمن خلال صمام التدرج التناسبي)صمام التناسب التدرج Gradient Proportioning Valve(حيث تسحب المحاليل من أوعية المحاليل وتخلط في الصمام بضغط منخفض ثم ينتقل المحلول المخلوط حسب متطلبات الفحص والفصل الى مضخة ذات ضغط عالية واحدة والذي يضخ إلى عامود الفصل .
سعره مرتفع لكن يمكن خلط الوجه المتحرك بكل سهولة ويسر ويعتبر من أفضل الأجهزة . HPLC

This system operates by mixing solvents to form the mobile phase through a Gradient Proportioning Valve (GPV). Solvents are drawn from their respective reservoirs and mixed within the valve at low pressure. The resulting mixed mobile phase—prepared according to the analytical and separation requirements—is then delivered to a single high-pressure pump, which propels it through the separation column.

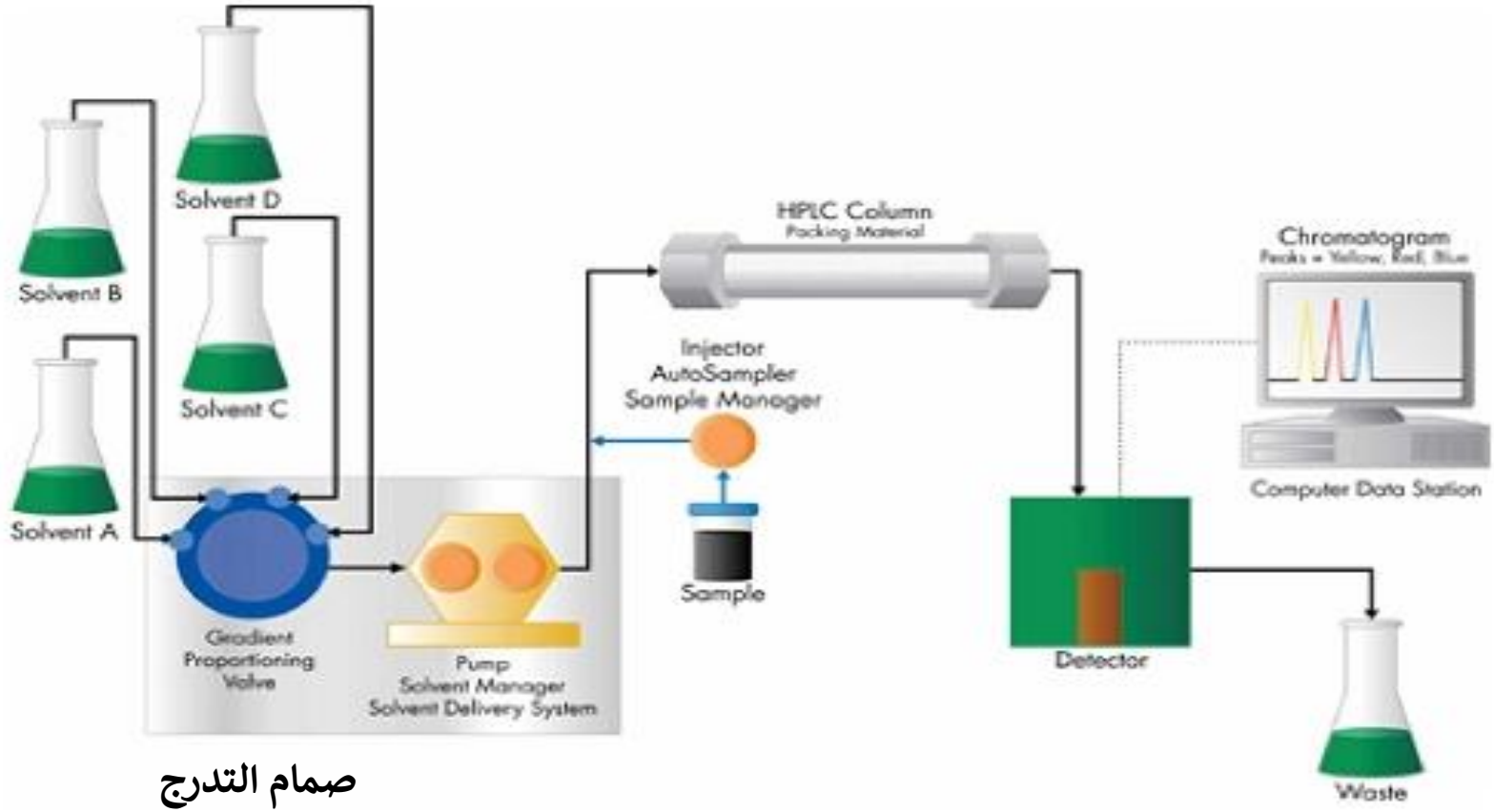
Advantages:

- Ease of mobile phase preparation: The system allows for simple, flexible, and automated mixing of multiple solvents via software-controlled gradient programs.
- Excellent performance: Considered one of the most versatile and widely preferred HPLC configurations for routine and advanced applications alike.

Disadvantage:

- Relatively high instrument cost: Although more economical than high-pressure gradient systems, it still represents a significant investment due to the precision components involved (e.g., proportioning valve, high-pressure pump, and control electronics).

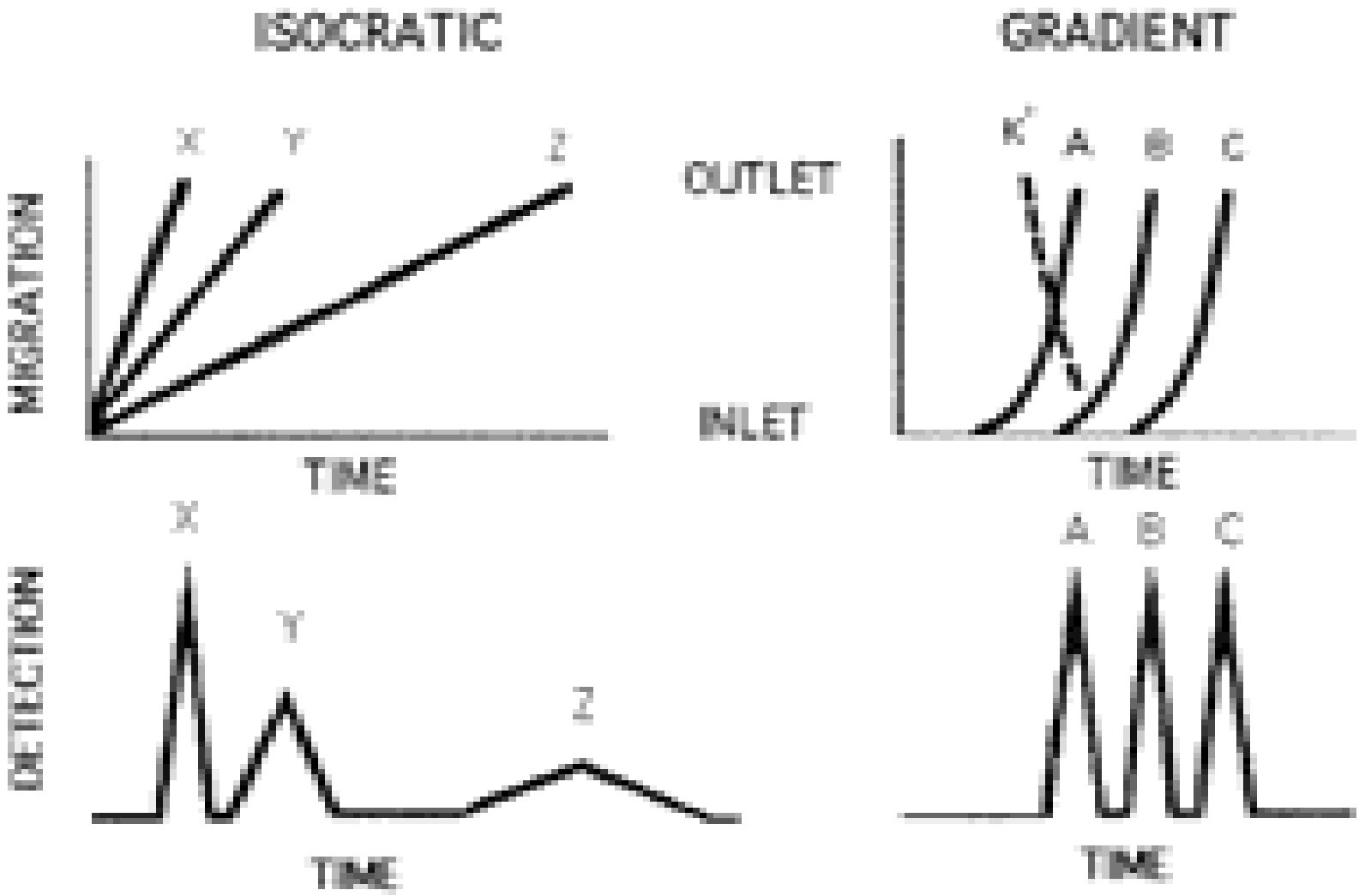
Gradient Elution (Low –Pressure Gradient System)



صمام التدرج التناسبي)صمام التناسب المتدرج(
Gradient Proportioning Valve

ما هي فائدة نظام GRADIENT؟

Benefits and Advantages of the Gradient Elution System in HPLC



فوائد ومزايا نظام التدرج (Gradient Elution System)**

يُعد نظام التدرج في أجهزة HPLC أداة قوية لتحسين كفاءة الفصل التحليلي، وتتمثل فوائده الرئيسية فيما يلي:

1 تحسين دقة الفصل (Resolution)

- يسمح بفصل مكونات الخليط المعقد التي تختلف اختلافًا كبيرًا في قطبيتها أو تقاربها للطور الثابت.
- يُقلل من تداخل القمم (Peak Overlap) ويُنتج قممًا أكثر حدة وتماثلاً.

2 تقليل زمن التحليل الكلي

- يُسرّع خروج المكونات شديدة الاحتفاظ (Strongly Retained Compounds) عن طريق زيادة قوة الإيلوشن تدريجيًا.
- يُجنب الحاجة إلى تشغيل طويل بثبات الظروف (كما في النظام المتساوي القياس Isocratic).

3 تحسين شكل القمم وحساسية الكشف

- يمنع ظاهرة "تذييل القمم (Peak Tailing)" للمركبات القطبية أو الأيونية.
- يُعزز نسبة الإشارة إلى الضوضاء (Signal-to-Noise Ratio)، مما يحسن حدود الكشف (LOD) والتقدير الكمي الدقيق.

مرونة عالية في تطوير الطرق التحليلية

- يُتيح برمجة تغييرات متعددة في تركيب المذيب، ومعدل التدفق، أو حتى درجة الحرارة خلال نفس التشغيل.
- يُناسب تحليل عينات متنوعة التركيب دون الحاجة إلى تغيير العمود أو إعادة ضبط الطريقة من الصفر.

5 تنظيف العمود تلقائيًا وتحضيره للتشغيل التالي

- يمكن إنهاء البرنامج التدرجي بغسل العمود بمذيب قوي لإزالة الشوائب المتبقية.
- يُعيد العمود إلى ظروف التوازن الأولية تلقائيًا، مما يضمن تكرارية النتائج .(Reproducibility)

6 كفاءة في استهلاك المذيبات في الأنظمة الحديثة

- تُمكّن البرمجة الدقيقة من خلط المذيبات حسب الحاجة الفعلية فقط، مما يُقلل الهدر مقارنة بالخلط اليدوي المسبق.

7 ملاءمة للعينات المعقدة والمتنوعة وهو مثالي لتحليل:

- المستخلصات النباتية والبيولوجية
- المنتجات الصيدلانية متعددة المكونات
- عينات البيئة مياه، تربة الملوثة بمركبات مختلفة الخواص
- **البروتينات، والببتيدات، والمركبات الأيضية المستقلبات في أبحاث البروتيوميات والأيضوميات الميتابولوميات**

Benefits and Advantages of the Gradient Elution System in HPLC

The gradient system in High Performance Liquid Chromatography (HPLC) is a powerful tool for enhancing analytical separation efficiency. Its key benefits include:

1 Improved Separation Resolution

- Enables the separation of complex mixture components that differ significantly in polarity or affinity for the stationary phase.
- Reduces peak overlap and produces sharper, more symmetrical peaks.

2 Reduced Total Analysis Time

- Accelerates the elution of strongly retained compounds by gradually increasing the elution strength of the mobile phase.
- Eliminates the need for prolonged runs under constant conditions (as required in Isocratic systems).

3 Enhanced Peak Shape and Detection Sensitivity

- Prevents peak tailing phenomena for polar or ionic compounds.
- Improves the signal-to-noise ratio (S/N), thereby enhancing limits of detection (LOD) and enabling more accurate quantitative analysis.

4 High Flexibility in Analytical Method Development

- Allows programming of multiple changes in solvent composition, flow rate, or even column temperature within a single run.
- Accommodates the analysis of samples with diverse compositions without requiring column replacement or complete method re-optimization.

5 Automatic Column Cleaning and Preparation for Subsequent Runs

- The gradient program can conclude with a column wash using a strong solvent to remove residual impurities.
- Automatically returns the column to initial equilibrium conditions, ensuring result reproducibility.

6 Efficient Solvent Consumption (in Modern Systems)

- Precise software-controlled programming enables on-demand solvent mixing, minimizing waste compared to pre-mixed manual preparation.

7 Suitability for Complex and Diverse Sample Types

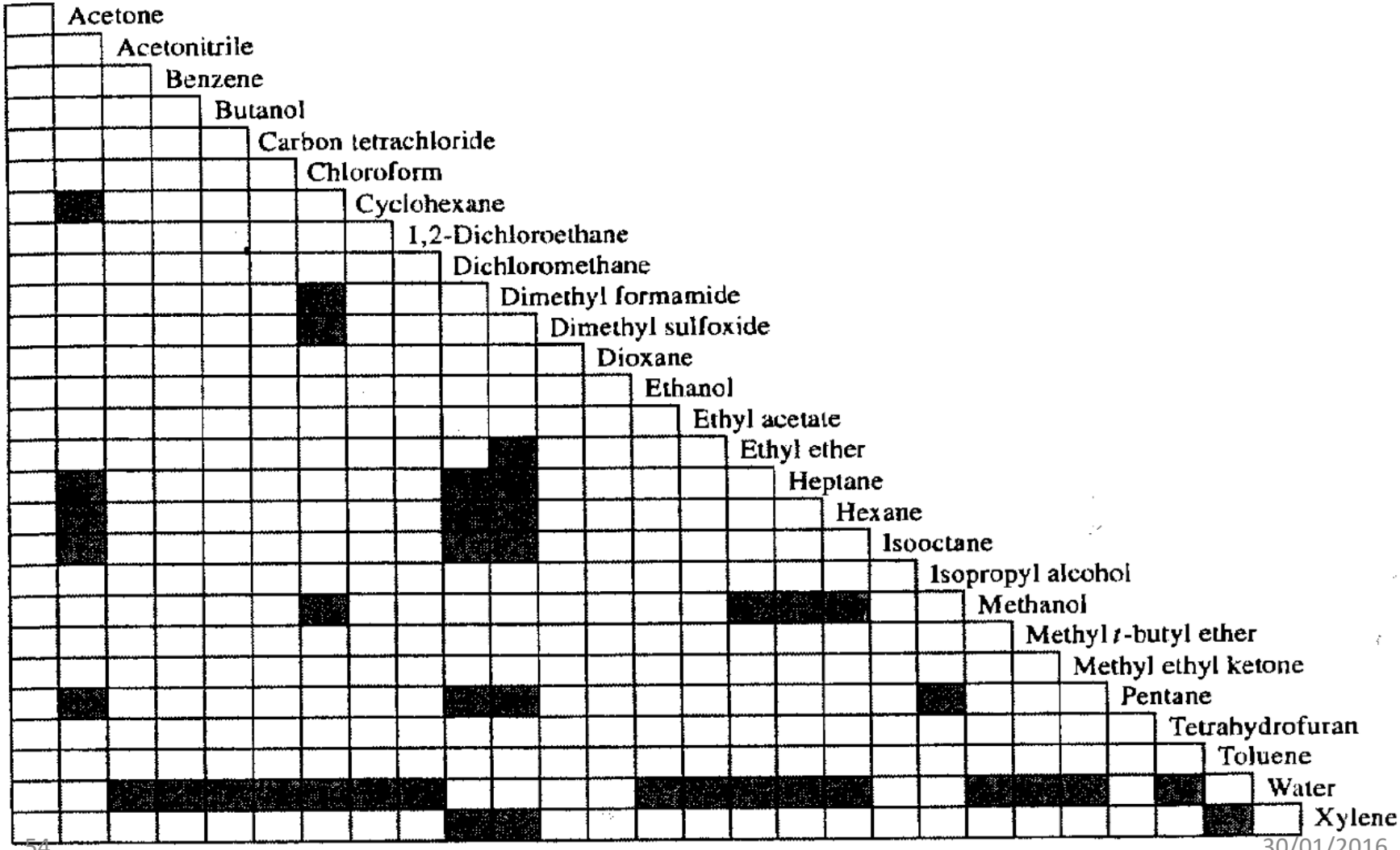
- Ideal for analyzing:
  - Plant and biological extracts
  - Multi-component pharmaceutical formulations
  - Environmental samples (water, soil) contaminated with compounds of varying properties
  - Proteins, peptides, and metabolites in proteomics and metabolomics research

جدول إمكانية مزج المحاليل والمذيبات ل mobile phase؟

Table of Solvent Miscibility for Mobile Phase Preparation

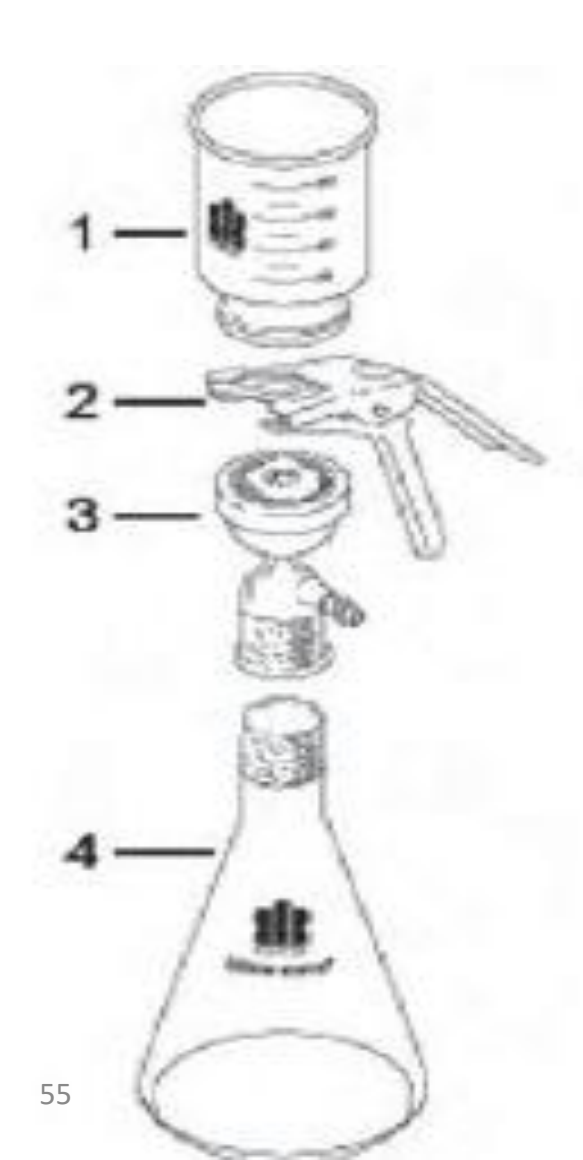
محاليل ومذيبات لها قابلية الخلط — Miscible

محاليل ومذيبات لا يمكن خلطها — Immiscible



نقاوة المحاليل والمذيبات المستعملة في جهاز HPLC

Purity of Solutions and Solvents Used in the Instrument



بسبب حساسية جهاز HPLC يجب استعمال مذيبات ومحاليل ذات نقاوة عالية تسمى HPLC grade بحيث لا تحتوي على أدنى أثر من الشوائب التي قد تؤثر على الفحص والجهاز, والصفة الغالبة لهذه المحاليل والمذيبات أنها مرتفعة الثمن ورغم ذلك لا يمكن استعمالها مباشرة إنما يجب تصفيتها وتنقيتها بواسطة نظام تصفية خاص Filtration يحتوي على Membrane Filter (0.45 µm) لا يتأثر بالمذيبات والمحاليل (وجهاز شفط خاص)انظر الشكل(

Due to the high sensitivity of HPLC instruments, it is essential to use solvents and solutions of ultra-high purity, designated as *HPLC-grade*. These must be completely free from even trace impurities that could interfere with the analysis or damage the instrument.

A characteristic feature of these HPLC-grade solvents and solutions is their relatively high cost. Nevertheless, they cannot be used directly without prior preparation: they must be filtered and purified using a specialized filtration system.

This filtration system typically includes:

- A membrane filter (0.45 µm pore size) that is chemically inert and resistant to the solvents and solutions being filtered.
- A dedicated vacuum aspiration device (see figure) to facilitate the filtration process.

نظافة العينات المستعملة في جهاز HPLC

Purity of Solutions and Solvents Used in the Instrument



بسبب حساسية جهاز HPLC يجب تحضير وتنقية العينات المراد فحصها بواسطة Syringe Filter يحتوي على Membrane Filter (0.45 µm) لا يتأثر بالمذيبات والمحاليل.
ولا يمكن حقن العينة مباشرة .

Due to the high sensitivity of HPLC instruments, samples intended for analysis must be carefully prepared and purified using a *Syringe Filter* equipped with a *Membrane Filter* (0.45 µm pore size) that is chemically inert and resistant to the solvents and solutions used.
Direct injection of unfiltered samples into the HPLC system is strictly prohibited.

نظام HPLCذا نظام التنقية او كرماتوجرافي التجهيز

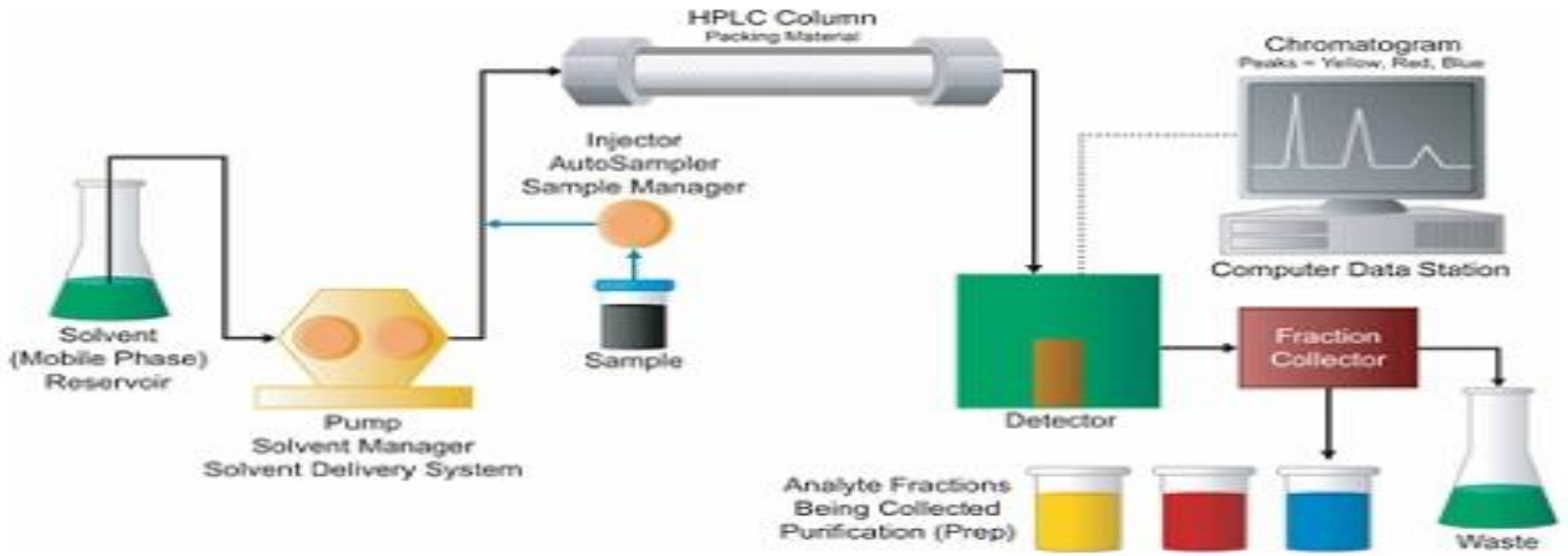
HPLC-SYSTEM PURIFICATION (PREPARATIVE CHROMATOGRAPHY))

يمكن الاستفادة من جهاز HPLCكجامع للمواد المفصولة وخاصة الغالية الثمن وذا نقاوة عالية بإضافة مجمع للمواد المفصولة Fraction Collectorحيث يمكن فصل المواد كل على حدا بعد فصلها عن الوجه المتحرك mobile Phase

The HPLC (High-Performance Liquid Chromatography) system can be effectively utilized for collecting separated compounds—particularly those that are high-value, costly, and require high purity—by equipping it with a **Fraction Collector**. This accessory enables the individual collection of target compounds immediately after their separation from the **mobile phase**."

HPLC-System Purification (Preparative

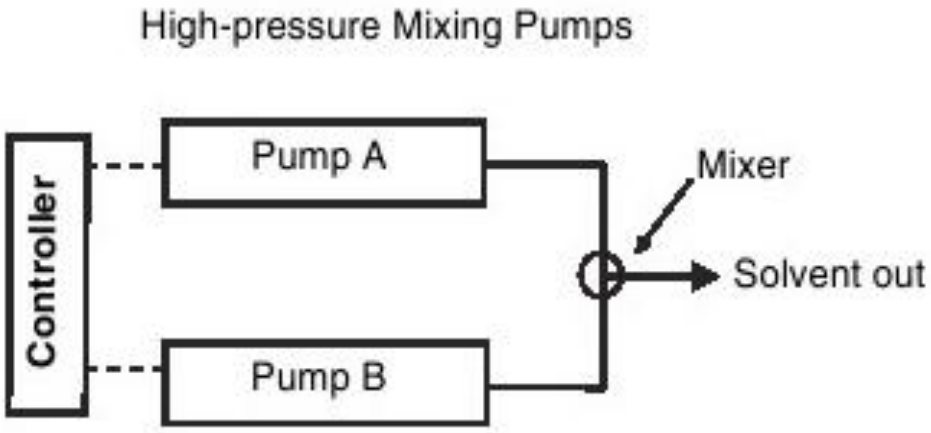
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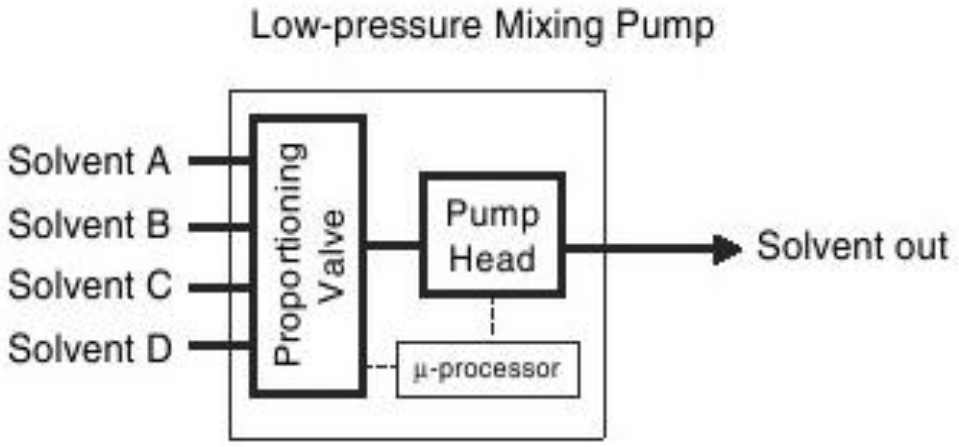
مجمع المواد المفصولة

رسم تخطيطي لمضخاط الخلط Mixing Pump
Schematic Diagram of Mixing Pump

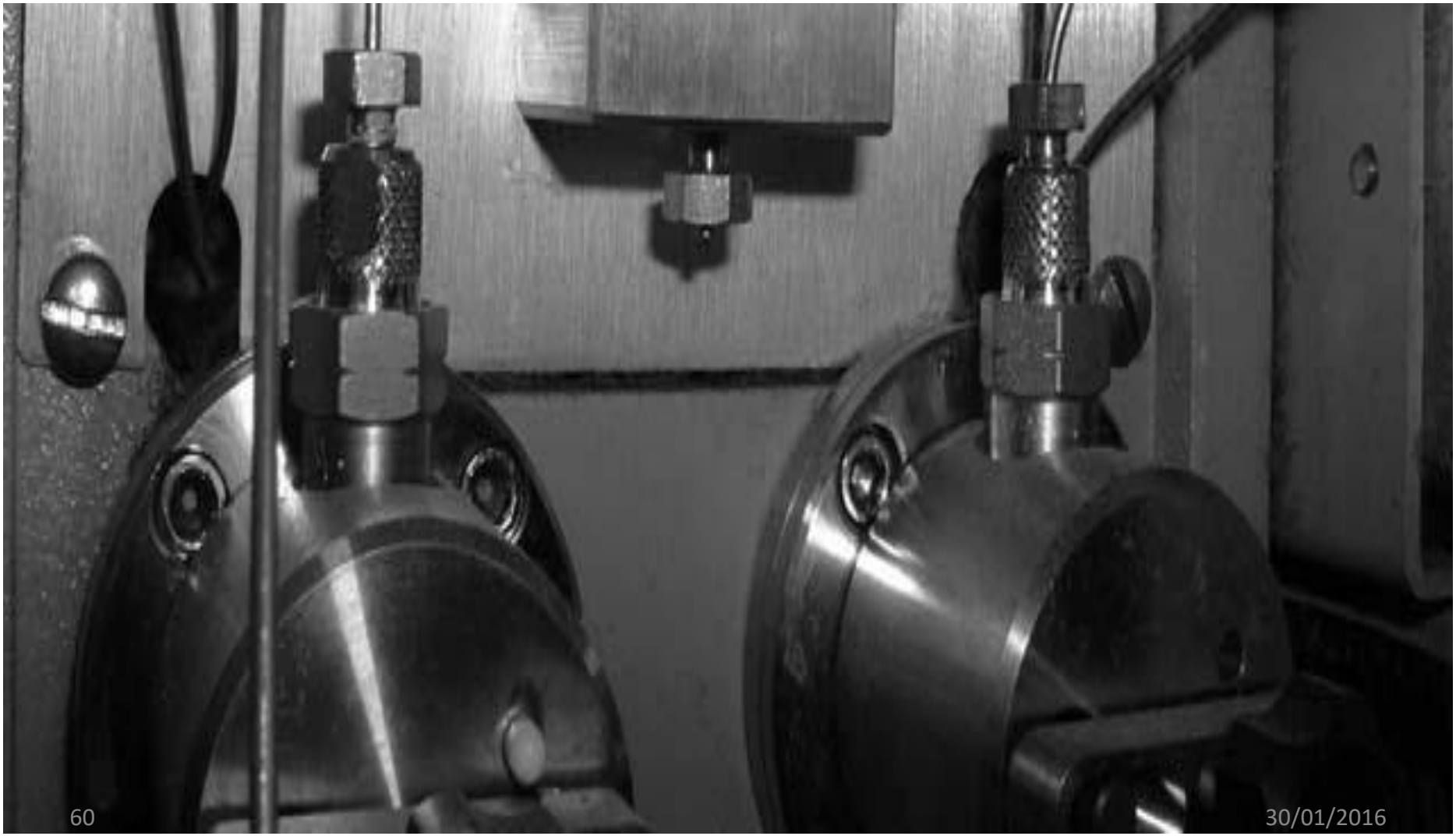
a)High-Pressure Mixing Pumps



b)Low-pressure Mixing Pumps



مضخة ذات الضغط العالي
A High- Pressure Pump



مضخة ذات الضغط العالي
A High- Pressure Pump



مضخة ذات الضغط العالي
A High- Pressure Pump

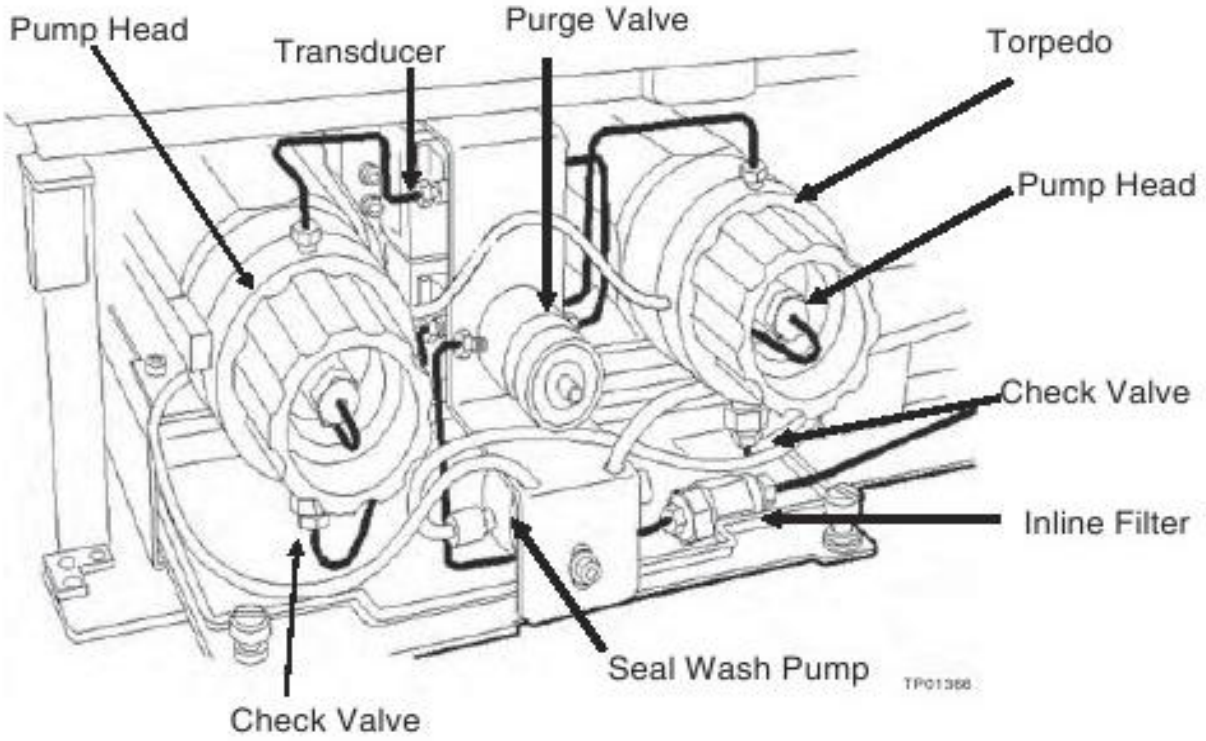


Figure 10.1. Diagram of the inside of the Waters Alliance 2690 pump module after opening the front panel, showing the two pump heads and various components. Note that while the layout on different manufacturers' pumps is not the same, most pumps consist of the same general components and convenience features such as the slide-out tray allowing easy assess for maintenance. Diagram courtesy of Waters Corporation.

مخطط جانبي لمضخة ذات ضغط عالي
Side View Diagram of a High-Pressure Pump

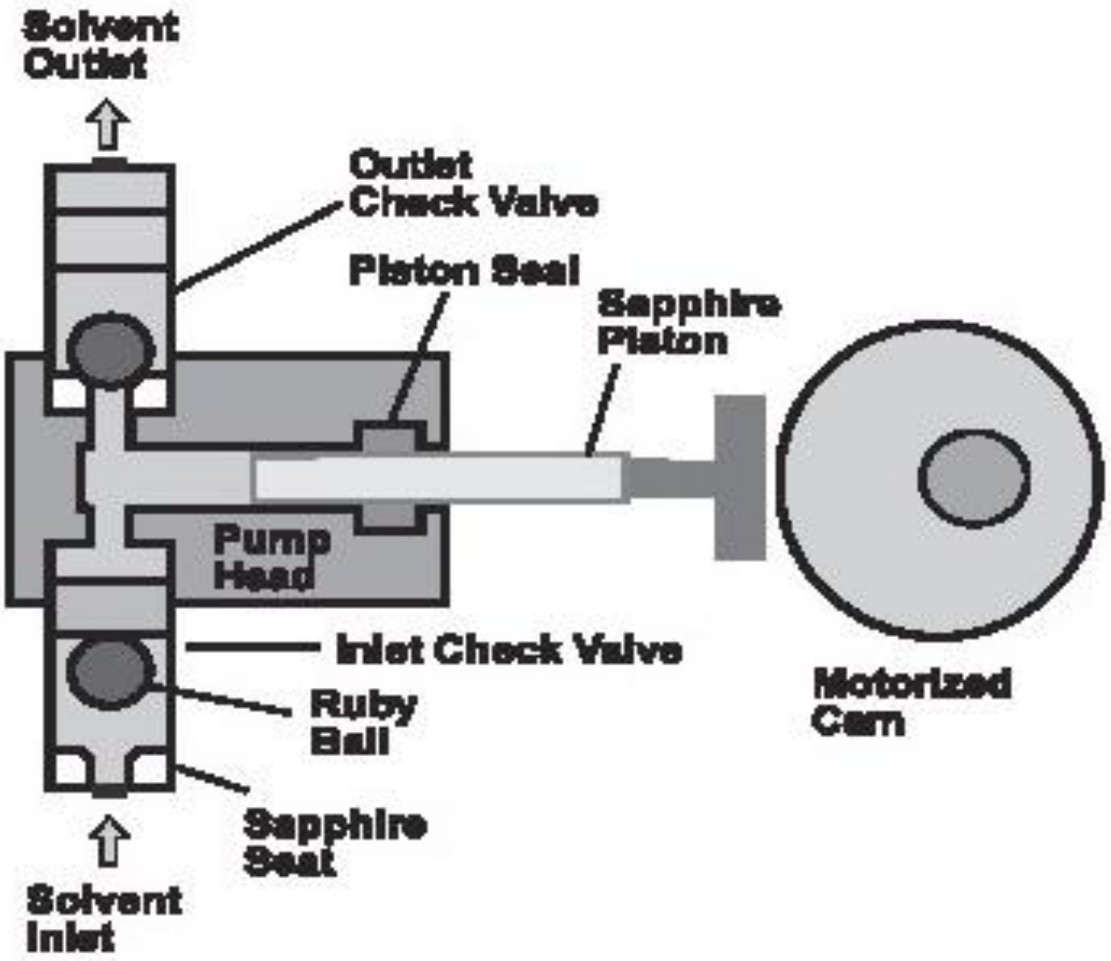


Figure 4.3. A schematic of a reciprocating single-piston pump.

Manual Injection المُحّقن اليدوي
6-Port Valve



Manual Injection المُحّقن اليدوي
6-Port Valve

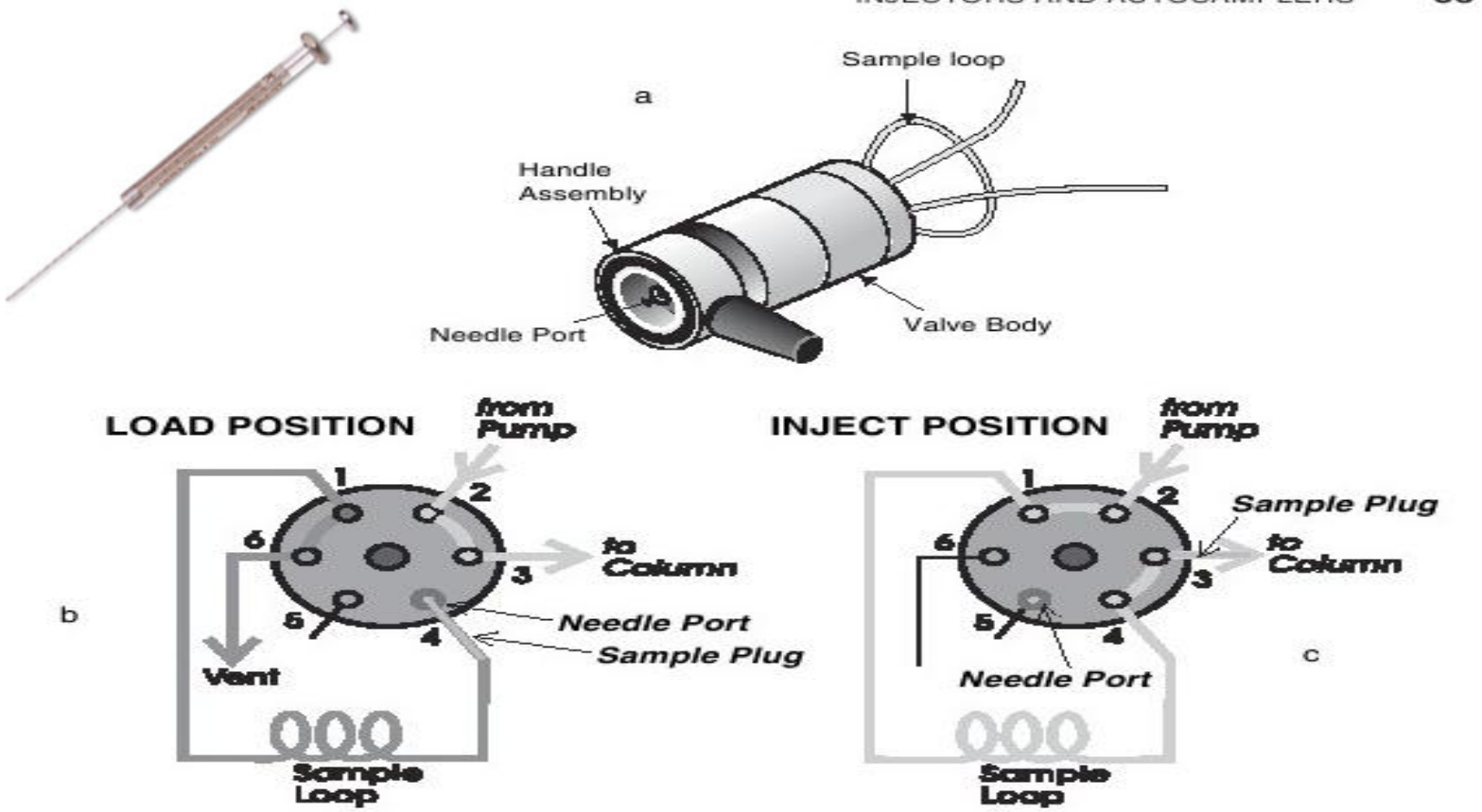


Figure 4.6. (a) A diagram of a manual injector valve. A schematic of a Rheodyne 7125 injector valve during the LOAD (b) and INJECT (c) cycle during a partial-loop injection operation.

Manual سرنجة للمُحّقن اليدوي
Injection 6-Port Valve



عملية حقن يدوي لمُحّقن من نوع 6-Port Valve (manual Injector

Manual Injection Process Using a 6-Port Valve Injector (Manual Injector)



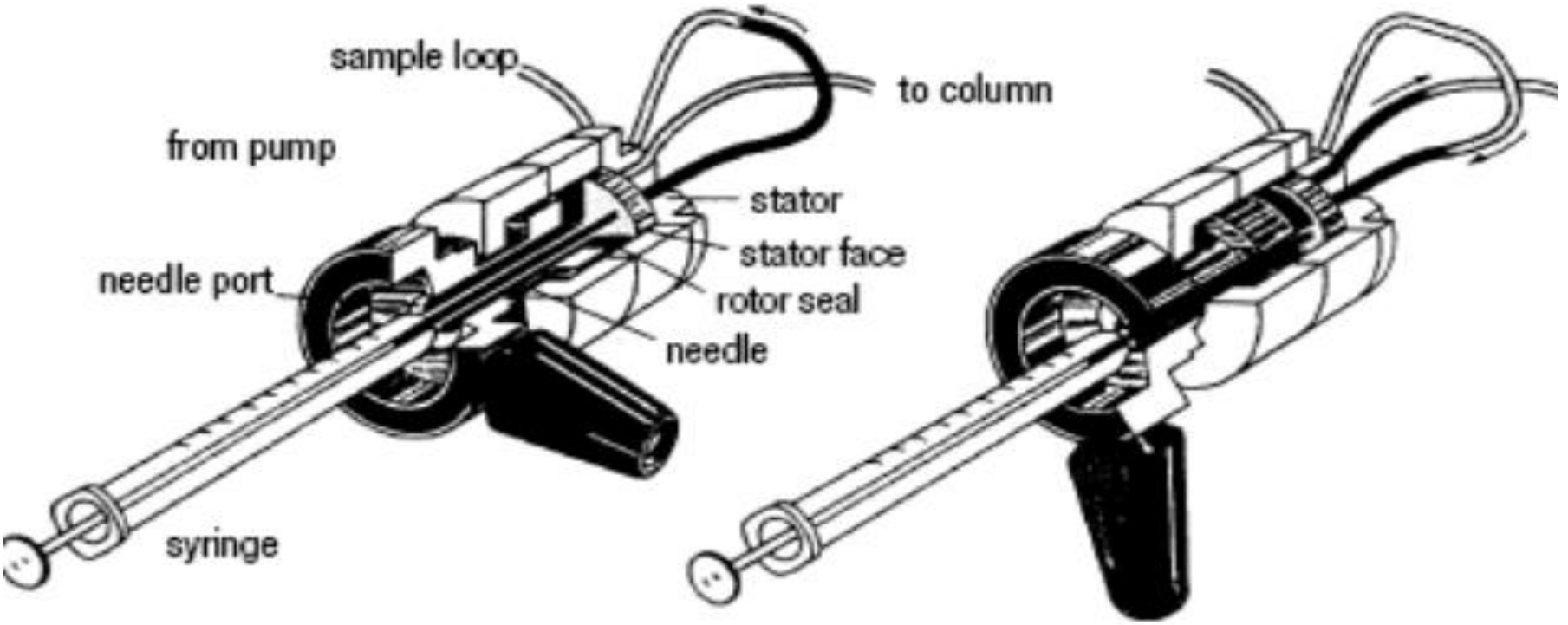


المُحّقن نوع 6-Port valveهو عبارة صمام يدوي لإدخال العينة في تيار الوجه المتحرك Mobile Phaseبواسطة سرنجة ذا طبيعة خاصة يكون رأسها مستوي يتم إدخال العينة في Loopفي حالة تسمى) Loadويكون ذراع المُحقن متجه على وضع In Load)بالكمية المراد فحصها والكمية الزائدة تذهب إلى الفضلات والوجه المتحرك mobile Phaseيكون من المضخة الى عامود الفصل وليس هناك أي احتكاك في هذه الحالة بين العينة والوجه المتحرك

أما في حالة تحريك ذراع الحقن إلى وضع In Injectفيتم تحويل الوجه المتحرك إلى Loopالمحتوي على العينة المراد فحصها فتسحب العينة من خلال تيار الوجه المتحرك إلى عامود الفصل لتحليلها

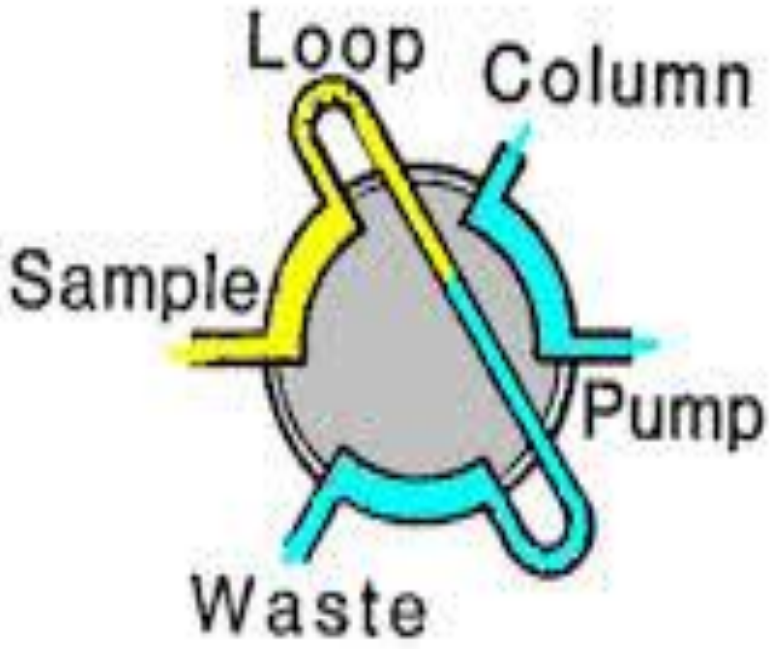
The 6-port valve injector is a manual rotary valve designed to introduce the sample into the mobile phase stream using a specialized syringe with a flat-tip needle. In the **LOAD** position (with the injector lever set to *Load*), the sample is introduced into the sample loop. The exact volume required for analysis fills the loop, while any excess is directed to waste. During this stage, the mobile phase flows directly from the pump to the separation column, with no contact between the sample and the mobile phase.

When the injector lever is rotated to the **INJECT** position, the mobile phase flow is redirected through the sample loop. This sweeps the loaded sample into the mobile phase stream and carries it to the separation column for analysis

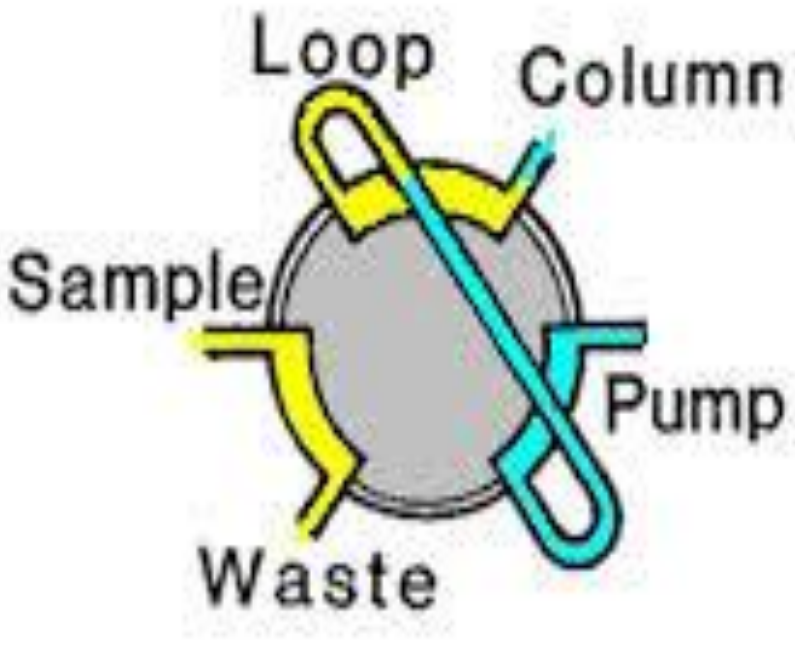


LEFT: In LOAD, mobile phase flows from the pump to the column via a rotor seal groove (not shown). Meanwhile sample is dispensed into the loop, displacing atmospheric-pressure mobile phase out a vent line (not shown).

RIGHT: In INJECT, sample is displaced from the loop to the column by mobile phase from the pump. The needle port is automatically isolated from pressure.


Loop
Column
Sample
Pump
Waste

LOAD


Loop
Column
Sample
Pump
Waste

INJECT

Autosampler المُحّقن المميكن

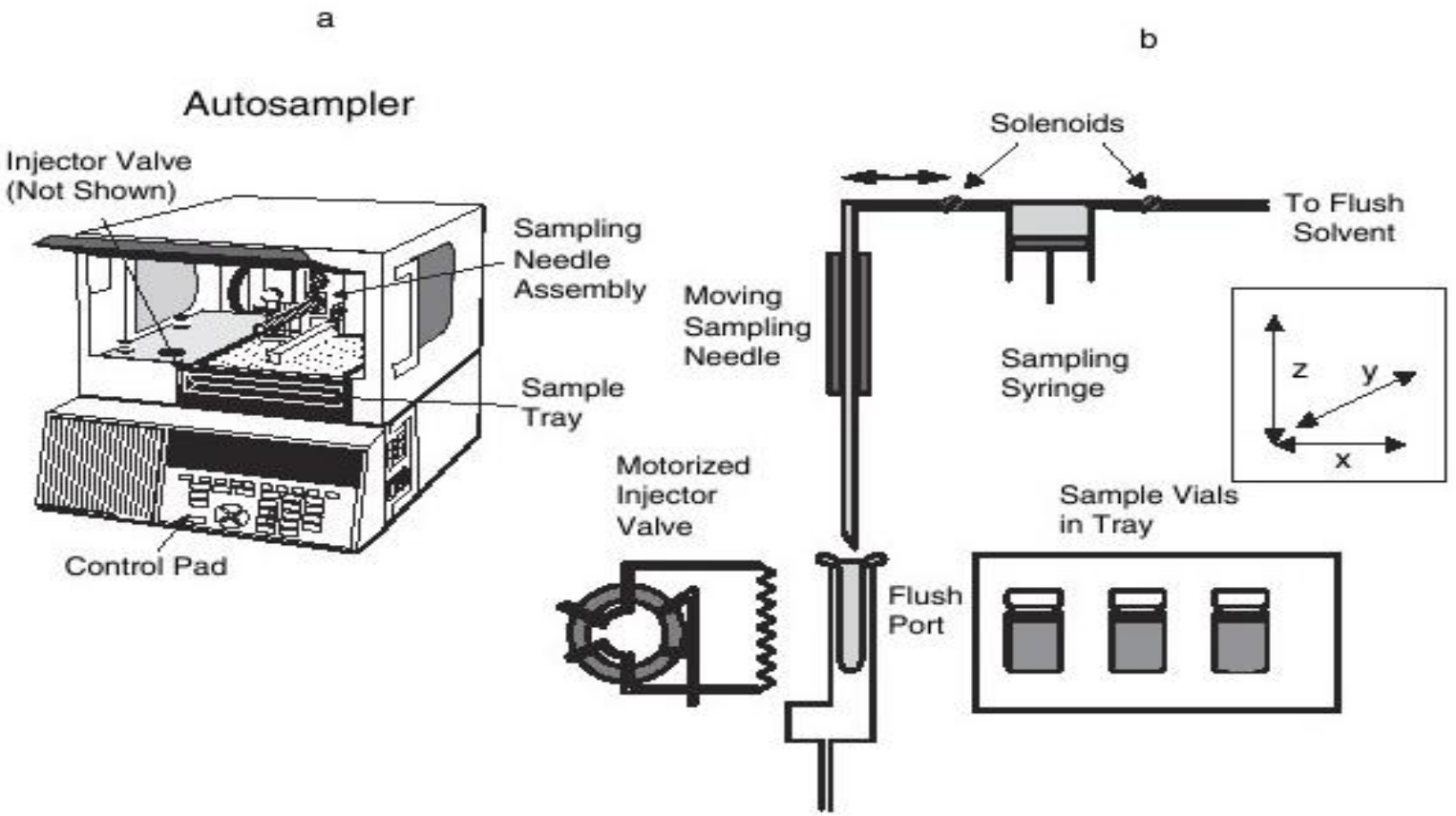


Figure 4.7. A drawing and a schematic of an x-y-z autosampler. Diagram courtesy of PerkinElmer, Inc.

تصنيف أعمدة HPLC حسب مادة التصنيع الخارجي
Classification of HPLC Columns Based on External Housing Material



1) Glass HPLC Column
2) PEEK HPLC Column
3) Stainless Steel Column

بعض صفات عواميد الفصل المستعملة اليوم
Key Characteristics of Modern Separation Columns Used Today

The general characteristics of a typical HPLC analytical column in use today are:

- 50–250 mm long × 2.0–4.6 mm i.d. packed with a 3- or 5-µm, C18 silica-based bonded phase
- Flow rate range of 0.5–2 mL/min using a mobile phase of methanol or acetonitrile in water or aqueous buffer
- Sample loading of 1 ng to 1 mg
- Operating pressure ranges 500–3,000 psi
- Has 4,000–20,000 theoretical plates
- Costs $150–$450
- Typical column life time of 500–2,000 injections or 3–24 months

كيف تصنف أعمدة الفصل من حيث المكونات وصفاتها؟
How are chromatographic columns classified according to their components and characteristics?

- Bonded Groups
  - C18, C8, C4, C3, amino, cyano, phenyl, polar-embedded (for RPC)
  - Diethylaminoethyl, sulfonate, quaternary ammonium, carboxylate (for IEC)
- Particle Size (d_p)
  - 2, 3, 5 µm (>10 µm for prep)
- Pore Size (d_{pore})
  - 60–200 Å
- Surface Area
  - 100–500 m^2/g
- Ligand (Bonded-Phase) Density
  - 2–4 µmole/m^2

Four fundamental HPLC separation principles
هناك أربعة مبادئ للفحوصات على ضوئها يتم اختيار المواد الصلبة (الوجه الثابت)عمود الفصل

There are four fundamental HPLC separation principles, based on which the solid packing materials (stationary phase) of the separation column are selected.

1)Normal-Phase Chromatography (NPC)

2)Reversed –Phase Chromatography (RPC)

3)Ion –Exchange Chromatography (IEC)

4)Size-Exclusive Chromatography (SEC)

توجد أربعة مبادئ أساسية لفصل المواد في كروماتوغرافيا السائل عالية الأداء ((HPLC، ويتم على أساسها اختيار مواد الحشو الصلبة (الطور الثابت) لعمود الفصل:

1) كروماتوغرافيا الطور الطبيعي
NPC-Normal-Phase Chromatography

٢) كروماتوغرافيا الطور المعكوس
RPC-Reversed-Phase Chromatography

٣) كروماتوغرافيا التبادل الأيوني
IEC-Ion-Exchange Chromatography

٤) كروماتوغرافيا الاستبعاد الحجمي
SEC-Size-Exclusive Chromatography

Normal-Phase Chromatography -NPC

في هذه الحالة الوجه الثابت Stationary Phase أي مادة عامود الفصل بولر (Polar) والوجه المتحرك Mobile Phase غير بولر non -Polar
حيث تسرع المواد المفصولة الغير بولر بسرعة اكبر من المواد البولر.

In this mode, the stationary phase (i.e., the column packing material) is polar, while the mobile phase is non-polar. Consequently, non-polar analytes elute faster than polar ones.

Reversed –Phase Chromatography -RPC

في هذه الحالة يكون الوجه الثابت Stationary Phaseأي مادة عامود الفصل غير بولر non -Polarوالوجه المتحرك Mobile Phaseبولر حيث تسرع المواد المفصولة بولر بسرعة اكبر من المواد الغير البولر (Polar)

In this mode, the stationary phase (i.e., the column packing material) is non-polar, while the mobile phase is polar. Consequently, polar analytes elute faster than non-polar ones

أشكال الفصل في عواميد HPLC حسب المادة المالئة للعامود
HPLC Separation Modes Based on Column Packing Materia

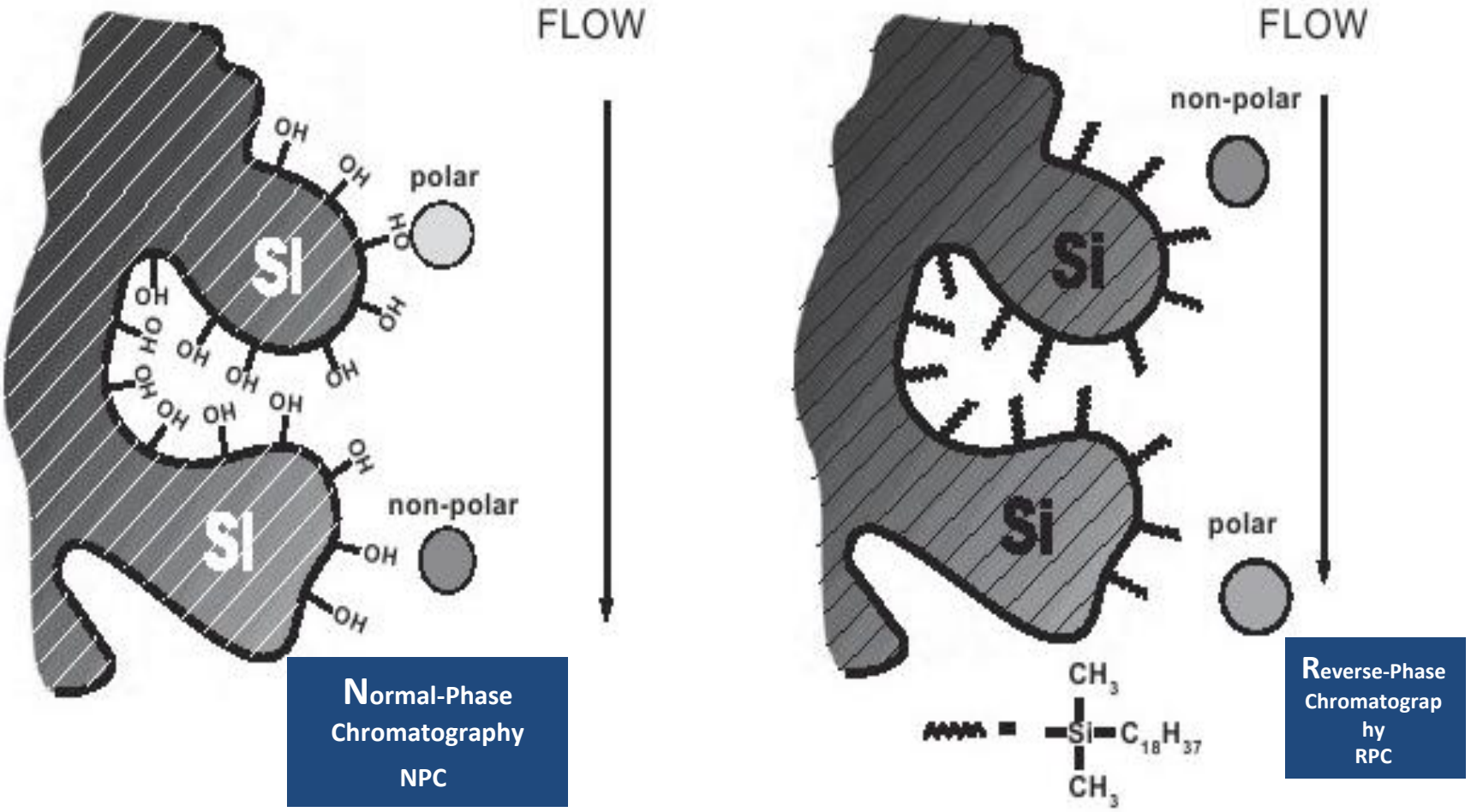
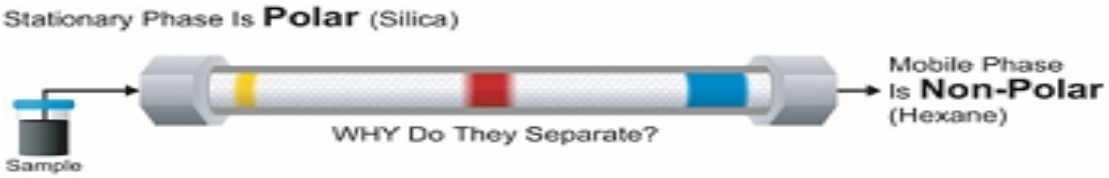
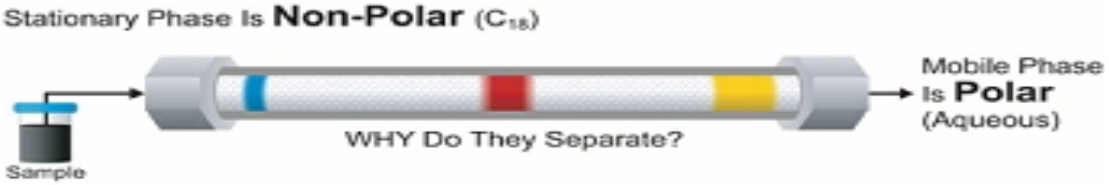


Figure 1.3. Schematic diagrams depicting separation modes of (a) normal-phase chromatography (NPC) and (b) reversed-phase chromatography (RPC).



Normal-Phase HPLC Chromatography



Reversed-Phase HPLC Chromatography

كروماتوجرفي الترشيح خلال الجيل
Gel Filtration Chromatography(SEC)
Size-exclusion Chromatography-SEC

يعتمد هذا التكنيك على مقدرة مواد مسامية معينة Porous Materialsمتواجدة في صورة جيل مثل مادة السفادكس Sephadexوالمعبأة في عمود أن تقوم بفصل المكونات الموجودة في المخلوط وذلك حسب حجم وشكل الجزيئات

Size and Shape of the Molecules

This technique relies on the ability of certain porous materials, present in gel form—such as Sephadex—and packed into a column, to separate the components of a mixture based on the size and shape of the molecules

كرماتوجرافي التبادل الأيوني
IEC)-Ion Exchange Chromatography)

وهذا التكنيك يعتمد على ميل الايونات المتواجدة في المحلول والمحملة بشحنات كهربائية *للاتجاه نحو ألأماكن المحملة بشحنة كهربائية معاكسة* Opposite Polarity بالوجه الثابت .

This technique relies on the tendency of ions in solution, which carry electrical charges, to migrate toward oppositely charged sites on the stationary phase.

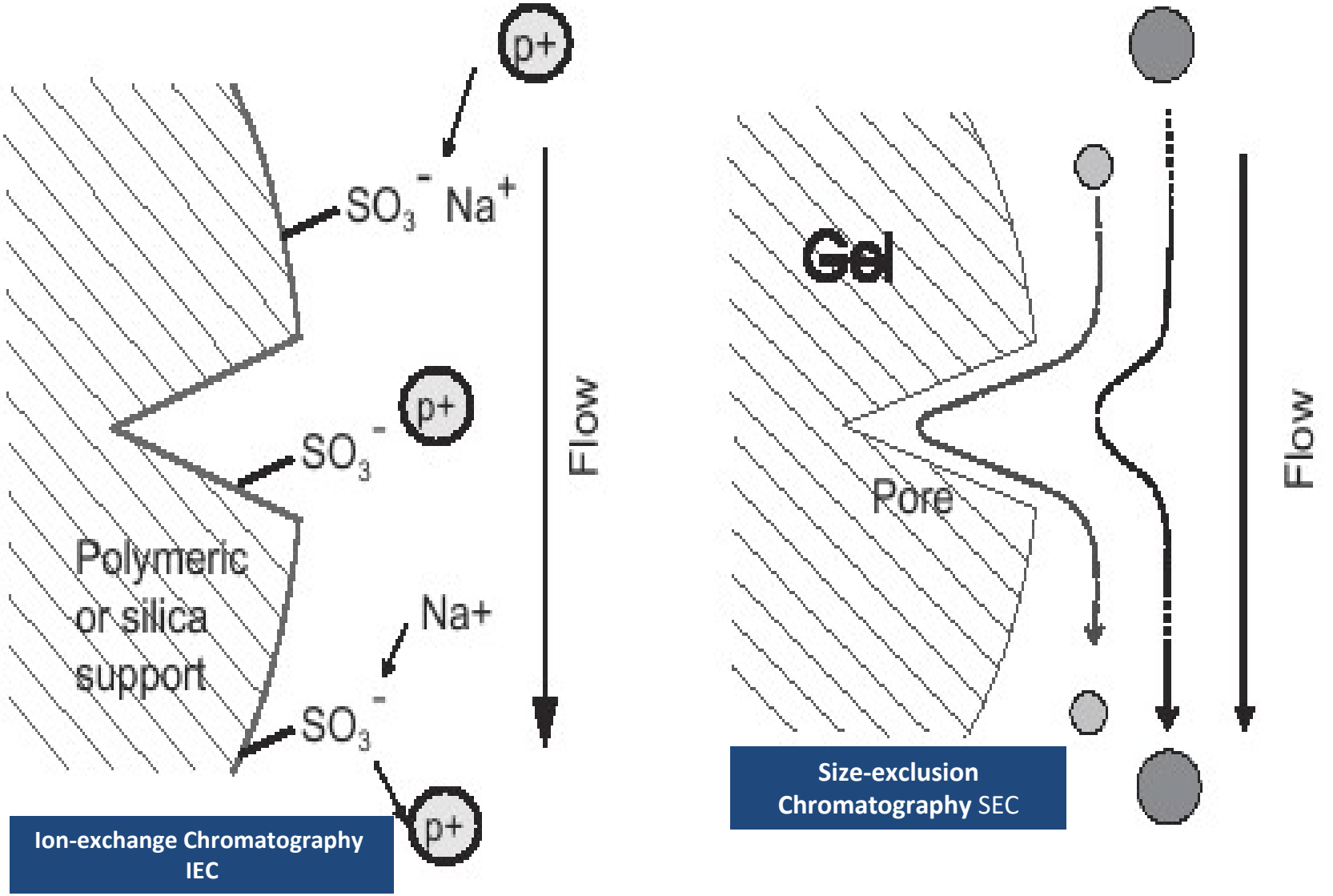


Figure 1.6. a. Schematic diagrams depicting separation modes of (a) ion-exchange chromatography (IEC), showing the exchange of analyte ion p^+ with the sodium counter ions of the bonded sulfonate groups; (b) size-exclusion chromatography (SEC), showing the faster migration of large molecules.

بعض اشكال التركيب الكيميائي للمواد المالئة لعامود الفصل

Some Forms of the Chemical Structure of Column Packing Materials for Separation Columns

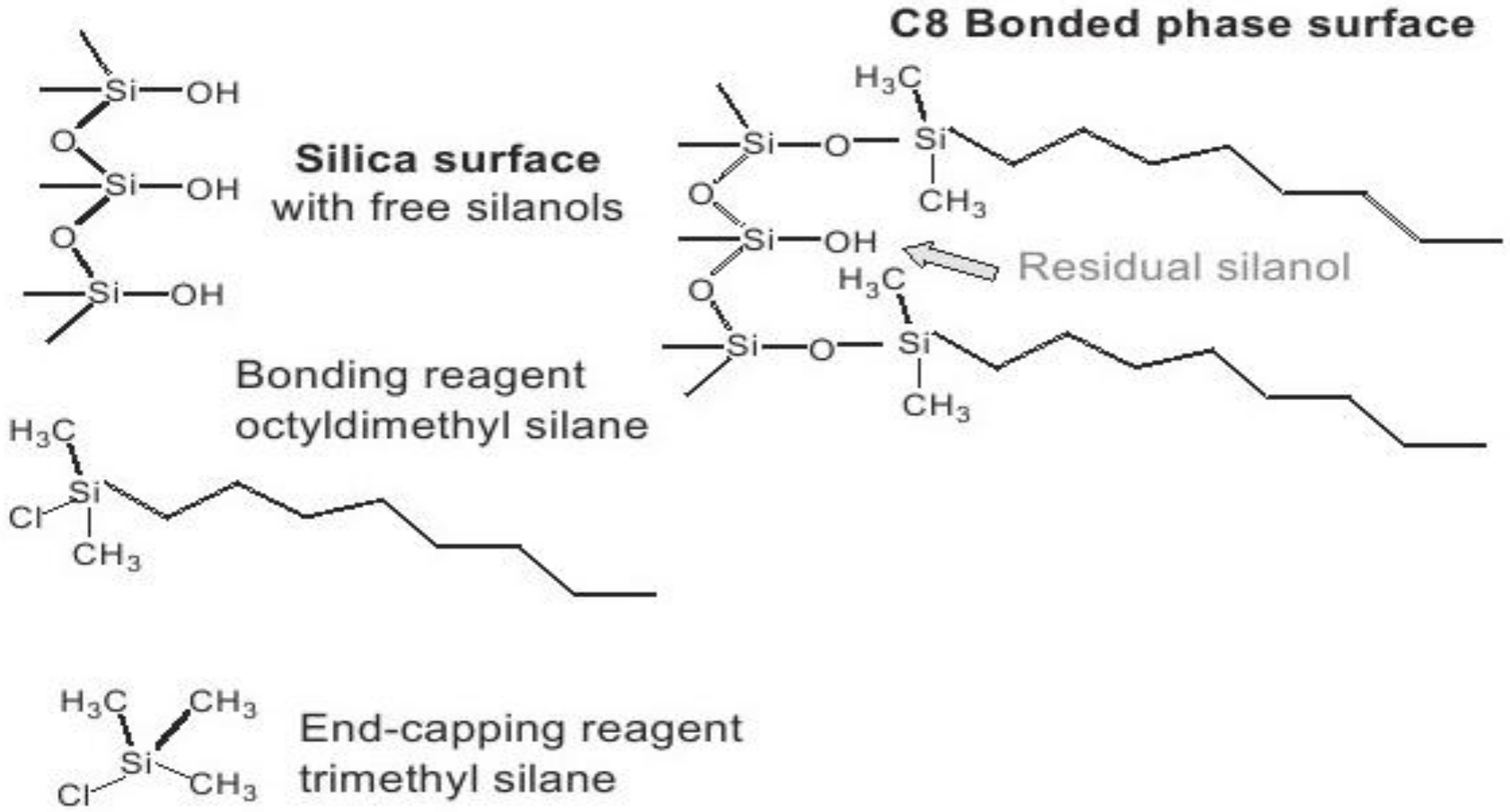


Figure 3.4. Structures of the typical bonding and end-capping reagents and diagram of the bonded silica with residual silanols.

بعض اشكال التركيب الكيميائي للمواد المالئة لعامود الفصل

Some Forms of the Chemical Structure of Column Packing Materials for Separation Columns

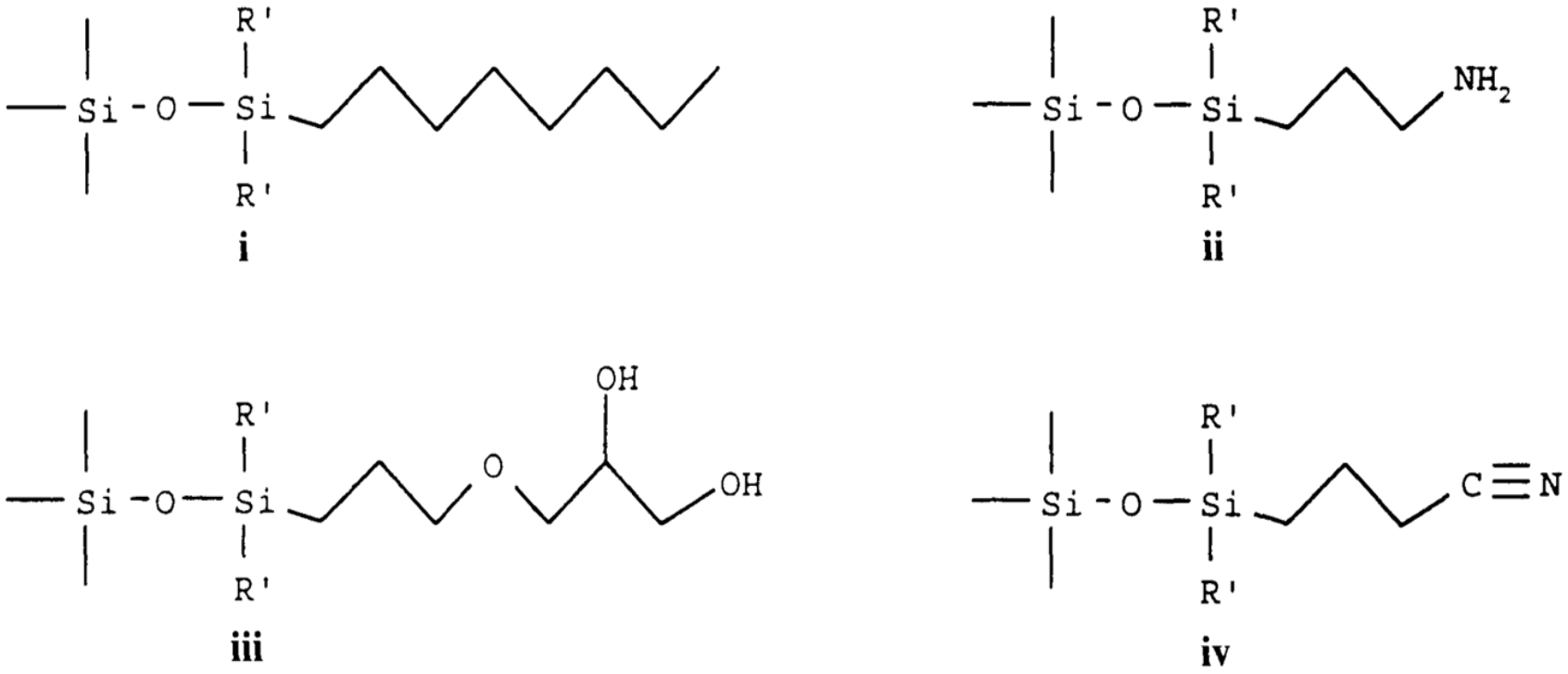


Figure 7 Chemical structure of functional ligands immobilized on silica surfaces: (i) octylsiloxane, (ii) aminopropylsiloxane, (iii) diol-substituted propylsiloxane, (iv) cyanopropylsiloxane.

بعض الامثلة لانواع عواميد HPLC حسب المادة المالئة
Examples of HPLC Column Types Classified by Packing Material"

Reversed-Phase (RPC)

C18	Octadecyl	$Si(CH_3)_2$- $(CH_2)_{17}$ - CH_3
C8	Octyl	$Si(CH_3)_2$- CH_2-CH_2- CH_2- CH_2- CH_2- CH_2- CH_2- CH_3
CN	Cyano	$Si(CH_3)_2$- CH_2- CH_2- CH_2- CN
ϕ	Phenyl	$Si(CH_3)_2$- C_6H_6
Polar-embedded (Amide)		$Si(CH_3)_2$- $(CH_2)_3NHCO(CH_2)_{14}CH_3$

Normal-Phase (NPC)

Si	Silica	Si - OH
NH_2	Amino	$Si(CH_3)_2$- CH_2- CH_2- CH_2- NH_2

Ion-Exchange (IEC)

SP	Sulfo propyl	$Si(CH_3)_2$ - CH_2- CH_2- CH_2- SO_3^-
CM	Carboxymethyl	$Si(CH_3)_2$-CO_2^-
DEAE	Diethylaminoethyl	$Si(CH_3)_2$ - CH_2-CH_2-CH_2-NH^+ $(C_2H_5)_2$
SAX	Triethylaminopropyl	$Si(CH_3)_2$-CH_2-CH_2-CH_2-$N^+(C_2H_5)_3$

Figure 3.7. Structures of some common bonded phases.

Mobile Phase Chromatographic Polarity Spectrum



Figure (R-1) Mobile Phase Chromatographic Polarity Spectrum

Chromatographic Polarity Spectrum by Analyte Functional Group
تحليل طيف الاستقطاب الكروماتوجرفي للمجموعات الوظيفية

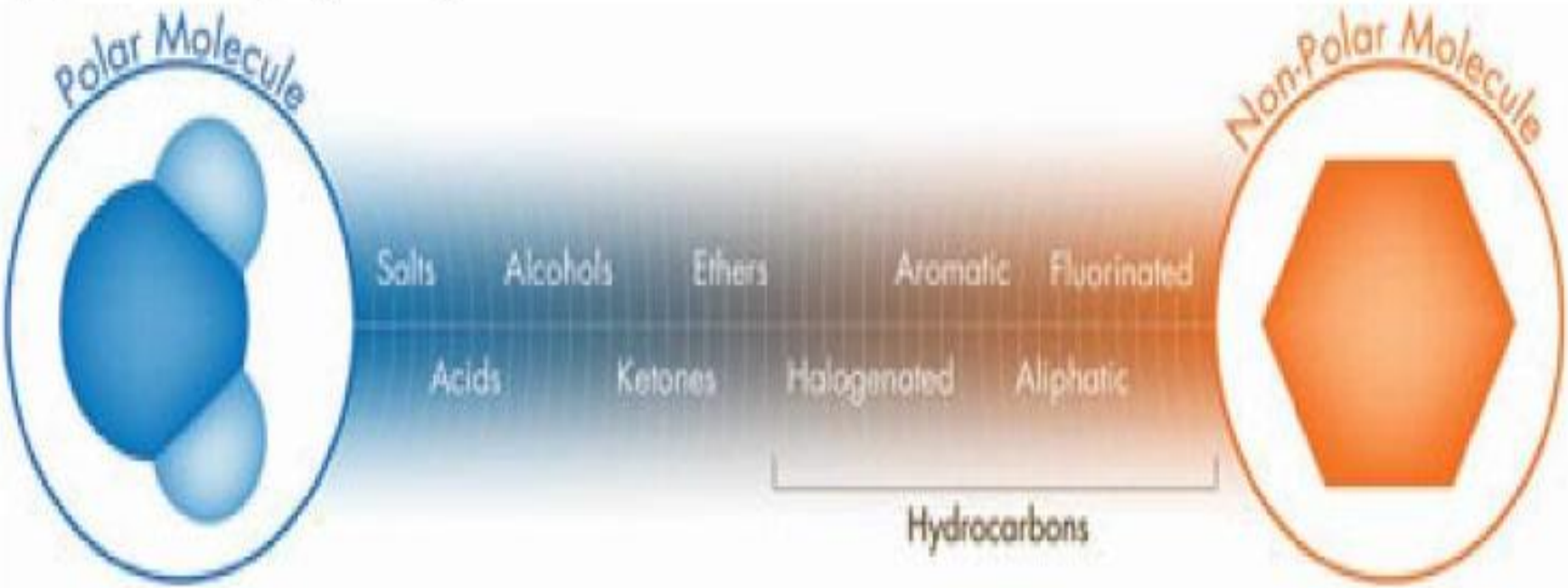


Figure (P): Chromatographic Polarity Spectrum by Analyte Functional Group

Chromatographic Polarity Spectrum Stationary Phase Particle
تحليل طيف الاستقطاب الكروماتوجرفي لمكونات الوجه الثابت



Figure (R-2) Stationary Phase Particle Chromatographic Polarity Spectrum

Compound Analyte Chromatography Polarity Spectrum
تحليل طيف الاستقطاب الكروماتوجرفي لمكونات العينة

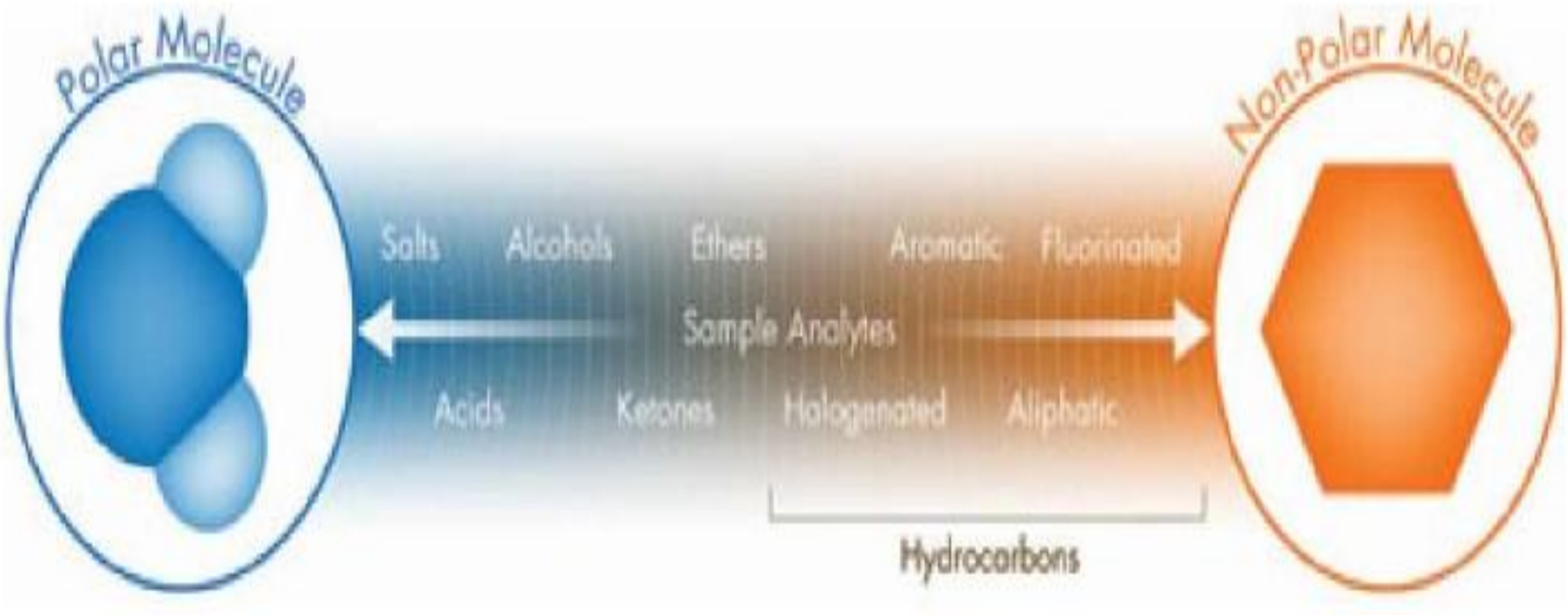


Figure (R-3) Compound/Analyte Chromatographic Polarity Spectrum

بعض صفات عواميد الفصل المستعملة اليوم
Key Characteristics of Modern Separation Columns Used Today

The general characteristics of a typical HPLC analytical column in use today are:

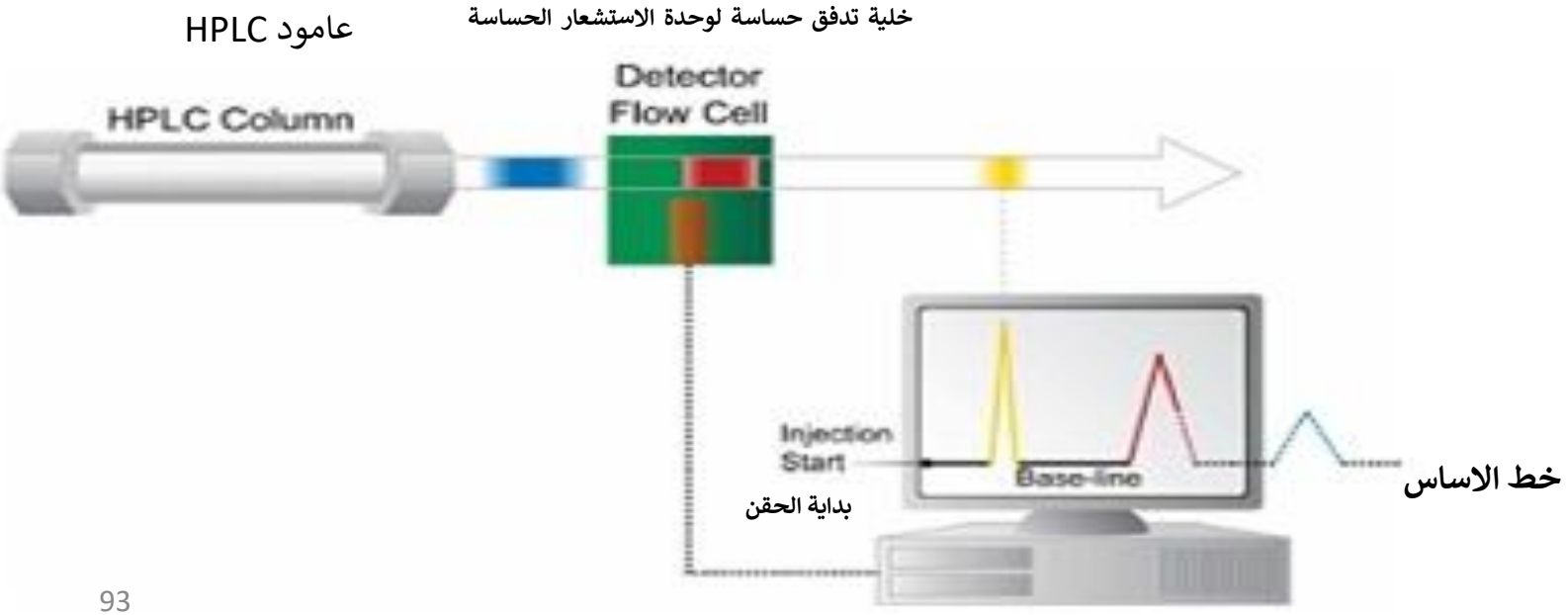
- 50–250 mm long × 2.0–4.6 mm i.d. packed with a 3- or 5-µm, C18 silica-based bonded phase
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- Costs $150–$450
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  - Diethylaminoethyl, sulfonate, quaternary ammonium, carboxylate (for IEC)
- Particle Size (d_p)
  - 2, 3, 5 µm (>10 µm for prep)
- Pore Size (d_{pore})
  - 60–200 Å
- Surface Area
  - 100–500 m^2/g
- Ligand (Bonded-Phase) Density
  - 2–4 µmole/m^2

نظام الاستشعار لوحدة الاستشعار الحساسة Detector

Sensing System of the Detector Module



Detector مثال على وحدة استشعار الحساسة

An Example of a Sensitive Detector Unit"

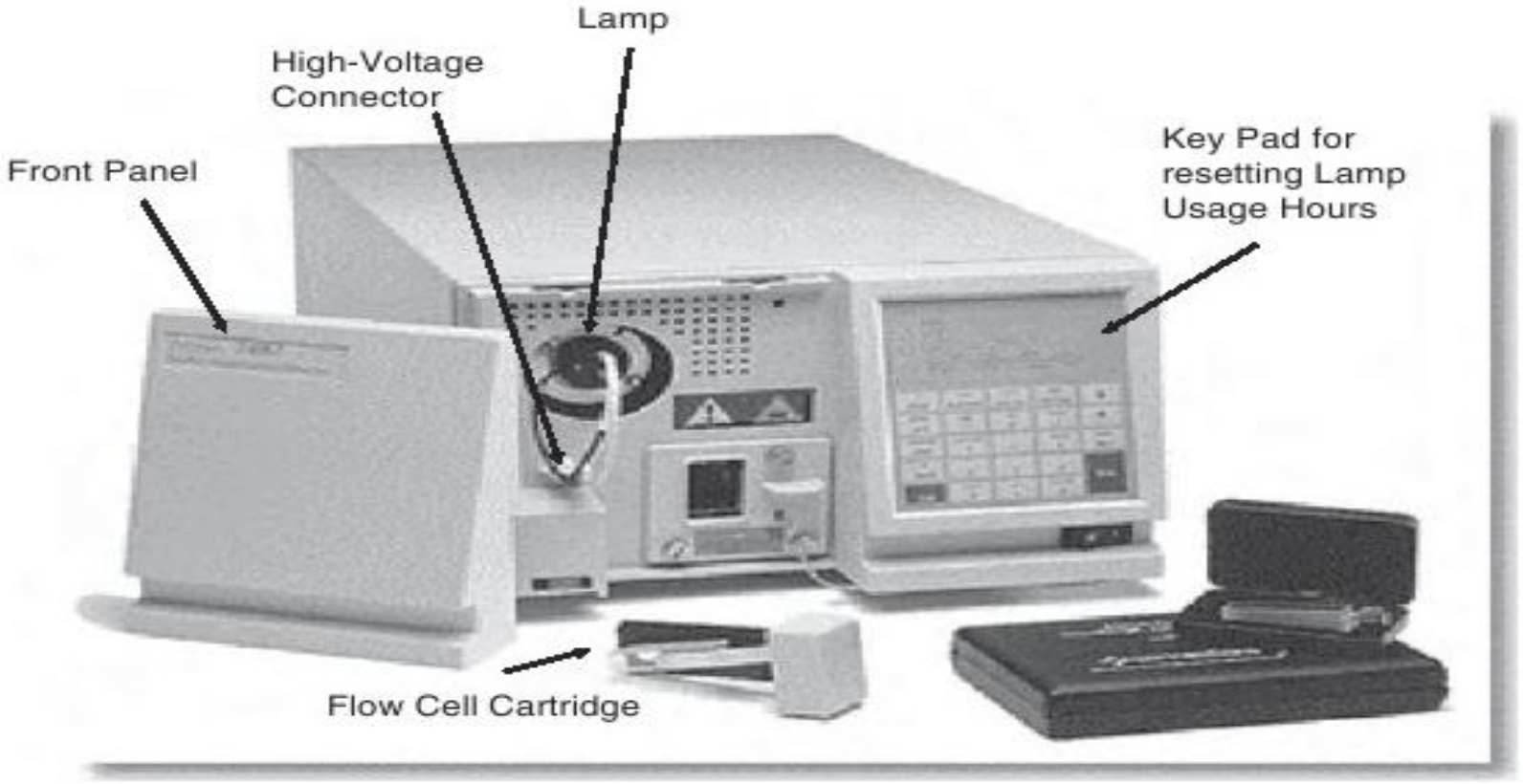


Figure 10.4. A picture of a Waters 2487 UV/Vis absorbance detector with the front panel removed showing the position of the UV lamp (source) and the flow cell cartridge. Picture courtesy of Waters Corp.

Flow Cell of the Detector

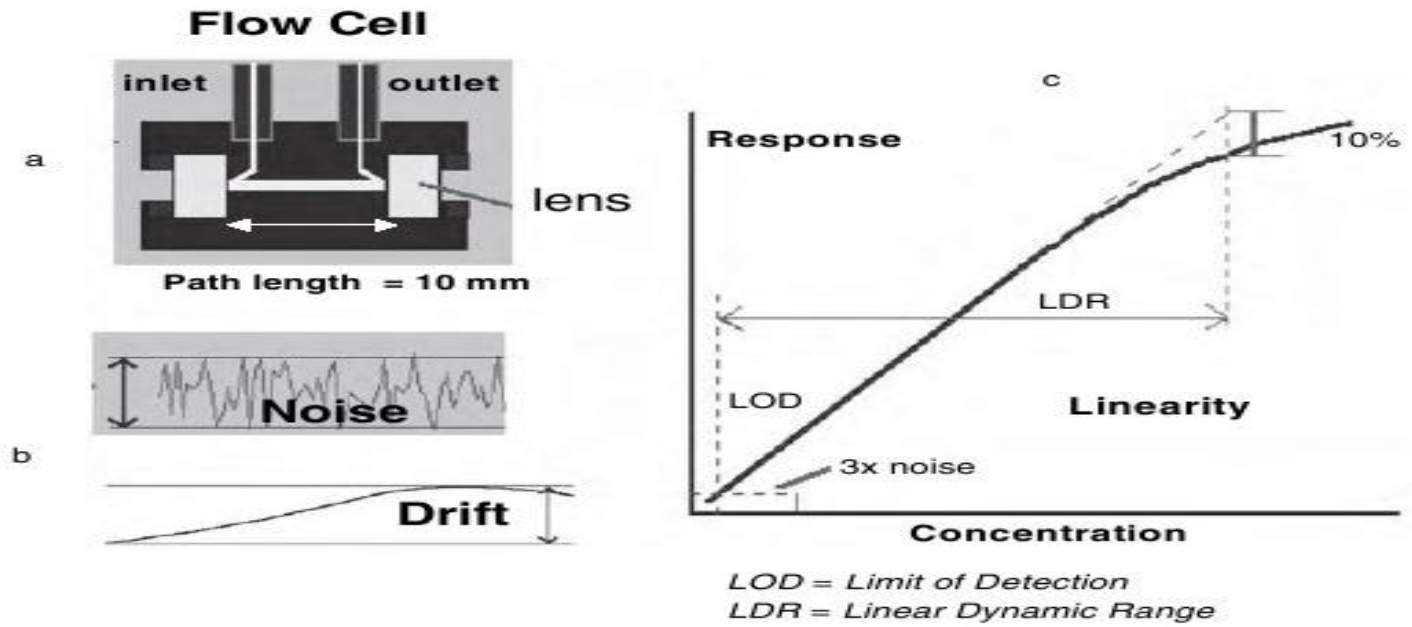


Figure 4.9. (a) Schematic diagram of a flow cell and (b) baseline chromatograms showing noise (magnified) and drift. (c) Chart of UV response with concentration of the analyte injected. Linearity range is from limit of detection (LOD) to the point deviating 10% from the linear response. Diagrams courtesy of Academy Savant.

Common HPLC Detector

Table 4.2. Common HPLC Detectors and Attributes

Detector	Analyte/attributes	Sensitivity
UV/Vis absorbance (UV/Vis)	Specific: Compounds with UV chromophores	ng–pg
Photo diode array (PDA)	Specific: Same as UV/Vis detectors, also provides UV spectra	ng–pg
Fluorescence (Fl)	Very specific: Compounds with native fluorescence or with fluorescent tag	fg–pg
Refractive index (RI)	Universal: polymers, sugars, triglycerides, organic acids, excipients; not compatible with gradient analysis	0.1–10 µg
Evaporative light scattering (ELSD)	Universal: nonvolatile or semivolatile compounds, compatible with gradient analysis	10 ng
Corona-charged aerosol (CAD)	Universal: use nebulizer technology like ELSD and detection of charges induced by a high-voltage corona wire	Low ng
Chemiluminescence nitrogen (CLND)	Specific to N-containing compounds based on pyro-chemiluminescence	<0.1 ng of nitrogen
Electrochemical	Very specific: Electro-active compounds (Redox)	pg
Conductivity	Specific to anions and cations, organic acids, surfactants	ng or ppm–ppb
Radioactivity	Specific, radioactive-labeled compounds	Low levels
Mass spectrometry (MS) MS/MS	Both universal and specific, structural identification; very sensitive and specific	ng–pg pg–fg
Nuclear magnetic resonance (NMR)	Universal, for structure elucidation and confirmation	mg–ng

تركيب بعد الوحدات الاستشعار الحساسة Detector
Schemantic of UV_VIS and PDA Detector

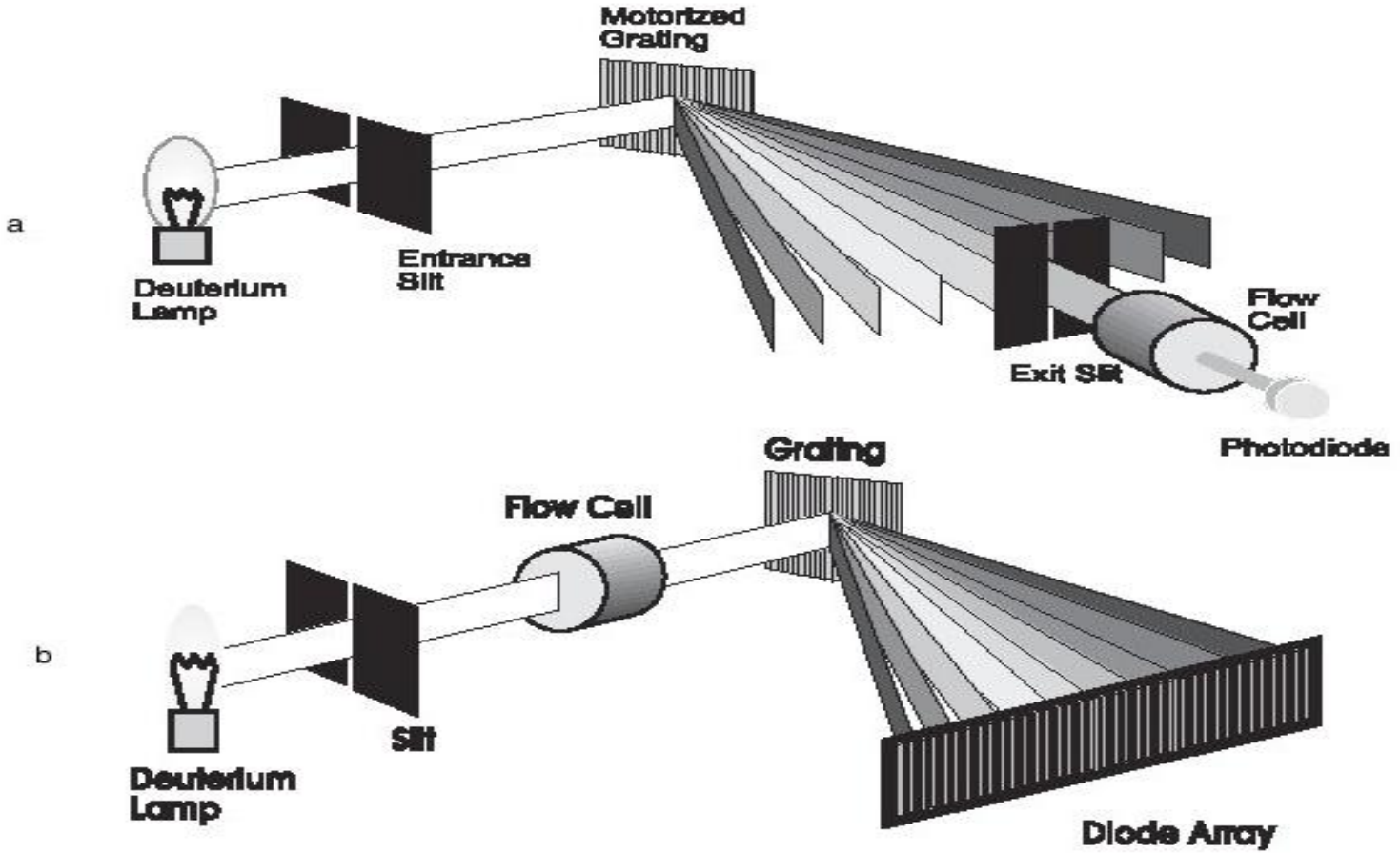
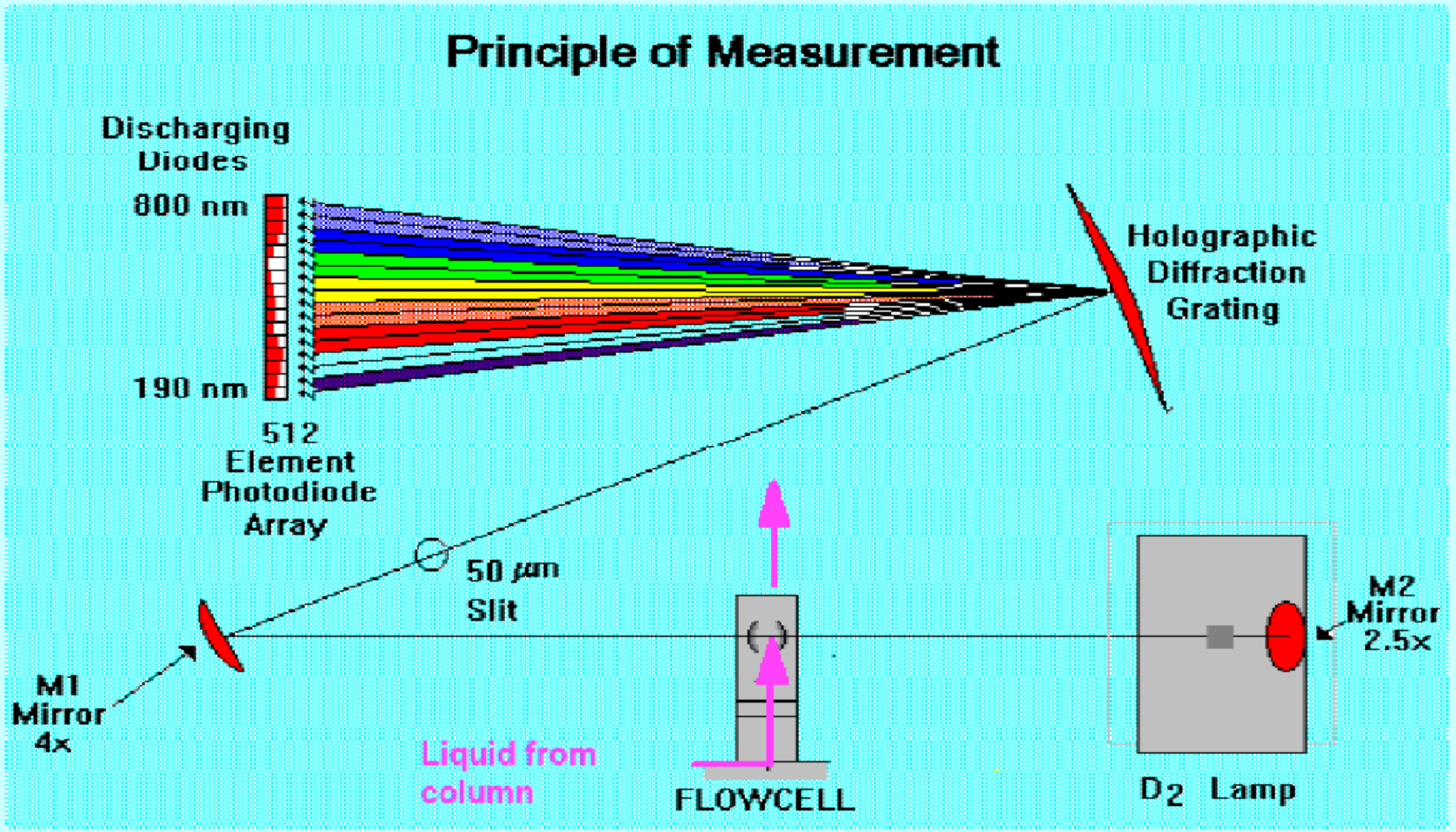


Figure 4.8. A schematic of a UV-Vis absorbance detector (a) and a photodiode array (PDA) detector (b). Diagrams adapted from and reprinted with permission from Academy Savant.

Photodiode array Detector



Dr. Shulamit Levin

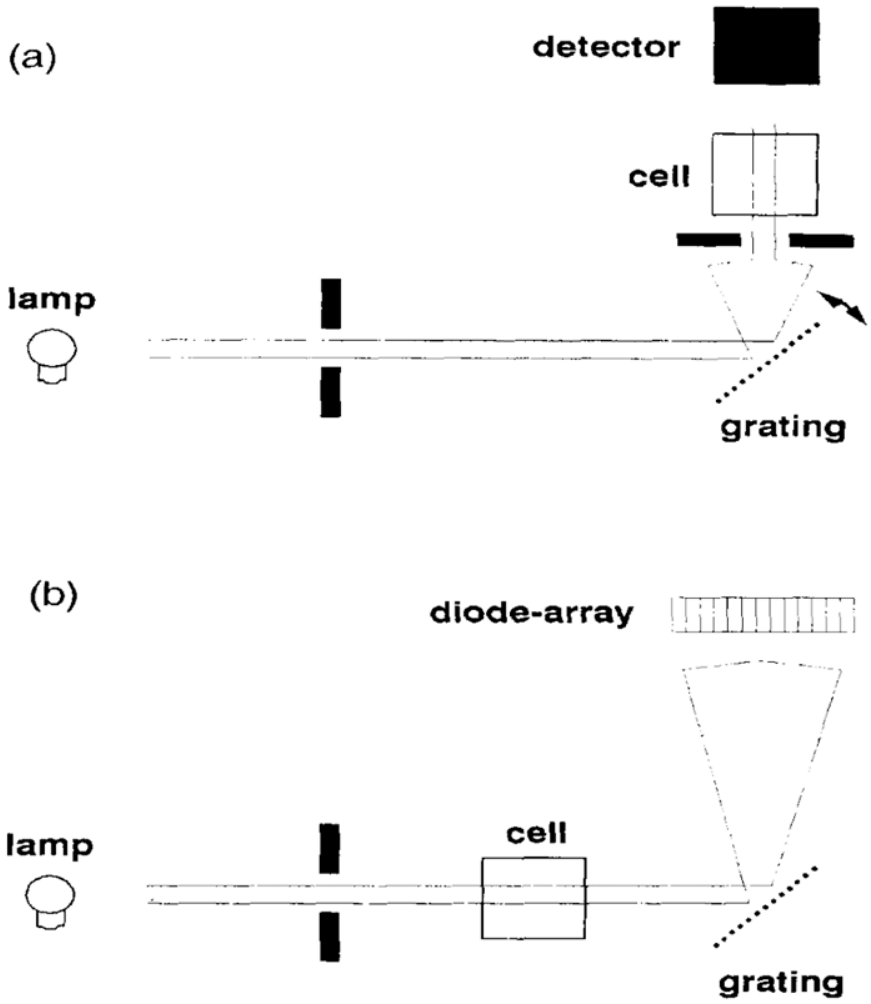


Figure 3 Optical arrangements for multiple wavelength detection: (a) rapidly scanning detector; (b) diode-array detector.

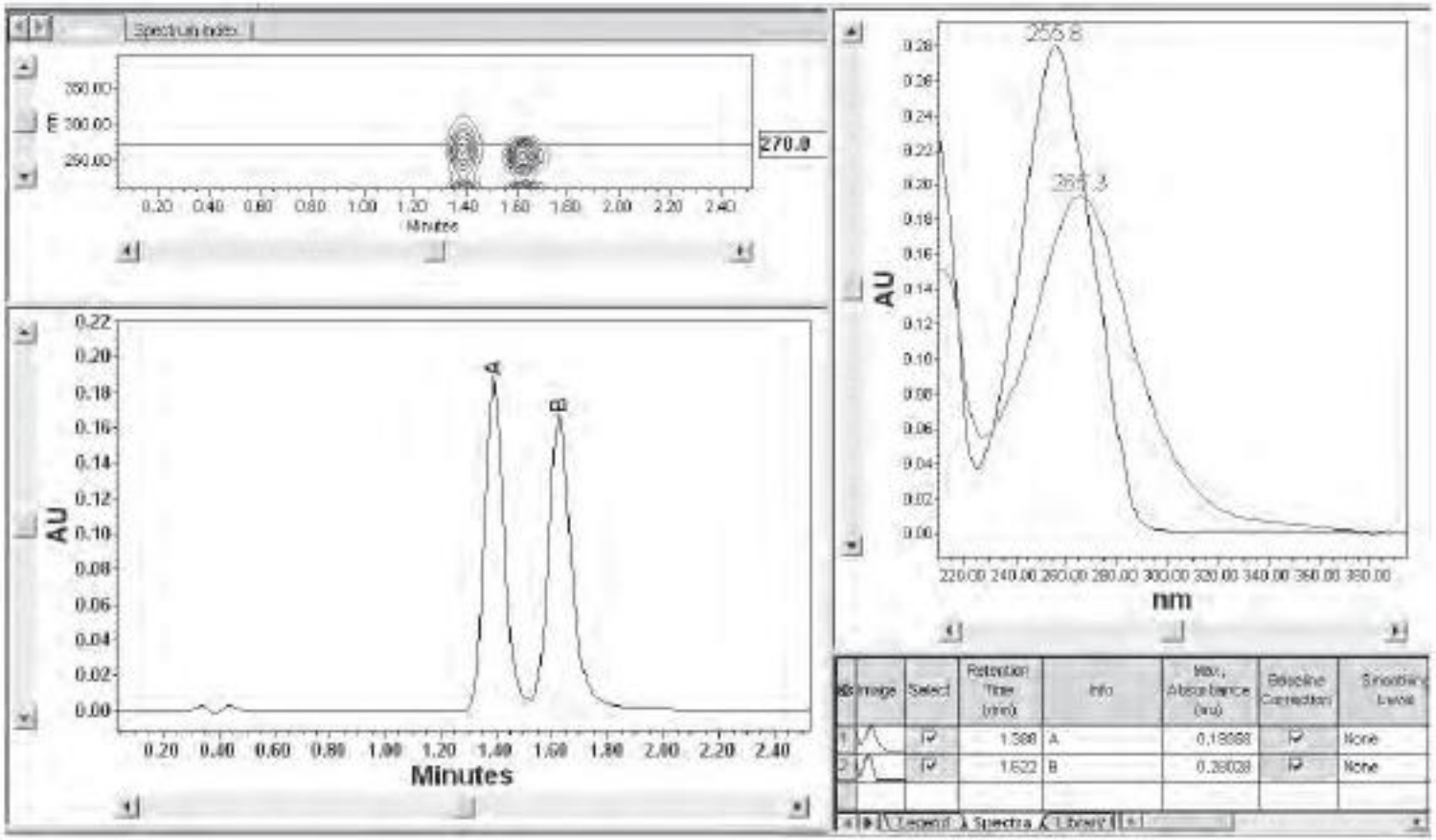


Figure 4.10. Waters EmPower screen showing a contour map, a chromatogram at 270 nm showing the separation of nitrobenzene and propylparaben. Spectra of these two components are shown in the right-hand panel annotated with their respective maximum absorbance wavelengths.

ما الفرق بين

UV-VIS Absorbance و Photodiode array (PDA)) Detectors)

UV-VIS Absorbance Detectorلا يمكنه
قراءة أكثر من wave lengthطول موجه وذلك انه
يعمل على تصفية كل الأمواج الضوئية ويأخذ هذه
الموجة واحدة للتحليل .
أما Photodiode array (PDA Detector)يمكنه
قراءة أكثر من موجة في نفس الوقت

الطيف Spectrum

1) في منطقة الطيف المنظور Visible Spectrum (400-750 نانومتر) ويكون مصدر هذا الطيف ال Tungsten Halogen lamp

2) في منطقة الطيف فوق البنفسجي Ultra-violet Spectrum من 185-400 نانومتر) ومصدر هذا الطيف Deuterium Lamp

تركيب بعض وحدات الاستشعار الحساسة Detectors

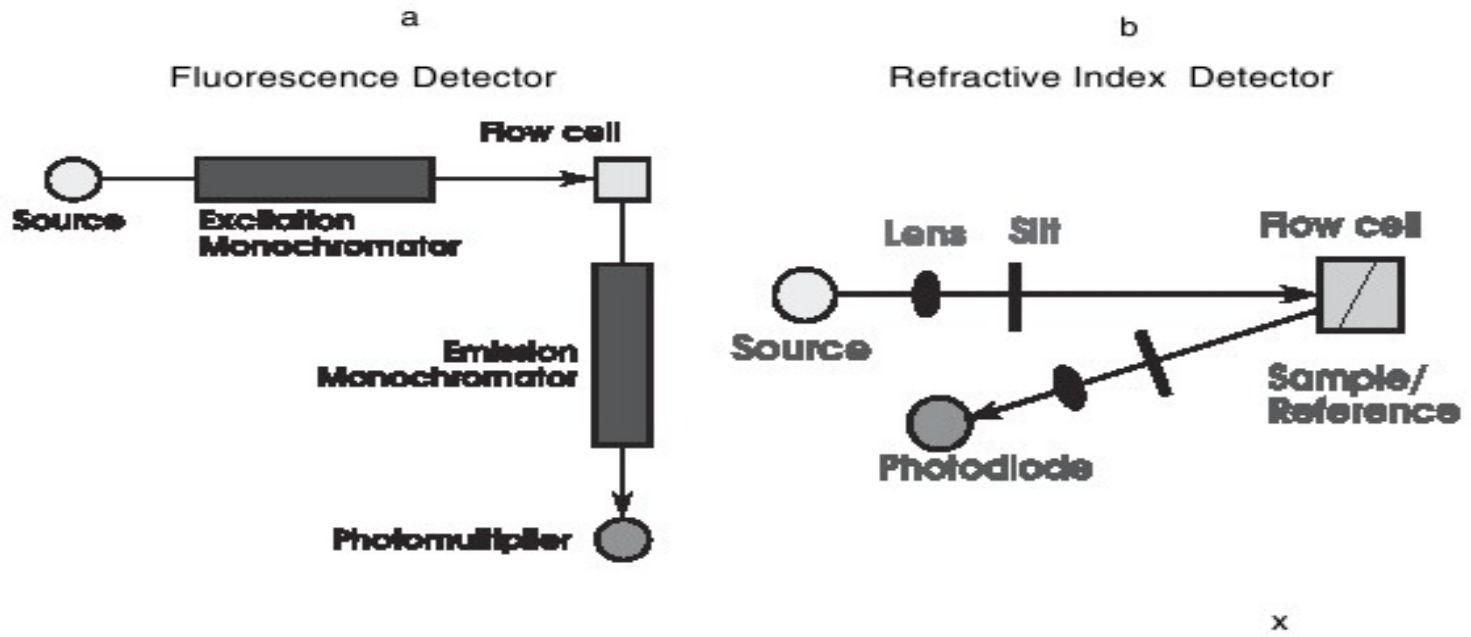


Figure 4.12. (a) Schematic diagram of a fluorescence detector with dual monochromators and a square flow cell. (b) Schematic diagram of a refractive index detector.

Degasser نظام طرد الغازات

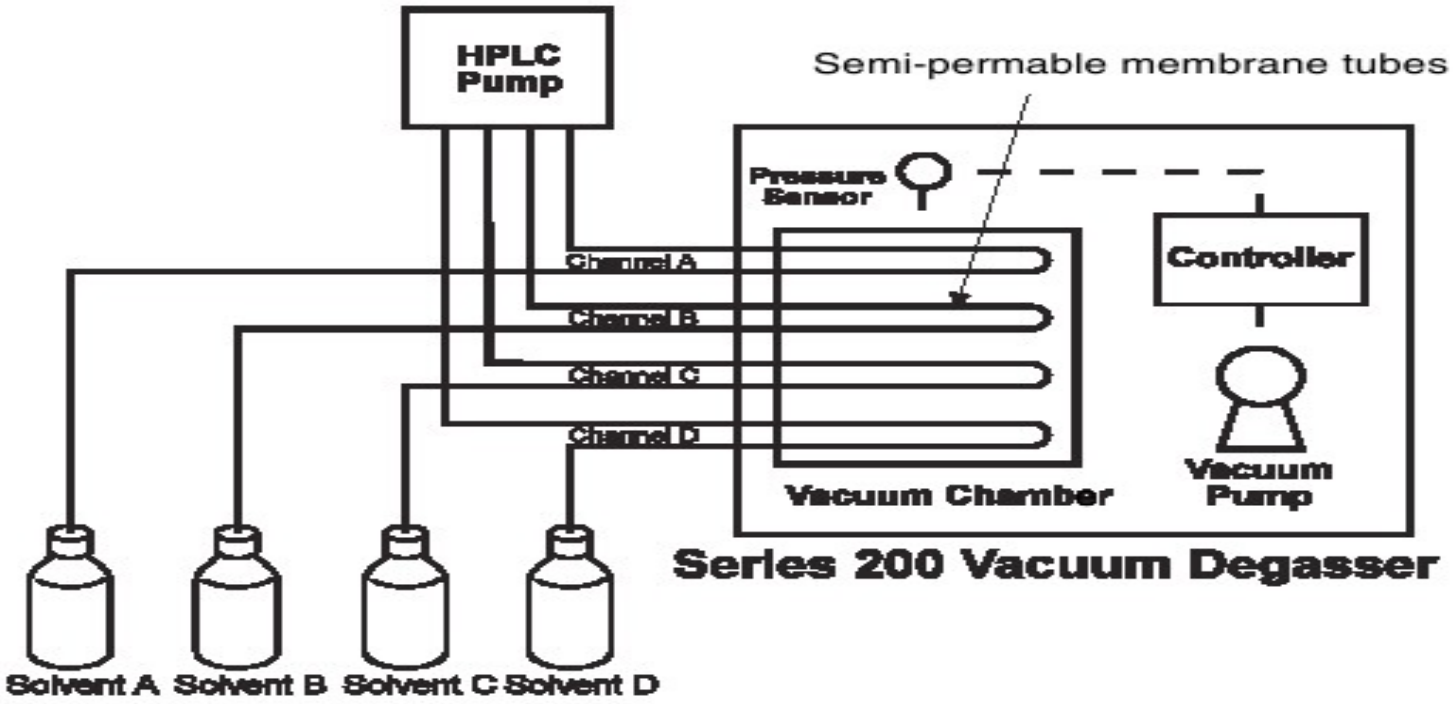


Figure 5.3. Schematic diagram of an online vacuum degasser. Older systems have relatively large semipermeable vacuum tubes of >10 mL. New systems have smaller tubes <1 mL. Note that for effective degassing, all four lines should be filled with solvents. Diagram courtesy of PerkinElmer, Inc.

What is Retention Time؟Retention Time ما هو

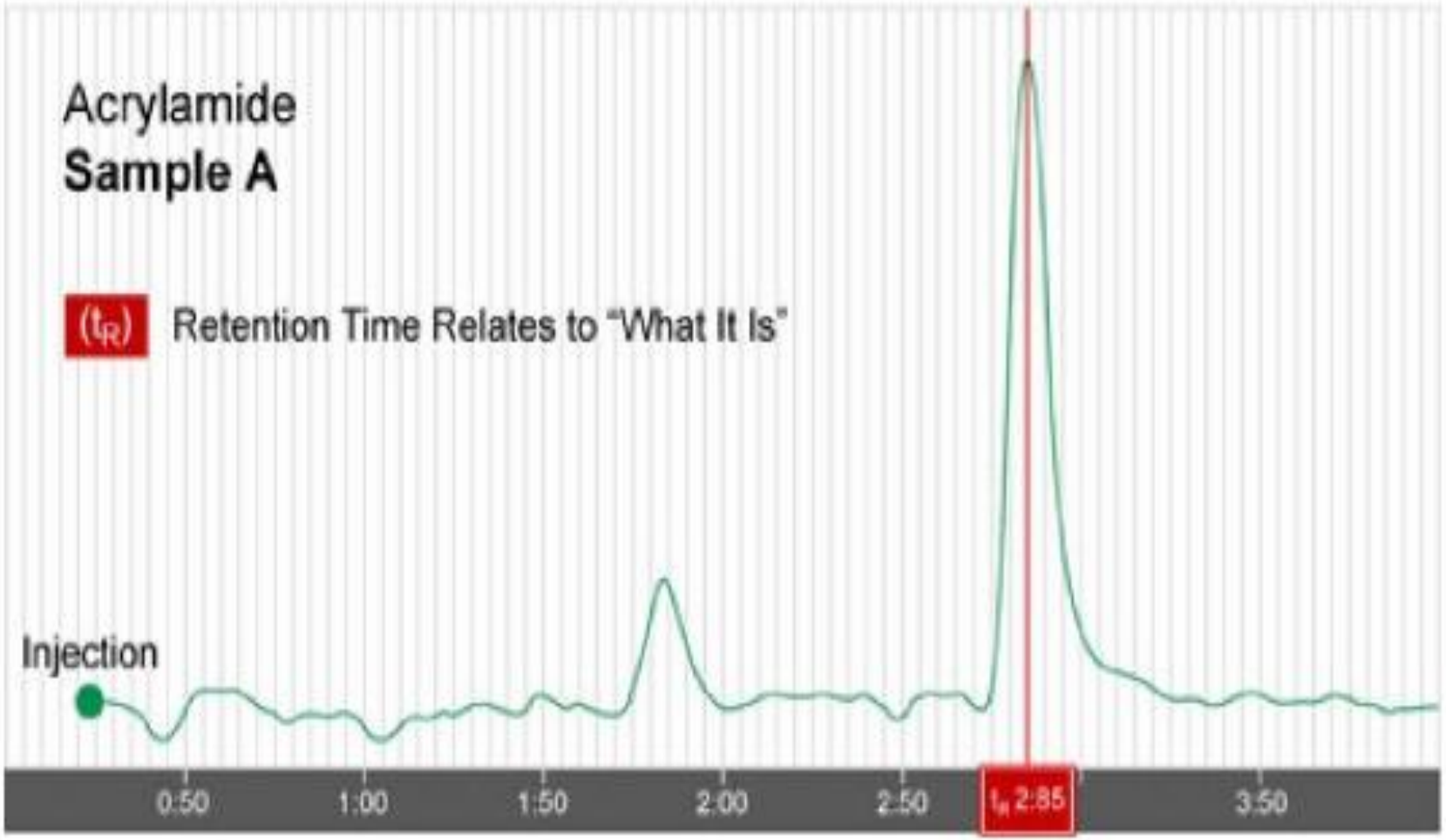


Figure (I-1) Identification

What is Retention Time?؟Retention Time ما هو

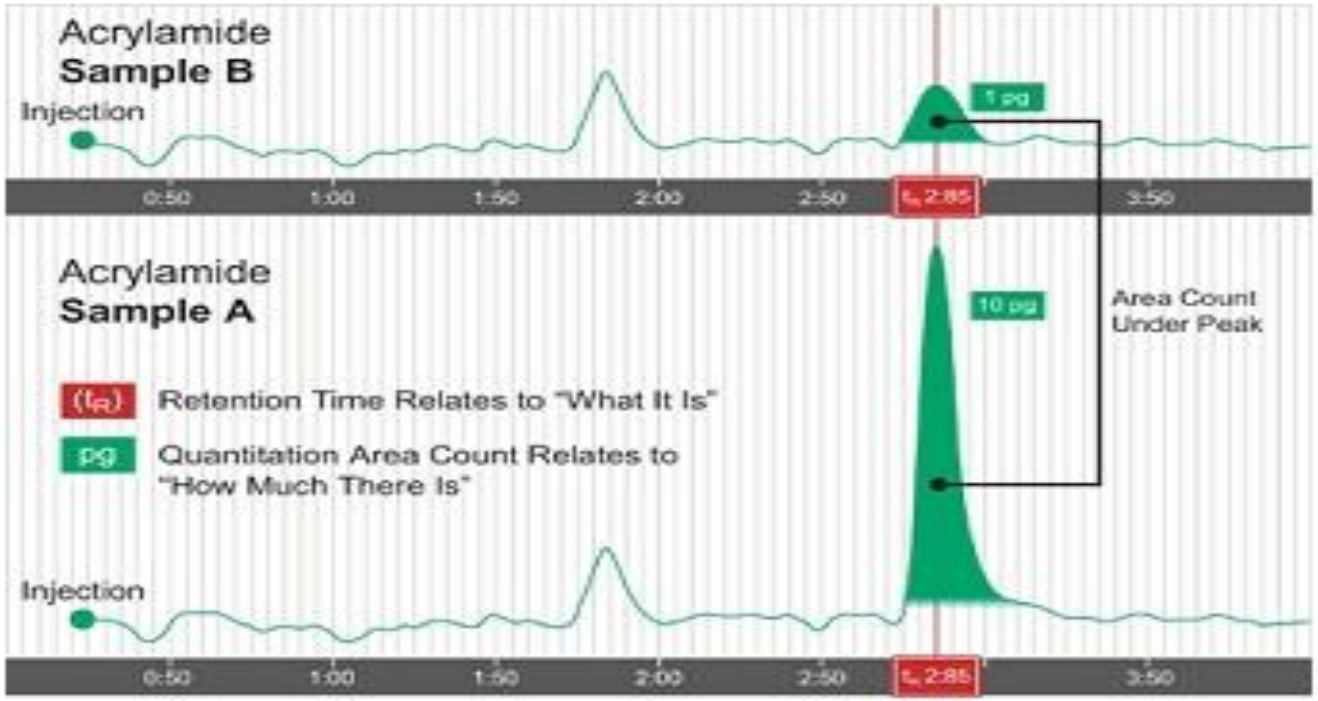
هو الزمن الذي تحتاجه المادة المفصولة لكي يقرأها Detectorوتبدأ من بداية الحقن والتي تعطي نوع من الإشارات الكهربائية ويحولها إلى مرتفعات سناميه Chromatographيمكن أن يقرأها

الإنسان عن طريق شاشة لكمبيوتر ويمثل القيمة النوعية للمادة المفصولة Qualitative Value

Retention Time is the time required for a separated analyte to travel from the point of injection through the chromatographic column until it is detected by the detector. During this process, the detector converts the presence of the analyte into electrical signals, which are then processed and displayed as chromatographic peaks on a computer screen. The retention time serves as a qualitative parameter for identifying the separated substance."

ماذا يعني Retention Time وماذا يعني Peak Area or High ؟

what does Peak Area (or Peak Height) mean?



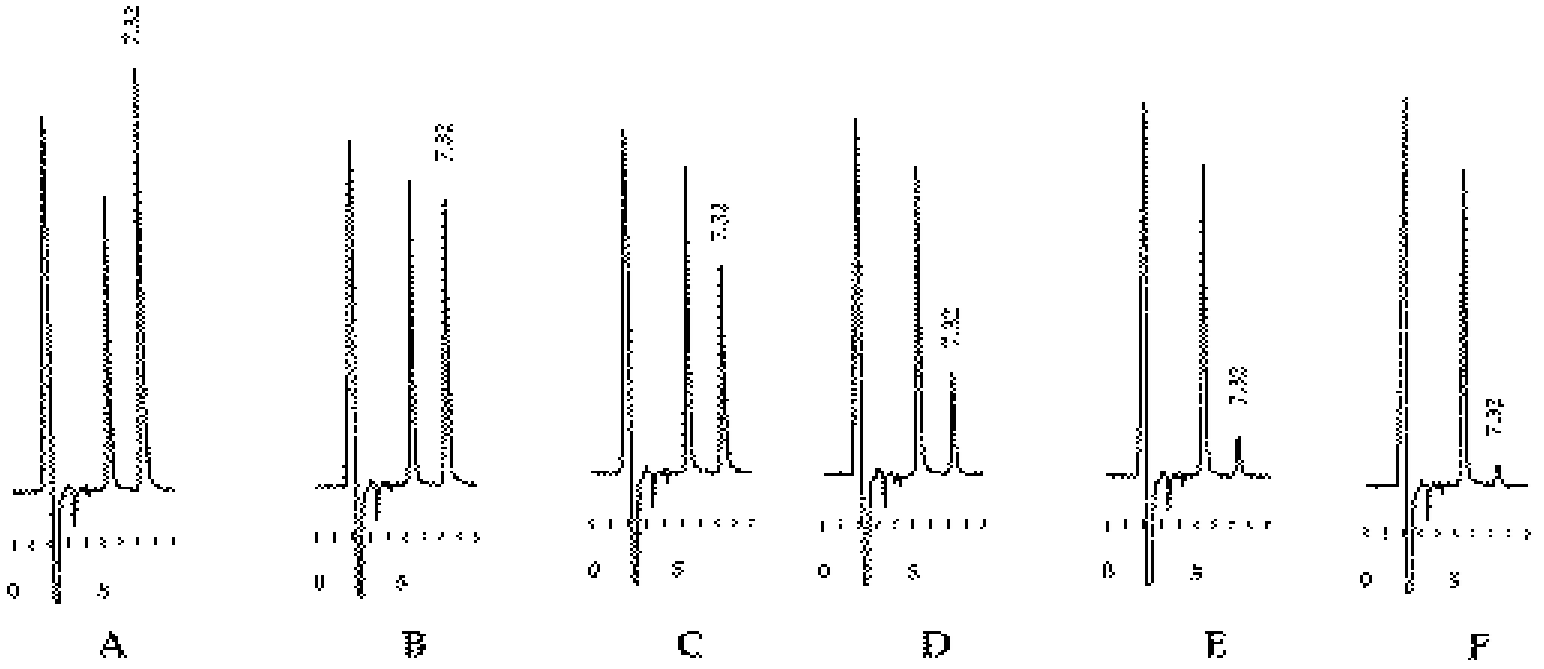
ماذا يعني Retention Time وماذا يعني Peak Area or High ؟

what does Peak Area (or Peak Height) mean?

عندما يتم حقن المادة من اجل الفصل يتكون إشارة الكهربائية ناتجة من Detector والتي تكون كثافتها يعتمد على كمية المادة المفصولة
فكلما كانت تركيز المادة المفصولة اكبر زاد ارتفاع السنام Peakوزاد حجمه وبمقارنته بمادة مرجعية Standardيمكن معرفة كمية المادة Quantitative Value

When a sample is injected for chromatographic separation, the detector generates an electrical signal whose intensity is proportional to the amount of the separated analyte. As the concentration of the analyte increases, both the peak height and peak area increase. By comparing these values with those obtained from a reference standard of known concentration, the quantity of the analyte in the sample can be determined—this represents the quantitative value of the analysis

تأثير تركيز المادة على ارتفاع السنام ومساحة السنام
Peak Area or Peak Height
The Effect of Analyte Concentration on Peak Height and Peak Area"



Pict. 1 HPLC chromatogram of a standard solution (7.32) at decreasing concentrations A) 100 µg/mL, B) 75 µg/mL, C) 50 µg/mL, D) 25 µg/mL, E) 10 µg/mL, and F) 5 µg/mL.

How can we verify that a new HPLC column is performing according to specifications?

مقياس كفاءة عمود الفصل

Calculation of chromatographic Resolution

انه عملية حسابية لتبيان كفاءة عامود الفصل لفصل المواد القريبة في وقت الخروج والتي توثر على القراءة ال Chromatogramفيجب ان تكون القيمة بين 1.5-2

"It is a mathematical calculation used to demonstrate the efficiency of the separation column in resolving compounds with similar retention times, which affects the chromatogram interpretation. Therefore, the resolution value should be between 1.5 and 2.0

يجب أن يكون حساب Resolutionللعامود الفصل ما بين 2 – 1.5

the resolution value should be between 1.5 and 2.0

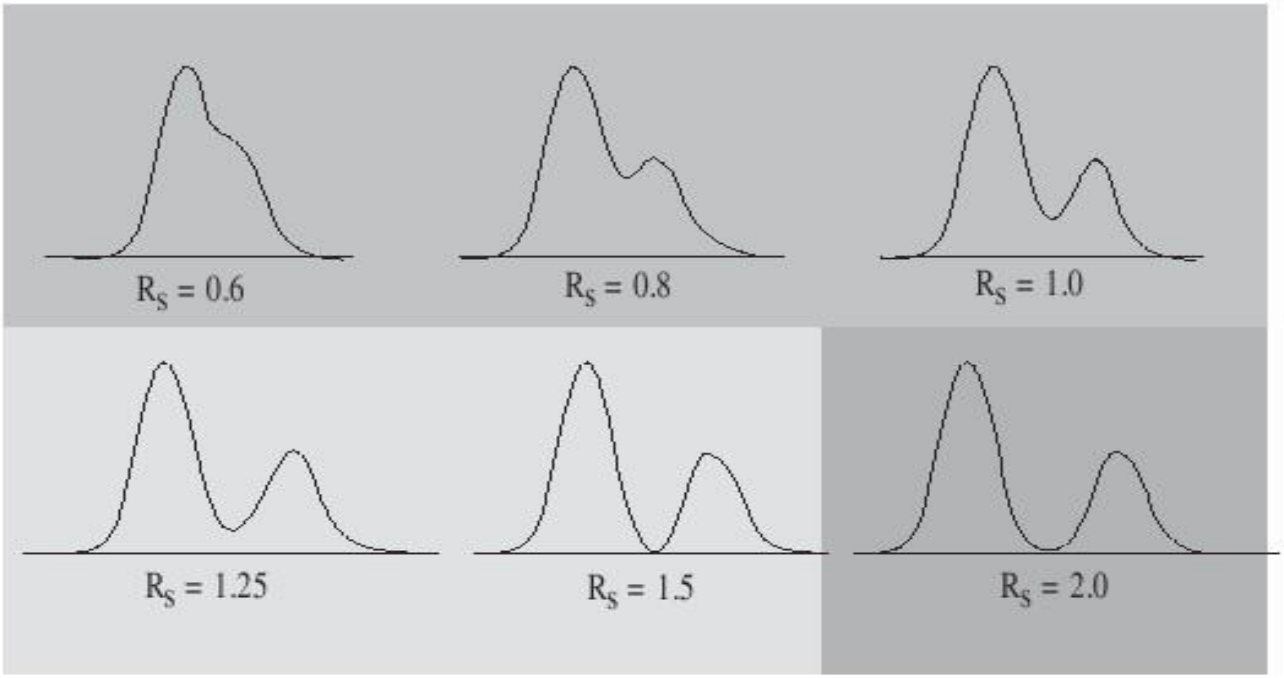


Figure 2.7. Diagrams showing two closely eluting peaks at various resolution values from 0.6 to 2.0. Figure reprinted with permission from Academy Savant.

Asymmetry factor, A_s = B/A

Tailing (T_f > 1.0) and

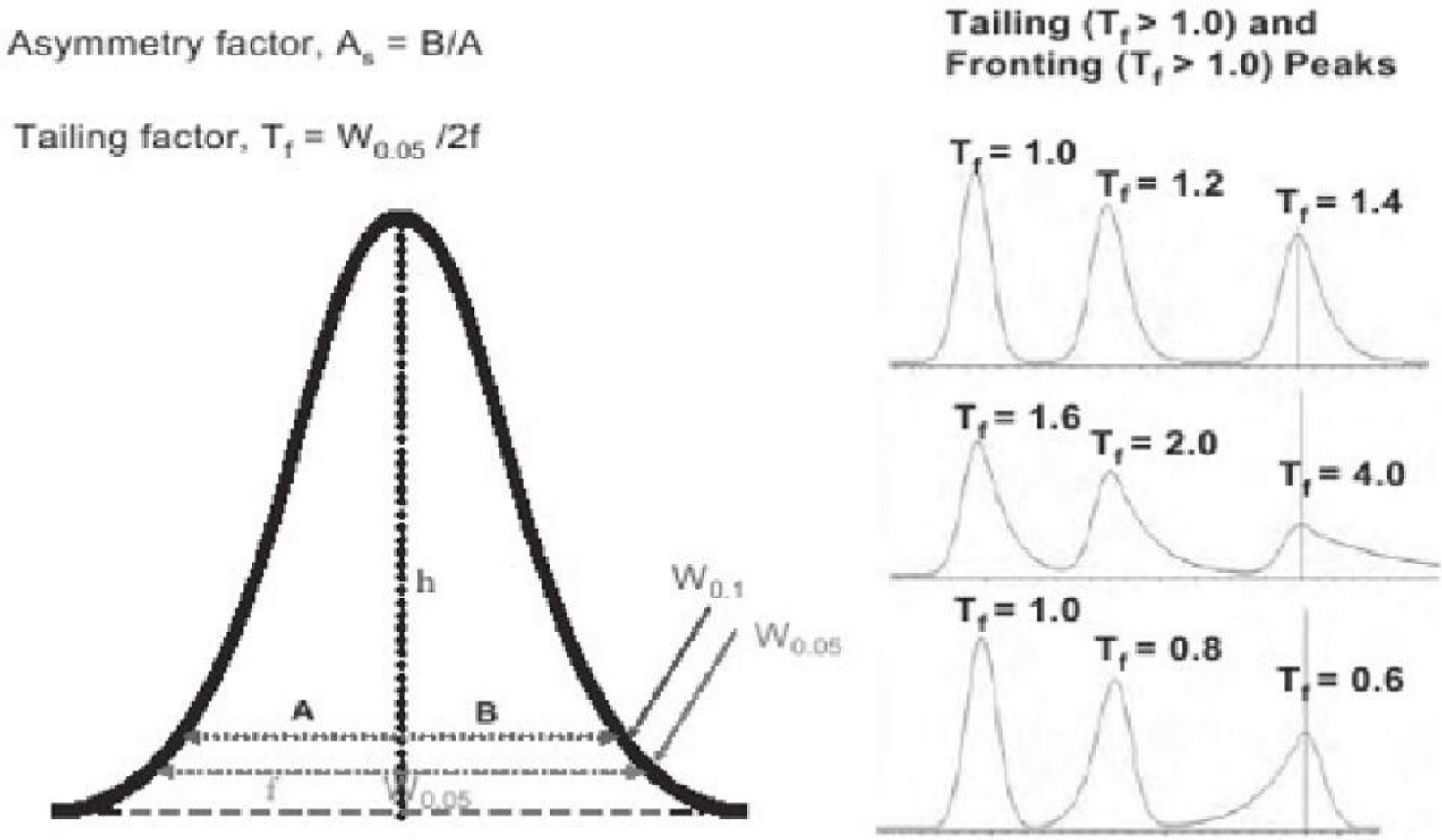
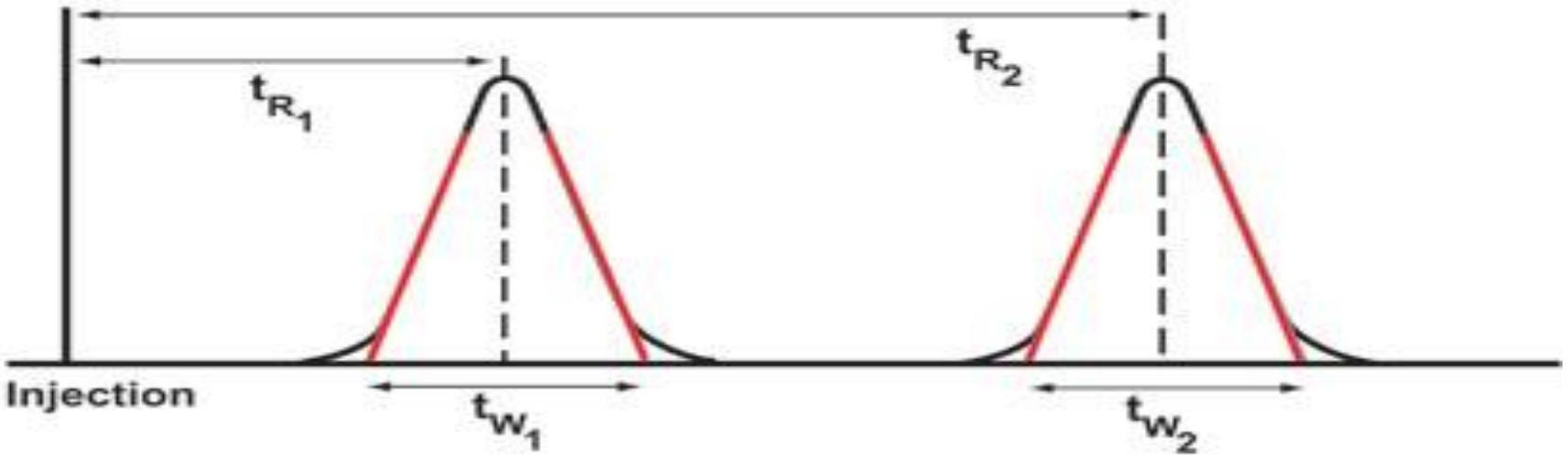


Figure 2.8. A diagram showing the calculation of peak asymmetry (A_s) and tailing factor (T_f) from peak width at 5% height ($W_{0.05}$) according to the USP. Inset diagrams show fronting and tailing peaks.

Calculation of chromatographic Resolution

$$R_s = (t_{R_1} - t_{R_2}) / 0.5\,(t_{w_1} + t_{w_2})$$



Note: t_{w_1} and t_{w_2} are obtained from the intersection of the tangents with the baseline.

For a symmetrical Gaussian peak the tangents are drawn at 0.6 times the peak height.

حساب Resolutionلعامود الفصل HPLC

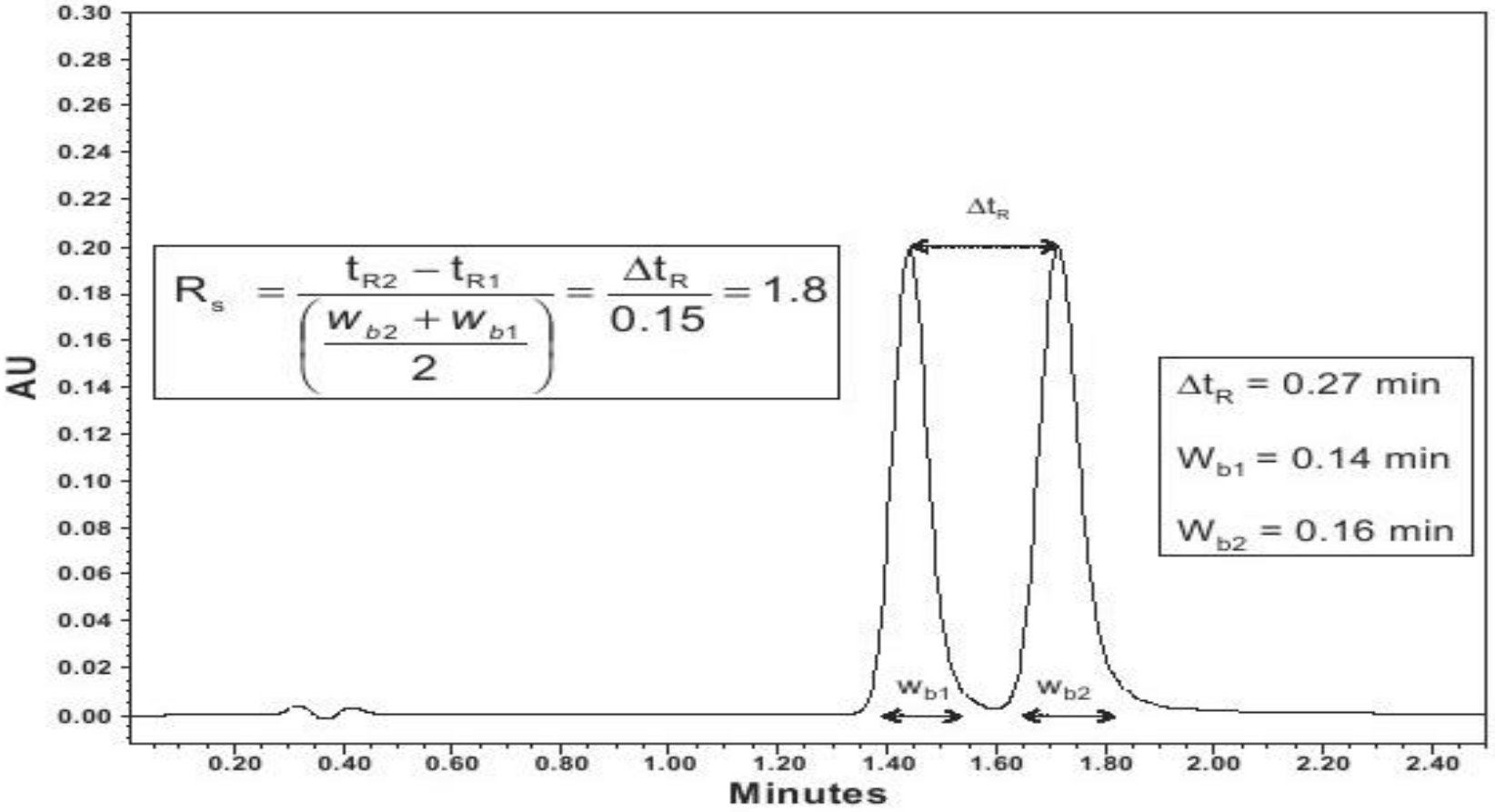
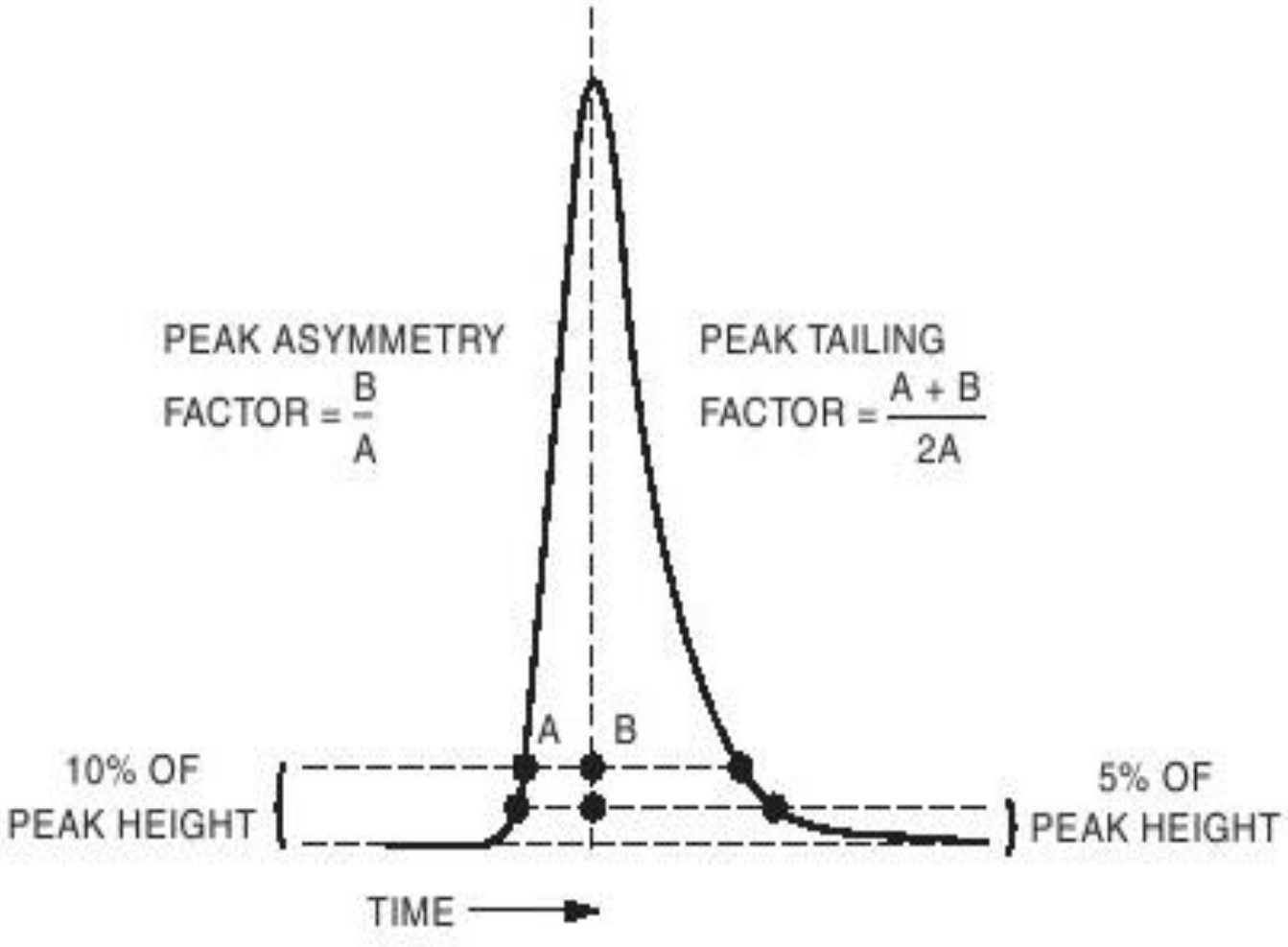


Figure 2.6. A chromatogram of two peaks with a resolution (R_s) of 1.8.

$$\text{Resolution, } R_s = \frac{t_{R2} - t_{R1}}{\left(\frac{w_{b1} + w_{b2}}{2}\right)} = \frac{\Delta t_R}{w_b}. \qquad \text{Eq. 2.16}$$

كيف نعرف ان عامود HPLC جديد ويعمل ؟

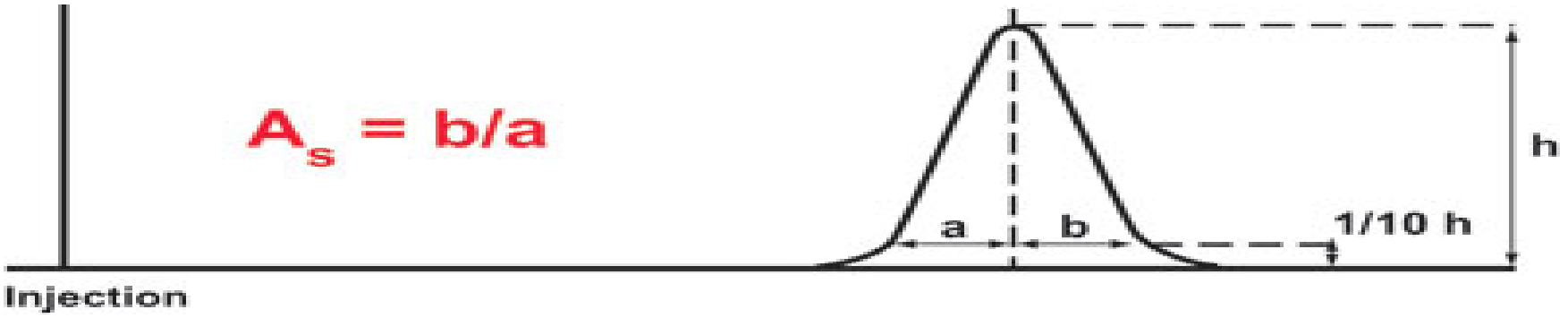
How can we verify that a new HPLC column is performing according to specifications?



Calculation of peak Asymmetry Factor (used by Tosoh)

Calculation of peak Asymmetry Factor (used by Tosoh)

Where:
As = peak asymmetry factor
b = distance from the point at peak midpoint to the trailing edge (measured at 10% of peak height)
a = distance from the leading edge of of peak to the midpoint (measured at 10% of peak height)



HPLC column is performance **Peak Asymmetry Factor As 10%**

How can we verify that a new HPLC column is performing according to specifications?

Peak **Asymmetry Factor** مقياس كفاءة عمود الفصل

Peak Asymmetry Factor (A_s, 10%)	Peak Tailing Factor (5%)
1.0	1.0
1.3	1.2
1.6	1.4
1.9	1.6
2.2	1.8
2.5	2.0

From Snyder, L.R., Kirkland, J.J., and Glajch, J.L., *Practical HPLC Method Development*, 2nd ed., John Wiley & Sons, New York, 1997. With permission.

To put these factors into context, column performance can be described as in the following table:

Peak Asymmetry Factor (A_s, 10%)	Column Performance
1.0–1.05	Excellent, new
1.2	Acceptable
2.0	Degraded, approaching poor
4.0	Not usable

From Snyder, L.R., Kirkland, J.J., and Glajch, J.L., *Practical HPLC Method Development*, 2nd ed., John Wiley & Sons, New York, 1997. With permission.

How can we verify that a new HPLC column is performing according to specifications?

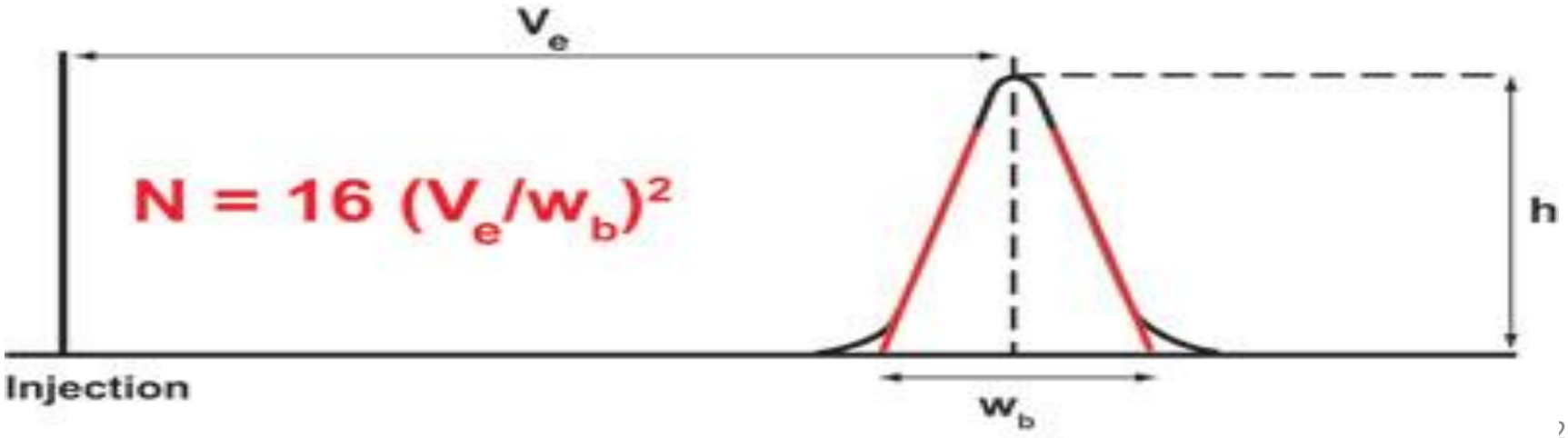
مقياس كفاءة عمود الفصل

1. Calculation of the number of Theoretical Plates (half-height method, used by Tosoh)
2. Calculation of the number of Theoretical Plates (USP method)
3. Calculation of peak Asymmetry Factor (used by Tosoh)
4. Calculation of Tailing Factor (USP method)
5. Calculation of the Height Equivalent to a Theoretical Plate (HETP)
6. Calculation of Reduced Plate Height (h)
7. Calculation of chromatographic Resolution

Calculation of the number of Theoretical Plates (USP method)

Calculation of the number of Theoretical Plates (USP method)

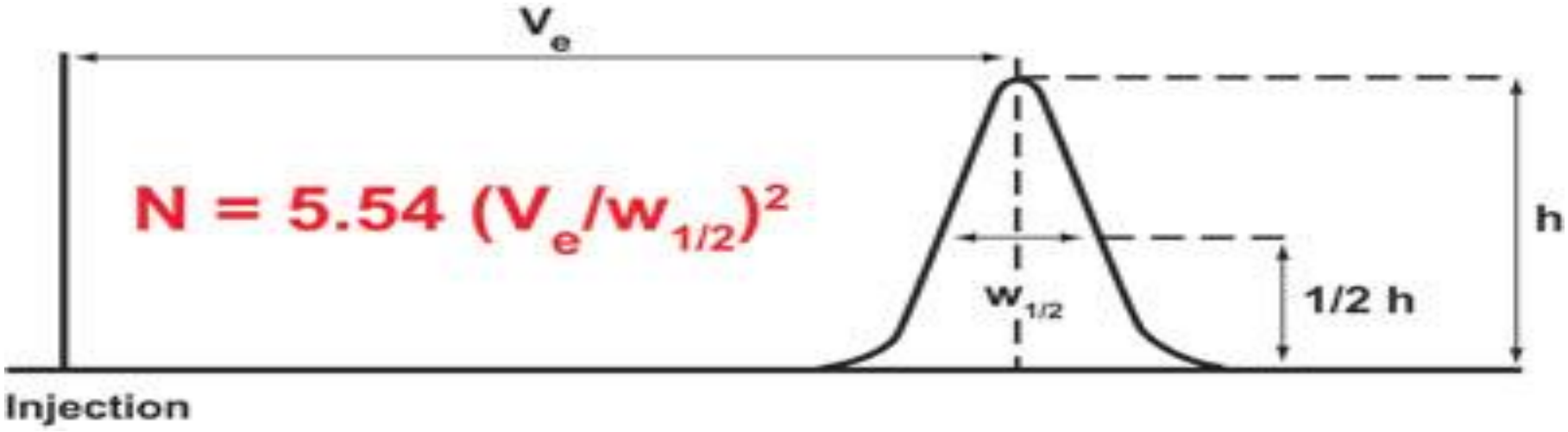
Where:
N = Number of theoretical plates
Ve = elution volume, retention time or retention distance (mL, sec, or cm)
h = peak height
wb = width of the peak at the base line (mL, sec, or cm)



Calculation of the number of Theoretical Plates (half-height method, used by Tosoh

Calculation of the number of Theoretical Plates (half-height method, used by Tosoh)

Where:
N = Number of theoretical plates
Ve = elution volume or retention time (mL, sec, or cm)
h = peak height
w1/2 = width of the peak at half peak height (mL, sec, or cm



Calculation of Tailing Factor (USP method)

Calculation of Tailing Factor (USP method)

Where:
T = tailing factor (measured at 5% of peak height)
b = distance from the point at peak midpoint to the trailing edge
a = distance from the leading edge of the peak to the midpoint

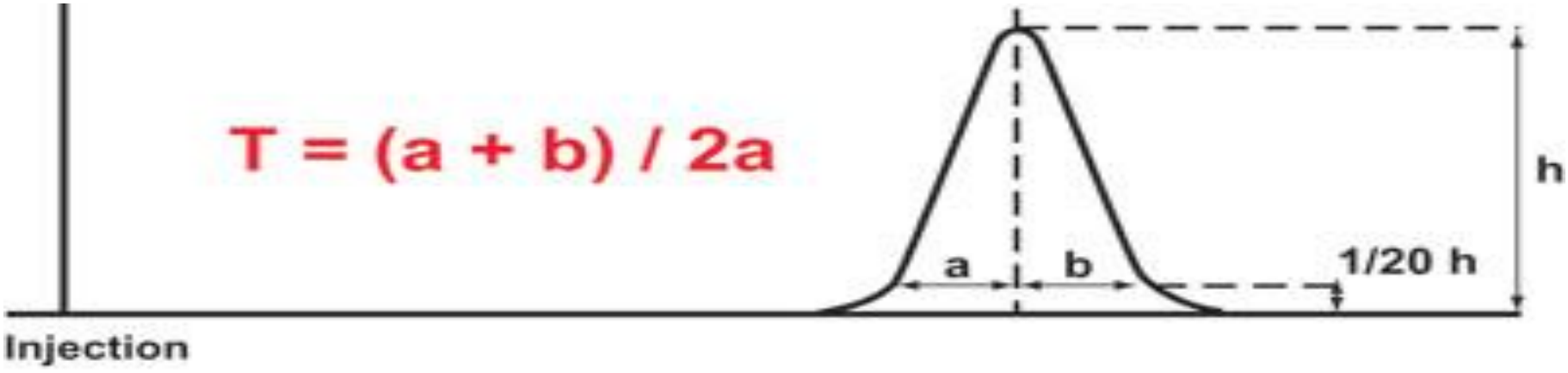


Plate Count (N)

By increasing the Plate Count (N) you will have a greater resolution.

Plate Count (N)	Plate Count in Equation	Resolution
1000	$(1000)^{1/2}$	31.6
2000	$(2000)^{1/2}$	44.7
3000	$(3000)^{1/2}$	54.7
5000	$(5000)^{1/2}$	70.7
10,000	$(10{,}000)^{1/2}$	100
15,000	$(15{,}000)^{1/2}$	122.4

By increasing the k' value you will have a greater resolution

k' Value	k' Term in Equation	Resolution
1	½	.50
2	2/3	.67
3	¾	.75
10	10/11	.91
20	20/21	.95

Calculation of the Height Equivalent to a Theoretical Plate (HETP)

Calculation of the Height Equivalent to a Theoretical Plate (HETP)

Where:
H = Height equivalent of a theoretical plate
L = Length of the column
N = Number of theoretical plates

Dimensions: when using HPLC or UHPLC columns, H is usually expressed in µm.

$$H = L / N$$

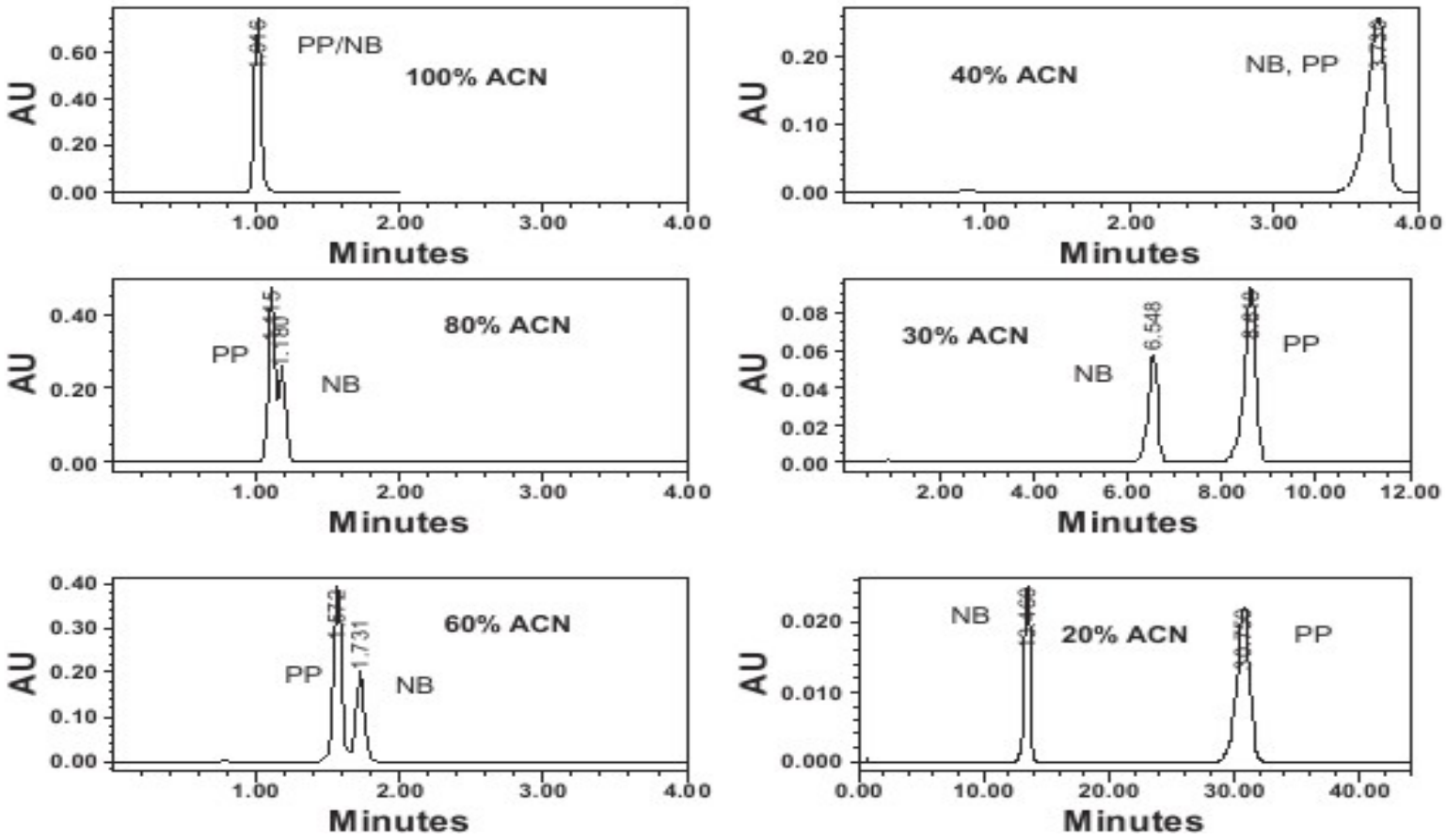
Calculation of Reduced Plate Height (h)

Giddings introduced dimensionless parameters for H and also for the linear velocity. Dimensionless parameters allow the direct comparison of the efficiency of two or more columns packed with different particle size packing materials. According to the theory, a well packed column should have a reduced plate height (h) in the range of 2-3 at a reduced velocity (v) of about 3. Here we only provide the formula for h.

$$h \equiv H / d_p$$

Where:
h = reduced plate height (sometimes referred to as the number of band widths)
H = height equivalent of a theoretical plate (μm)
dp = mean particle size (μm)

اثر نسب خلط مكونات الmobile phase على فاعلية فصل المواد المفحوصة؟
What is the effect of mobile phase composition ratios on the separation efficiency of analyzed compounds?"



على ماذا تعتمد كفاءة الفصل لعامود HPLC؟
"What factors determine the separation efficiency of an HPLC column?

١) طول عمود HPLC كلما زاد طول عامود HPLC زادت كفاءة الفصل
٢)حجم مواد الوجه الثابت Stationary Phase كلما قل حجم مواد الوجه الثابت زادت كفاءة الفصل في عمود HPLC
٣) قطر مسامات مواد الوجه الثابت (Pore Diameter) كلما قل قطر المسامات زاد مساحة السطح (Surface Area) للوجه الثابت وبتالي زادت كفاءة الفصل في عمود HPLC

1. **HPLC Column Length:** The longer the HPLC column, the higher the separation efficiency.
2. **Stationary Phase Particle Size:** The smaller the particle size of the stationary phase, the higher the separation efficiency in HPLC columns.
3. **Pore Diameter of the Stationary Phase:** The smaller the pore diameter, the greater the surface area of the stationary phase, and consequently, the higher the separation efficiency in HPLC columns.

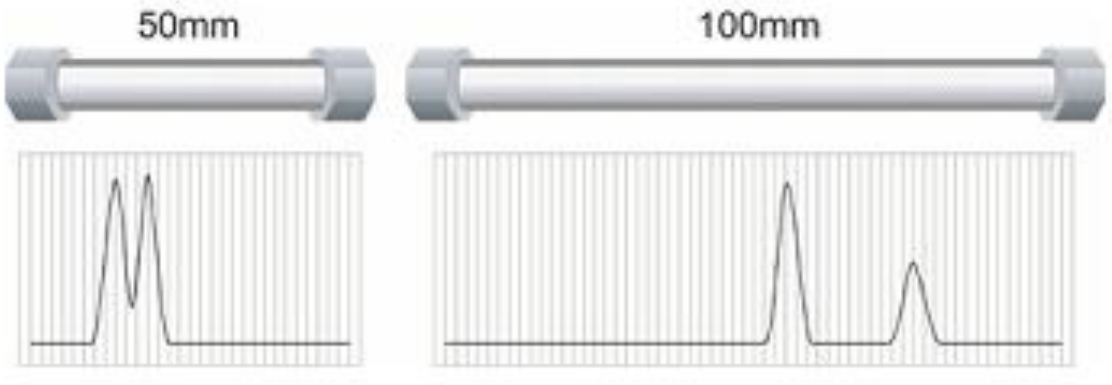
ما هي فائدة إضافة مادة End-Capping إلى عمود HPLC
"What are the advantages of applying End-Capping treatment to the stationary phase of an HPLC column?"

إذا أضيفت مادة End-Capping إلى أعمدة الفصل من نوع C18 أو C8 فانه يمكن الحصول على سلاسل اقصر داخل عامود الفصل على تحسين الاستقرار الهيدروليكي مما يقلل مساحة الذيل Peak Tailing ما يزيد كفاءة الفصل وقراءة وتحليل ال Peak بدقة اكبر ويزيد رقم ألٌ Resolution لعامود الفصل

"If an End-Capping agent is applied to C18 or C8 separation columns, it introduces shorter chains within the column that improve hydraulic stability. This reduces peak tailing, enhances separation efficiency, allows for more accurate peak reading and analysis, and increases the Resolution value of the column."

كلما زاد طول عامود HPLC زادت كفاءة الفصل

The longer the HPLC column, the higher the separation efficiency.



مثال على تأثير طول العمود على عملية الفصل
An Example of the Effect of Column Length on the Separation Process"

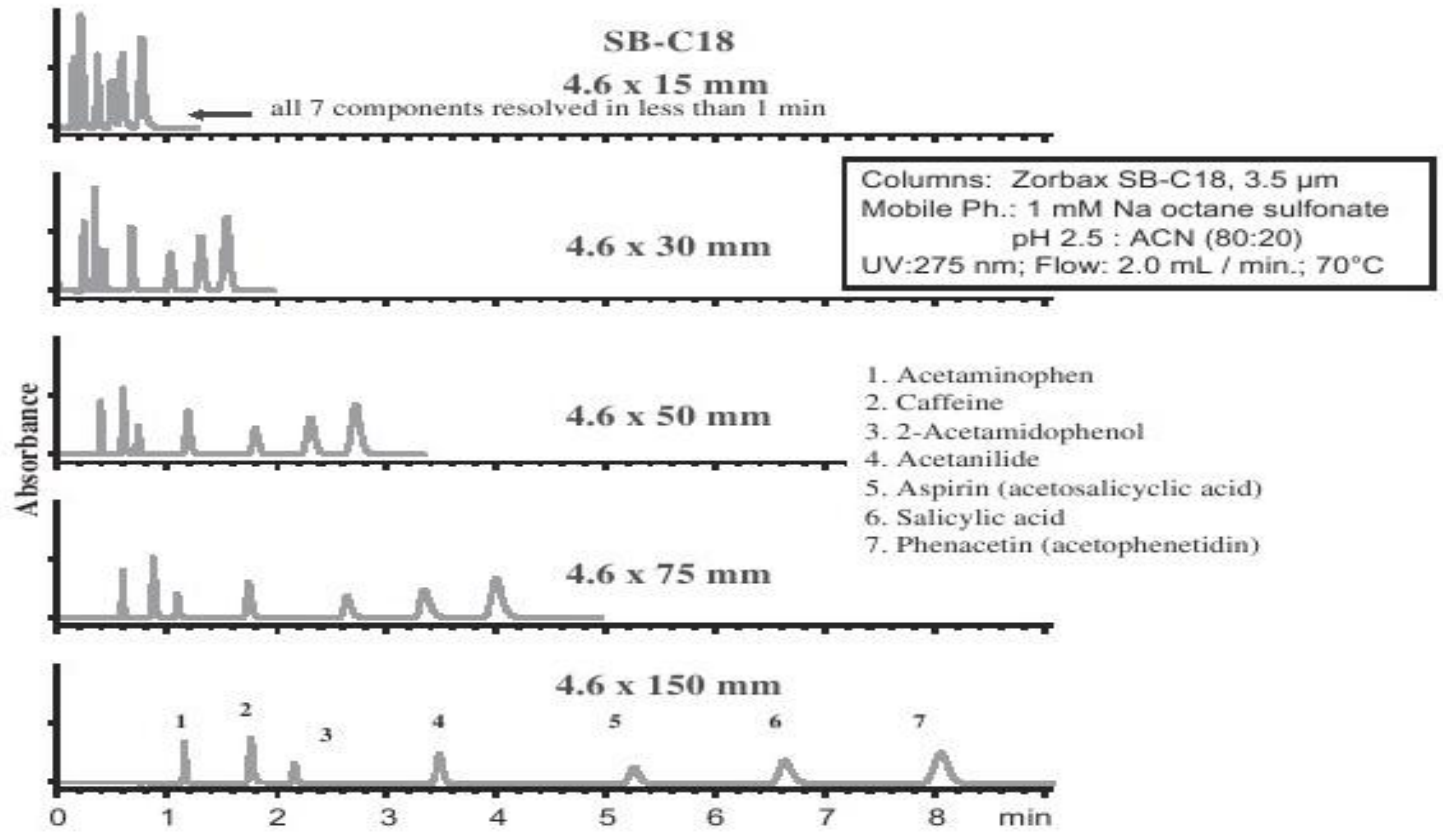
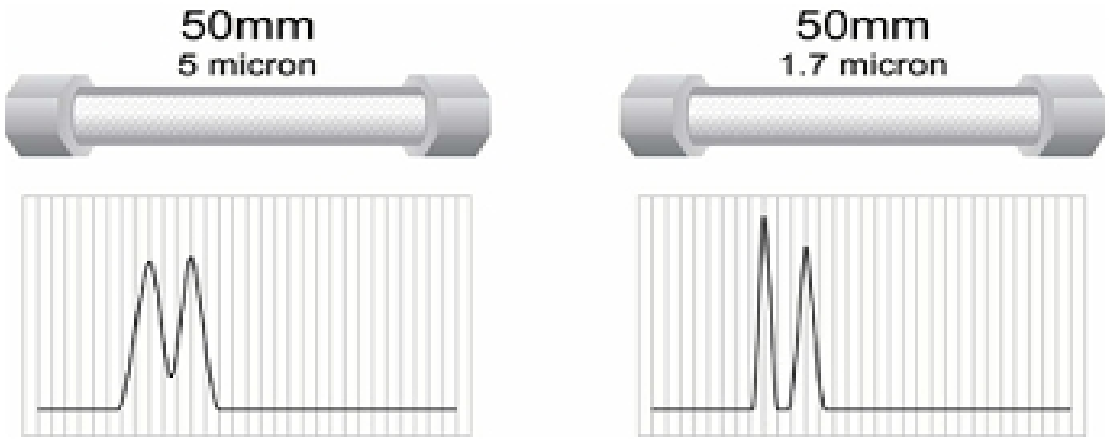


Figure 3.3. Chromatograms illustrating the effect of column length on efficiency, resolution, and sensitivity. Chromatograms courtesy of Agilent Technologies.

كلما قل حجم مواد الوجه الثابت زادت كفاءة الفصل في عامود HPLC
The smaller the stationary phase particle size, the higher the separation efficiency in an HPLC column."



Stationary Phase تأثير حجم مواد الوجه الثابت

stationary phase particle size efficiency

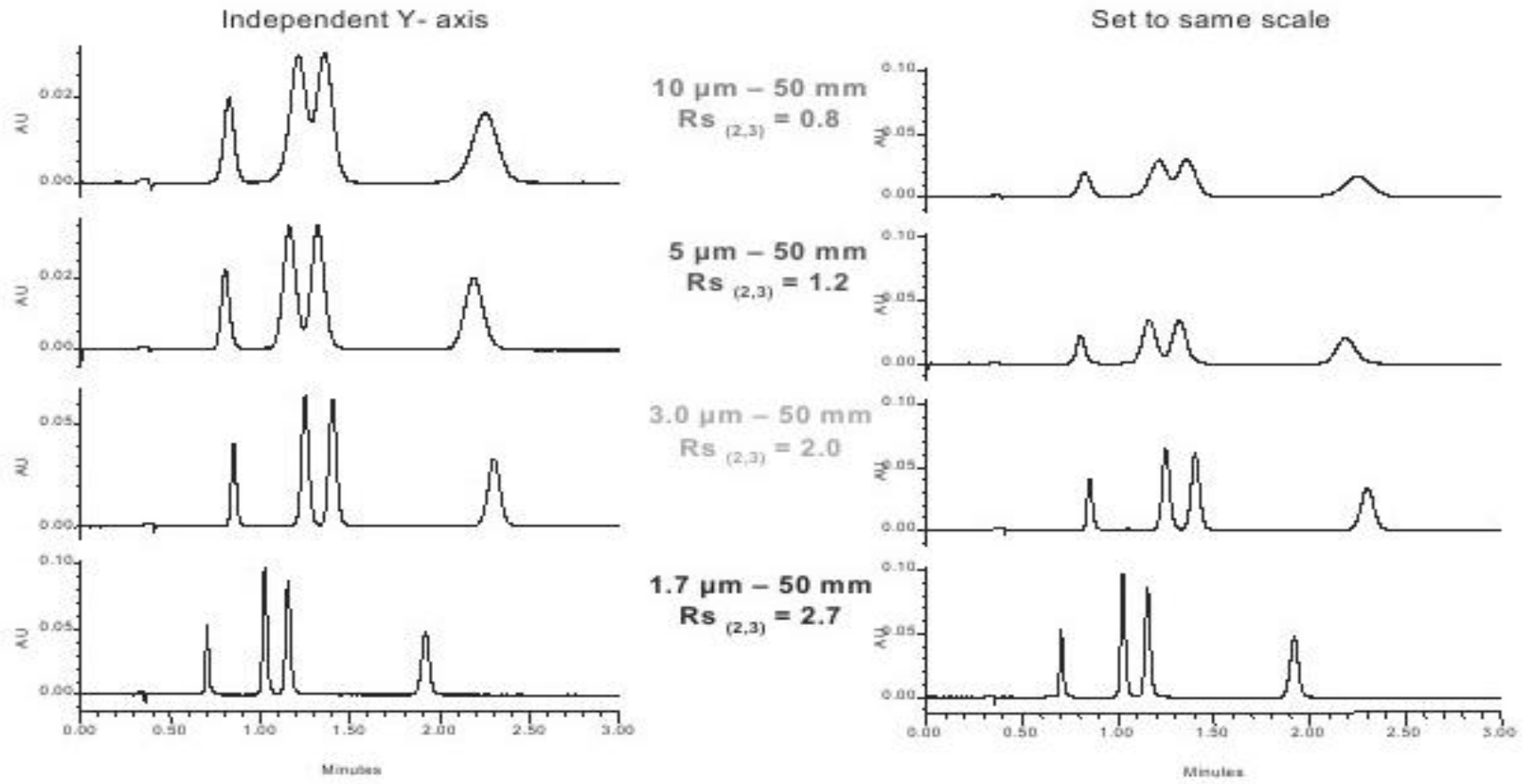
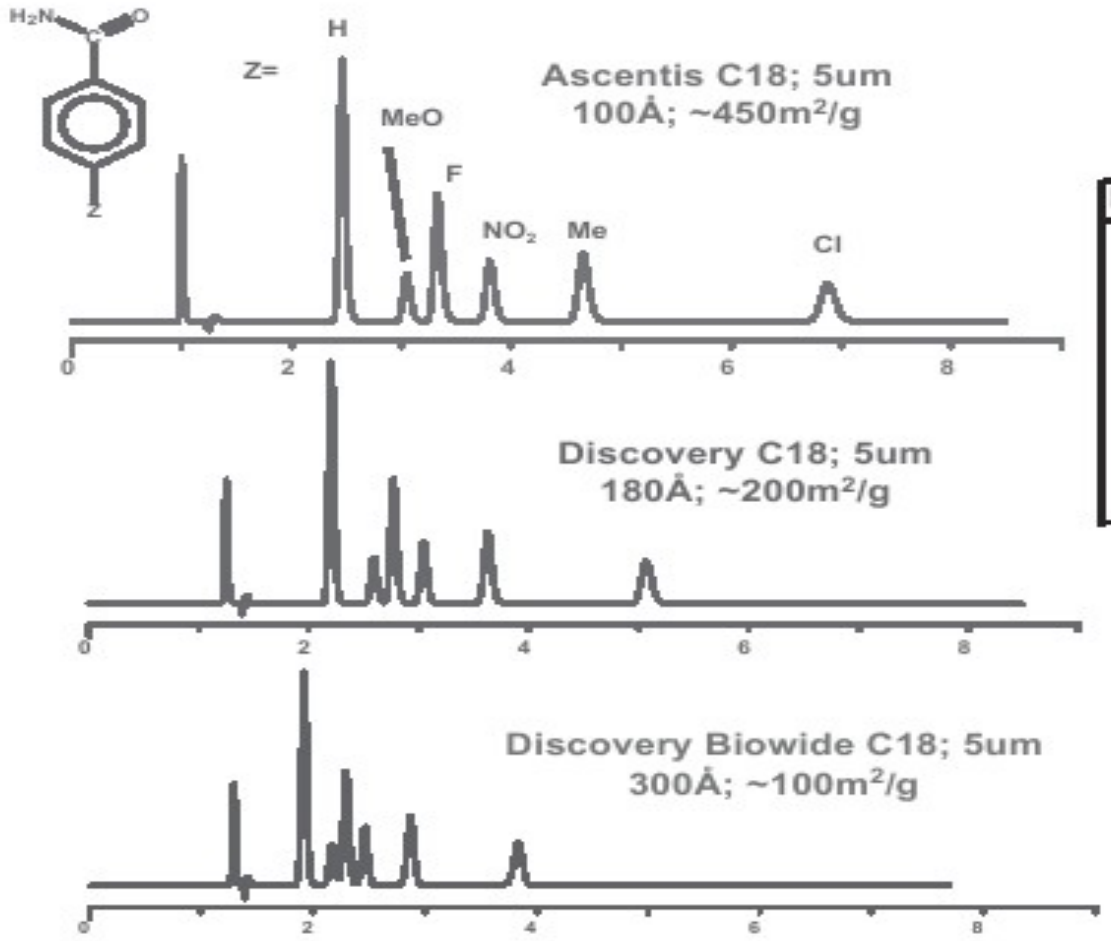


Figure 3.5. Chromatogram illustrating the effect of decreasing particle size (while keeping L = 50 mm) on efficiency, resolution, and sensitivity. Chromatograms courtesy of Waters

كلما قل قطر مسامات الوجه الثابت زادت فعالية الفصل في عامود HPLC

"The smaller the pore diameter of the stationary phase, the higher the separation efficiency in an HPLC column."



Pore Diameter	Surface Area
60	500
100	450
120	310
180	200
200	200
300	100
500	50
1000	25

كلما زادت مساحة المواد المكونة للوجه الثابت زادت فعالية الفصل في عمود HPLC

The larger the surface area of the stationary phase material, the higher the separation efficiency in an HPLC column."

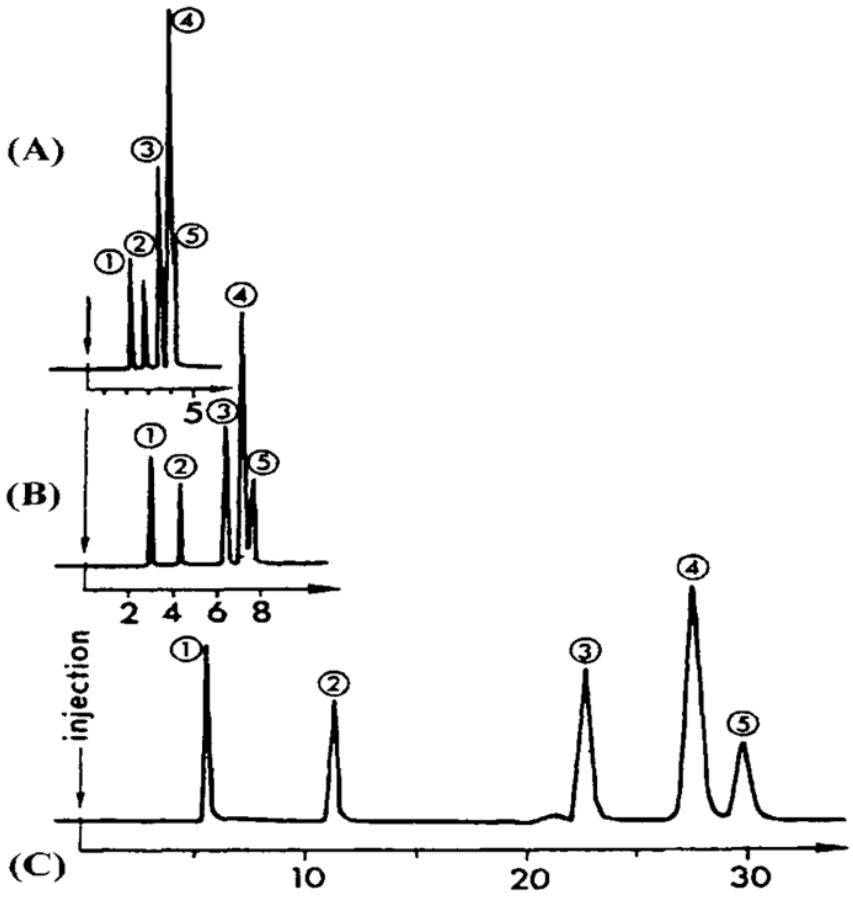


Figure 4 Variation of aromatic hydrocarbon retentions versus specific surface area of silica (Spherosil): Column: length 10 cm; internal diameter 0.4 cm. Stationary phase: Spherosil 4.6 μm, specific surface area: A = 470 $m^2\ g^{-1}$; B = 590 $m^2\ g^{-1}$; C = 900 $mg^2\ g^{-1}$. Mobile phase, dry hexane: Flow rate, 0.9 mL min^{-1}; room temperature; solutes: 1, toluene; 2, naphthalene; 3, biphenyl; 4, anthracene; 5, phenanthene. (From Ref. 42.)

اقسام الجهاز

🚀 المضخة الثنائية :(Binary Pump)قلب الجهاز النابض الذي يدفع المذيبات بدقة متناهية وضغط عالٍ لضمان تدفق ثابت.

💉 حاقن العينة :(Injector)البوابة الذكية التي تُدخل حجمًا محددًا من العينة)مثل 20ميكرولتر (إلى النظام دون فقدان الضغط.

🗂️ العمود الكروماتوجرافي :(Column)ساحة الفصل الحقيقية حيث تتفكك المكونات بناءً على انجذابها للطور الثابت)مثل C18).

🔍 الكاشف :(PDA Detector)العين الحساسة التي ترصد المركبات الخارجة وقيس امتصاصها للأشعة 190-800 نانومتر (وتحولها لإشارات.

🫧 مزيل الغازات :(Degasser)الحارس الصامت الذي يزيل فقاعات الهواء من المذيبات لمنع التشويش على النتائج وحماية المضخة.

💻 نظام التحكم والبرمجيات :العقل المدبر الذي يبرمج طريقة التحليل، يتحكم في الظروف، ويحلل البيانات لرسم الكروماتوجرام.

🌡️ فرن الأعمدة) :(Column Oven)إن وُجد (يحافظ على درجة حرارة ثابتة للعمود لضمان ثبات أزمنة الاحتفاظ ودقة الفصل.

🔗 خلية التدفق :(Flow Cell)الجسر الدقيق داخل الكاشف حيث تمر العينة أمام الضوء لقياس الامتصاص لحظيًا بحجم دقيق.

🔄 نظام الطور المتحرك :المذيبات المُحضَّرة والمفلترة التي تحمل العينة عبر النظام وتلعب دورًا محوريًا في كفاءة الفصل.

🎯 التكامل الوظيفي :يعمل كل مكون بتناغم تام وفق معادلة :ضخ ← حقن ← فصل ← رصد ← تحليل =نتيجة موثوقة ودقيقة ✨ !

بعض اشكال مشاكل خط الأساس baseline

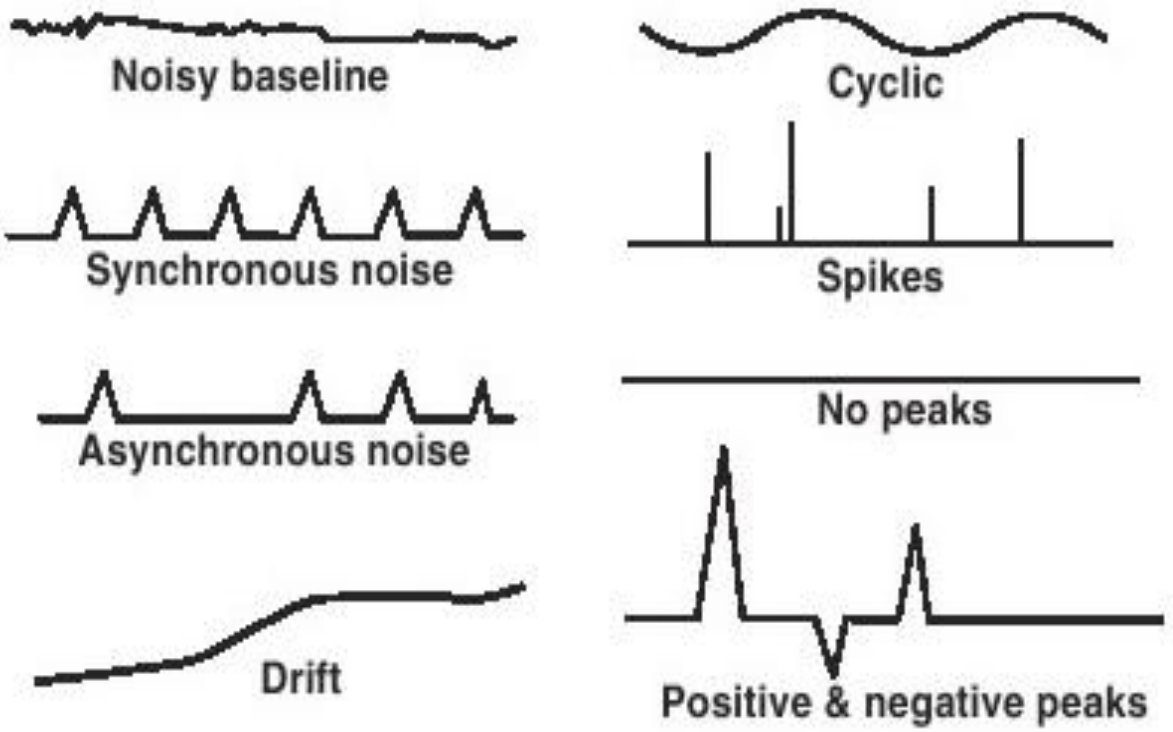


Figure 10.6. Examples of various baseline problems. Figure courtesy of Waters Corporation.

بعض اشكال noiseوتأثيرها على Chromatogram

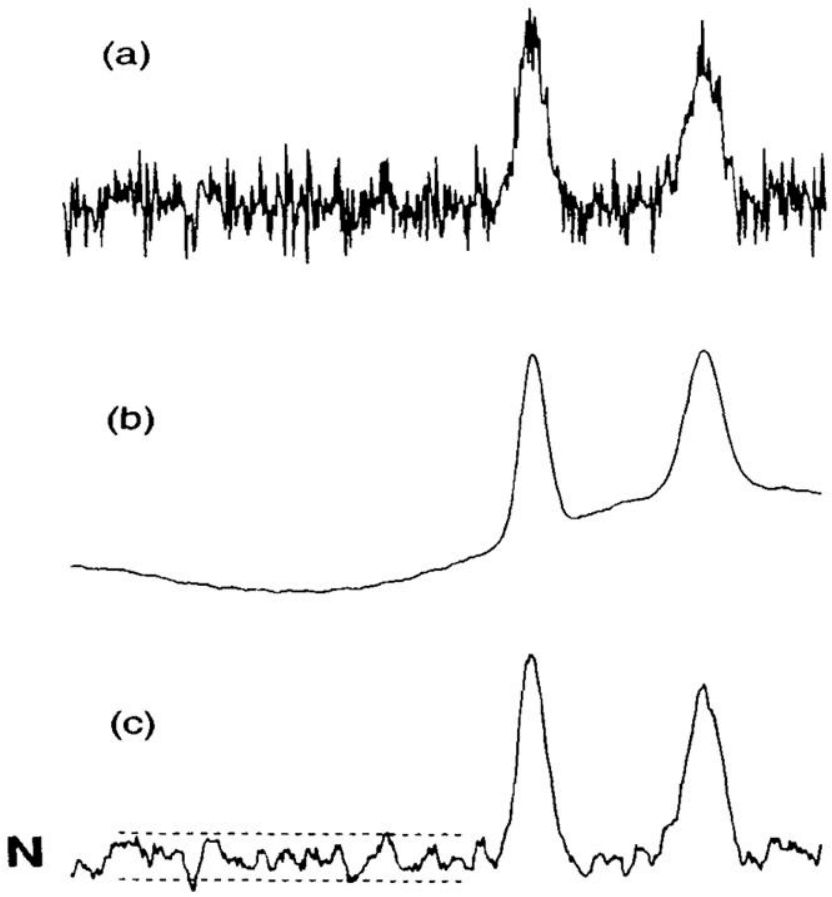


Figure 1 Types of noise in chromatographic signals: (a) high-frequency noise; (b) low-frequency noise or drift; (c) midfrequency noise and determination of the peak-to-peak noise level (*N*).

أنواع noise التي قد تنتج عن Detectors

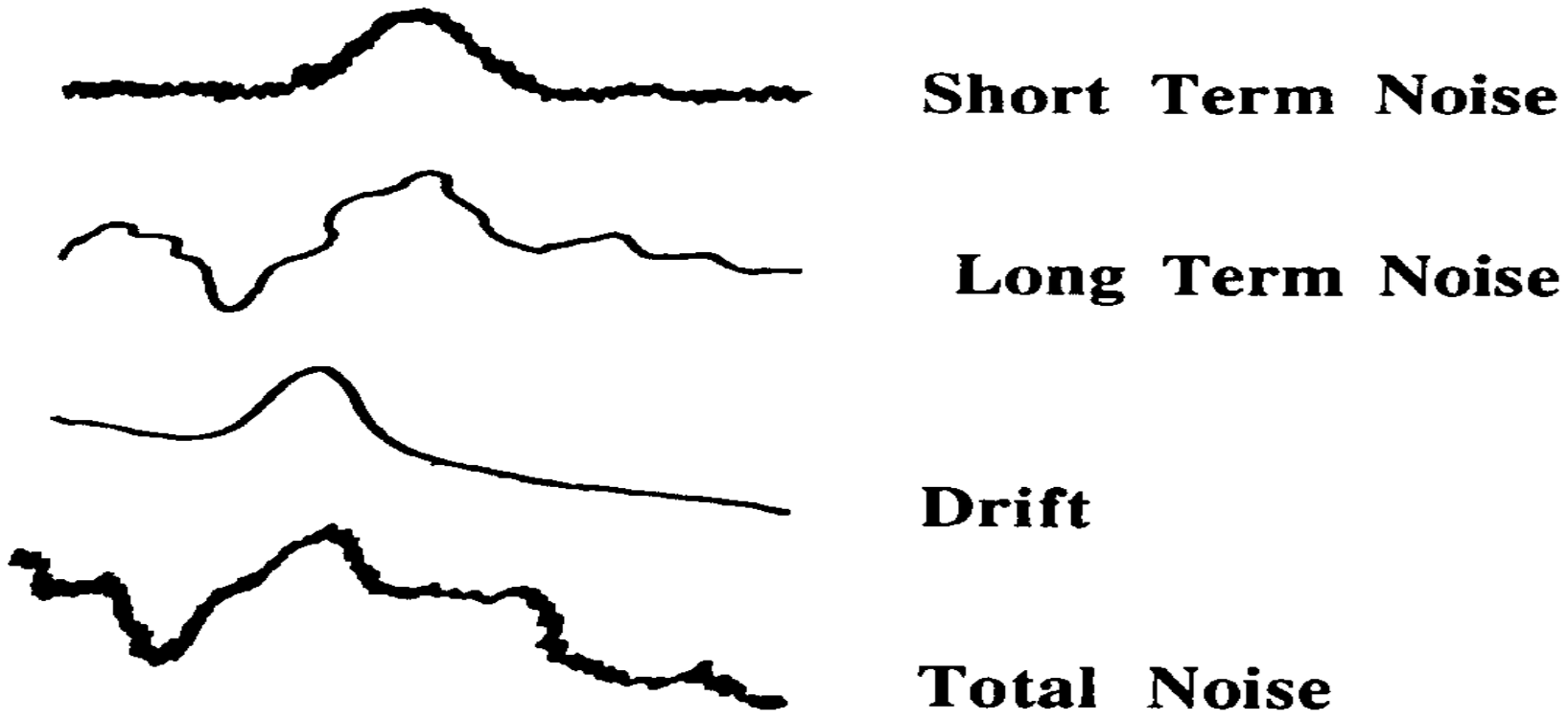


Figure 3 Different types of detector noise.

كيف يمكن من شكل Chromatographمعرفة مشاكل الجهاز؟

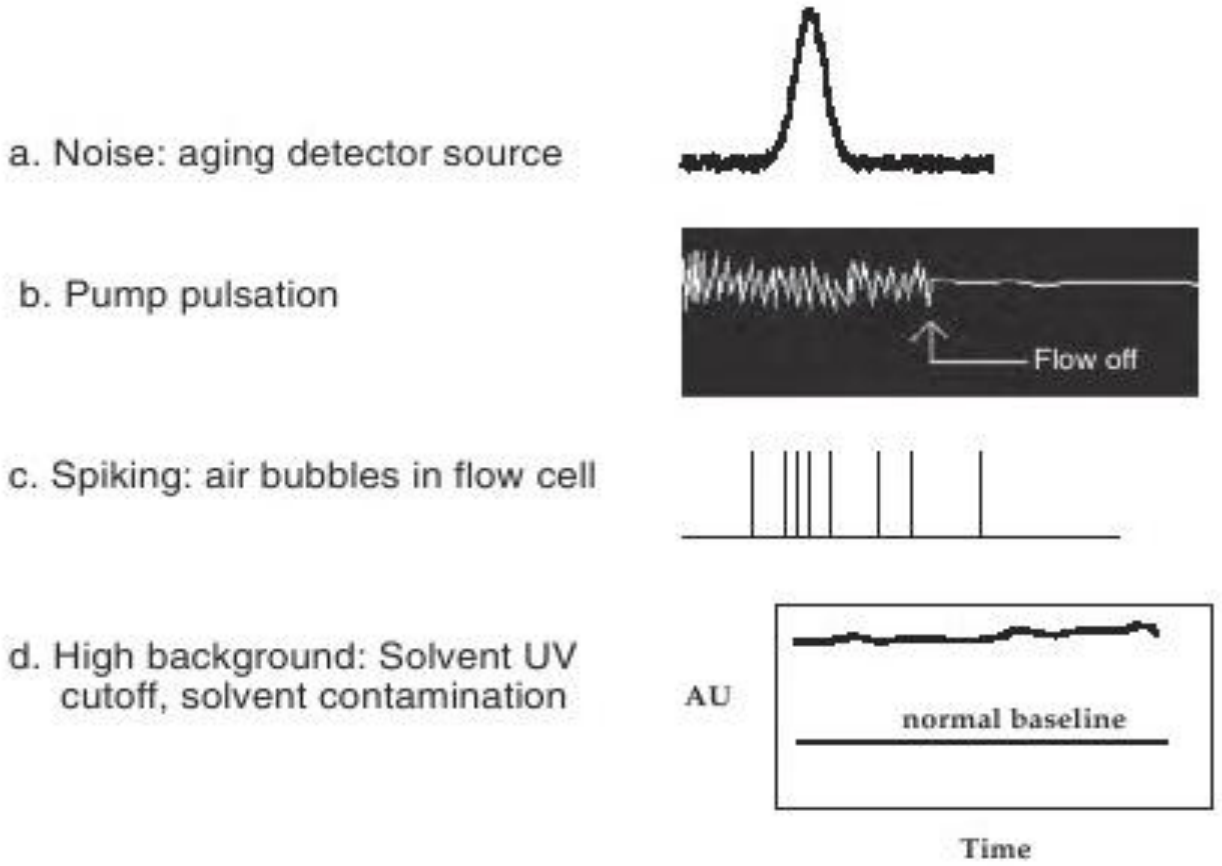


Figure 10.7. Examples of various baseline problems and possible causes.

بعض مشاكل Peaksوحلها ؟

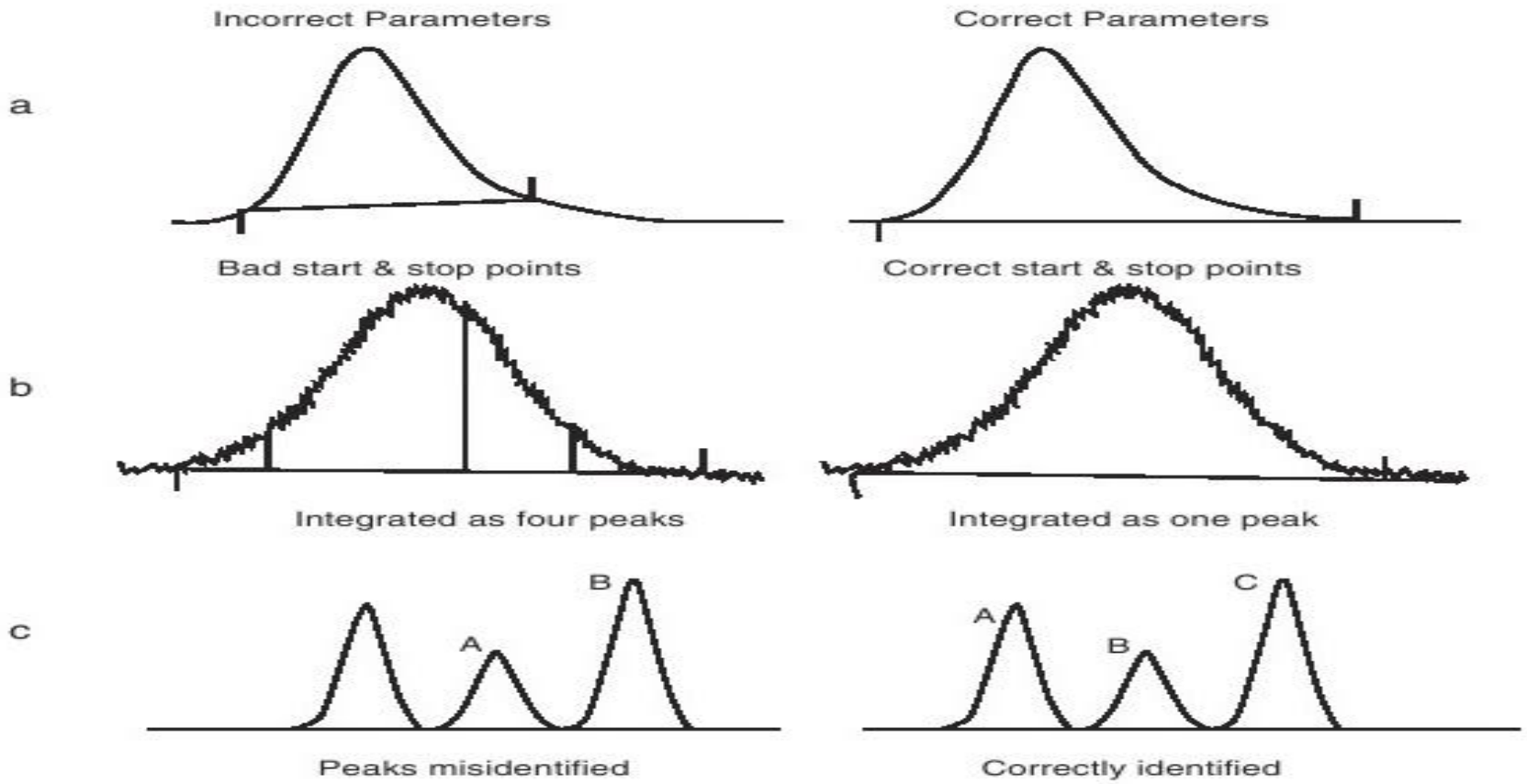


Figure 10.14. Example of various peak integration problems caused by setting incorrect integration parameters such as bad baseline and area thresholds (a and b) or wrong retention time windows (c). Figure courtesy of Waters Corporation.

بعد مشاكل جهاز HPLCومسبباتها؟

Table 10.4. Summary of Symptoms of Common HPLC Problems and Probable Causes

	Symptom	Most probable cause
Pressure	Too high	Column, filter
	Too low	Leaks from column, pump
Baseline	Noisy	Old lamp, mobile phase
	Drifting (wandering)	Degassing, mobile phase, contaminated column
Peak	No peak	Detector, injector
	Bad peaks	Column, mobile phase, bad column connections
Data performance	Retention time	Degassing, column, temperature changes
	Peak area	Autosampler, column, integration, lamp
	Accuracy	Sample prep, autosampler

Definitions

- %RSD
- Asymmetry Factor
- Baseline Noise
- Capacity Factor
- CV
- Dead Time
- Dwell Volume
- Extra-column Volume
- Flat-top
- Internal Standard
- Peak Width
- Plate Number
- Resolution
- Retention Time
- Separation Factor
- Shoulder
- System Suitability
- Tailing Factor

Definition: CV

CV ("Coefficient of Variation") is a common synonym for "%RSD" of a number of determinations: the standard deviation (SD) divided by the mean, expressed as a percentage.

as a percentage:

$$\%\boldsymbol{RSD} = \%\boldsymbol{CV} = \frac{\boldsymbol{SD}}{\bar{\boldsymbol{x}}}$$

Definition: %RSD

%RSD ("Percent Relative Standard Deviation") is a common synonym for the "CV" of a number of determinations: the standard deviation divided by the mean, expressed as a percentage.

$$\%\boldsymbol{RSD} = \%\boldsymbol{CV} = \frac{\boldsymbol{SD}}{\bar{\boldsymbol{x}}} \times \mathbf{100}$$

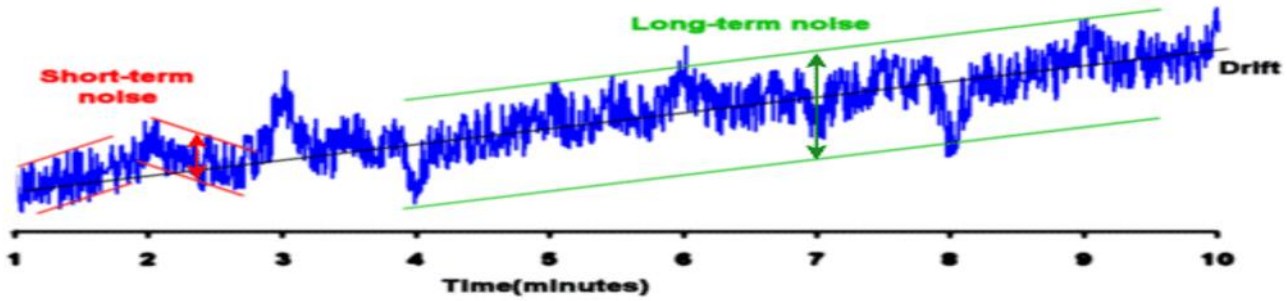
Baseline noise

Baseline noise: Random, short-term fluctuation in the detector signal when no analyte is present. Determines signal-to-noise ratio S/N, LOD, and LOQ.

Short-term noise : Random noise with a time constant of seconds or less. Affects instantaneous precision and peak-height measurement.

Long-term noise : Random or low-frequency noise with a time constant from a few seconds to a few minutes. Affects baseline stability during a run.

Drift : A steady, systematic movement of the baseline with a time constant from several minutes to hours. It is not true random noise; it affects accuracy and baseline correction.

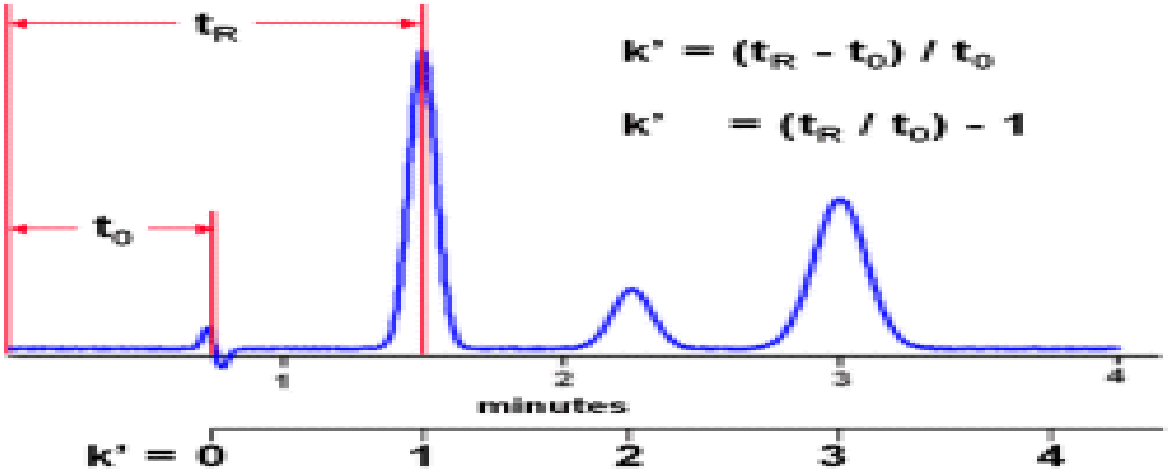


Definition: Dead time

The **dead time** of a column is symbolized by either **tM** (ASTM or IUPAC terminology) or **t0** (USP terminology). In either case, it is the elution time of an unretained component. It is also equal to the time required to flush one column volume of eluant through the column:

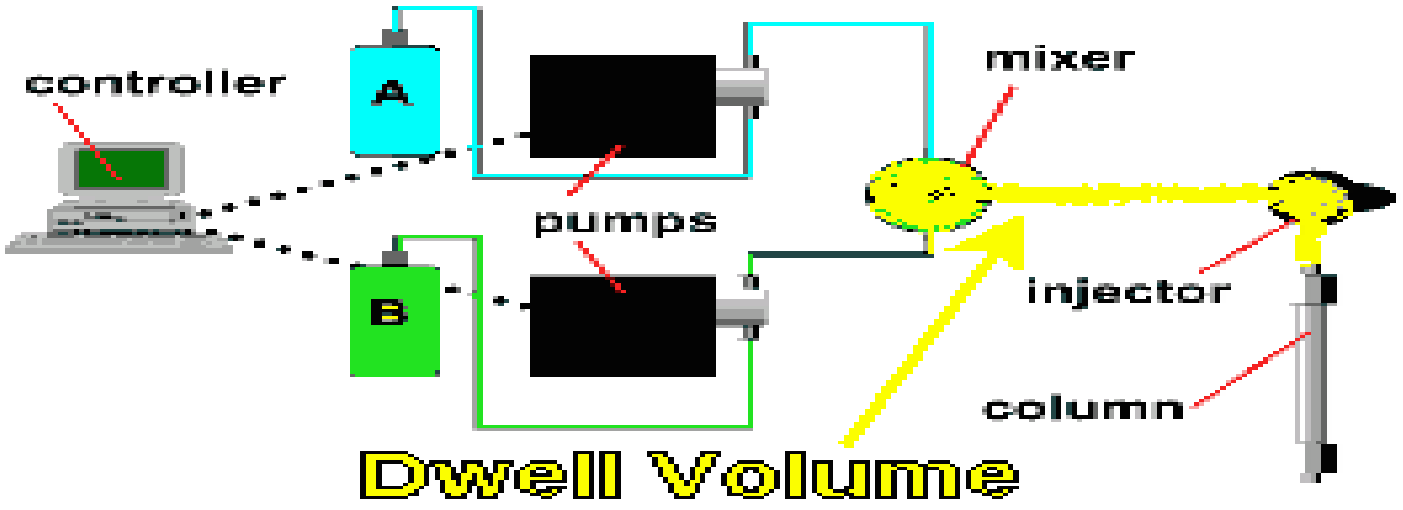
t0 = V0/F = Vm/F = tm

where **V0** (**Vm**) is the volume of liquid contained in the column, and **F** is the flow rate



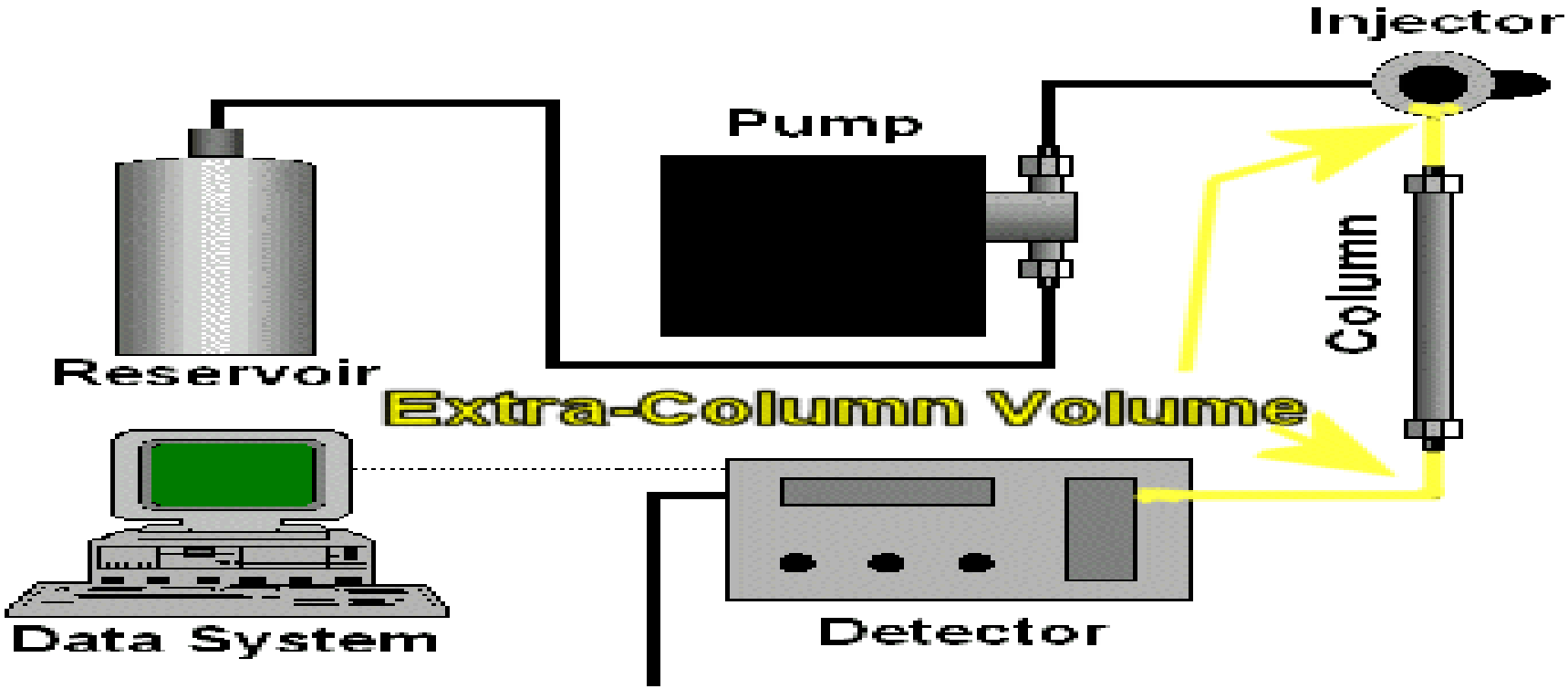
Definition: Dwell volume

The **dwell volume** (also called "gradient delay volume") of a gradient or on-line mixing system is the volume of liquid contained in the system between the point where the gradient is formed and the point where the mobile phase enters the column. This volume includes the mixer, transfer lines, and any swept volume (including the sample loop) in the injection system. In a low-pressure-mixing ("one pump") system, the dwell volume will also include the proportioning valve and pump liquid-end components.



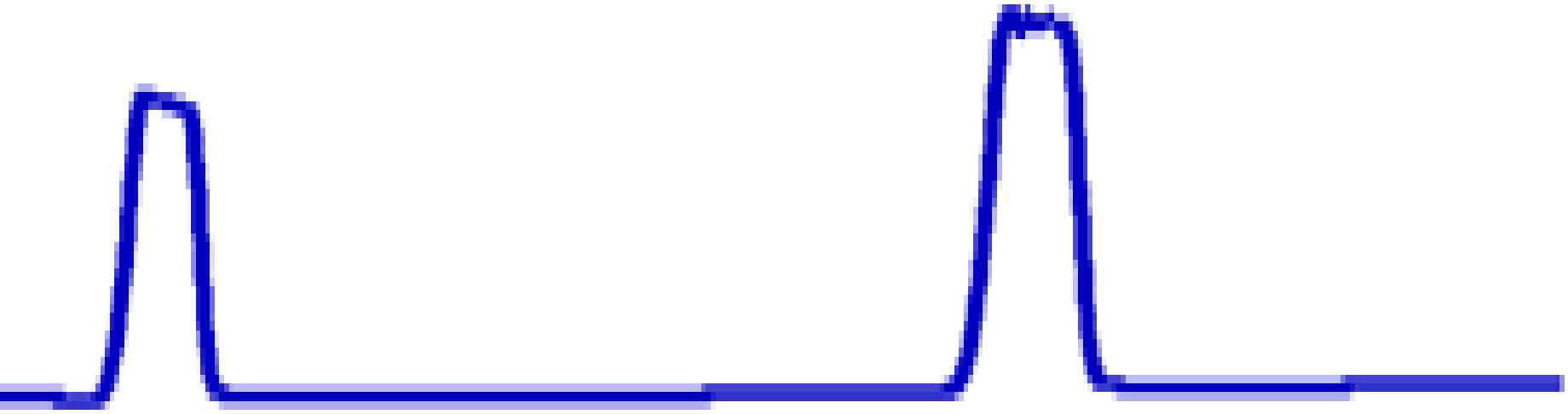
Definition: Extra-column volume

The **extra-column volume** of an HPLC system includes the entire volume of liquid in the system between the point where the sample is introduced and the column inlet plus the volume between the column outlet and the outlet of the detector cell.



Definition: flat-top

In some cases, peaks may take on a "flat-top" or mesa appearance

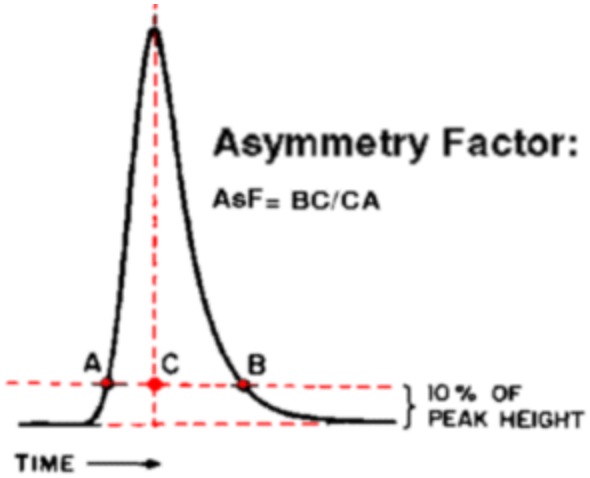


Definition: internal standard

An **Internal Standard** is a compound purposely added to both samples and standards at a known concentration in order to provide a basis for comparison in quantitation. Computationally, the only difference between "internal standardization" and "external standardization" is that the former works with area ratios and concentration ratios rather than areas and concentrations. Internal standardization can improve precision when the dominant sources of error are related to sample preparation or injection; because such errors affect both the internal standard and the analyte peak in the same way, they will tend to cancel out when the ratio of areas is calculated. Internal standardization can degrade precision when the dominant sources of error are related to separation, peak shape, or integration; because such errors affect the internal standard peak and the analyte peak differently, they will tend to combine when the ratio of areas is calculated.

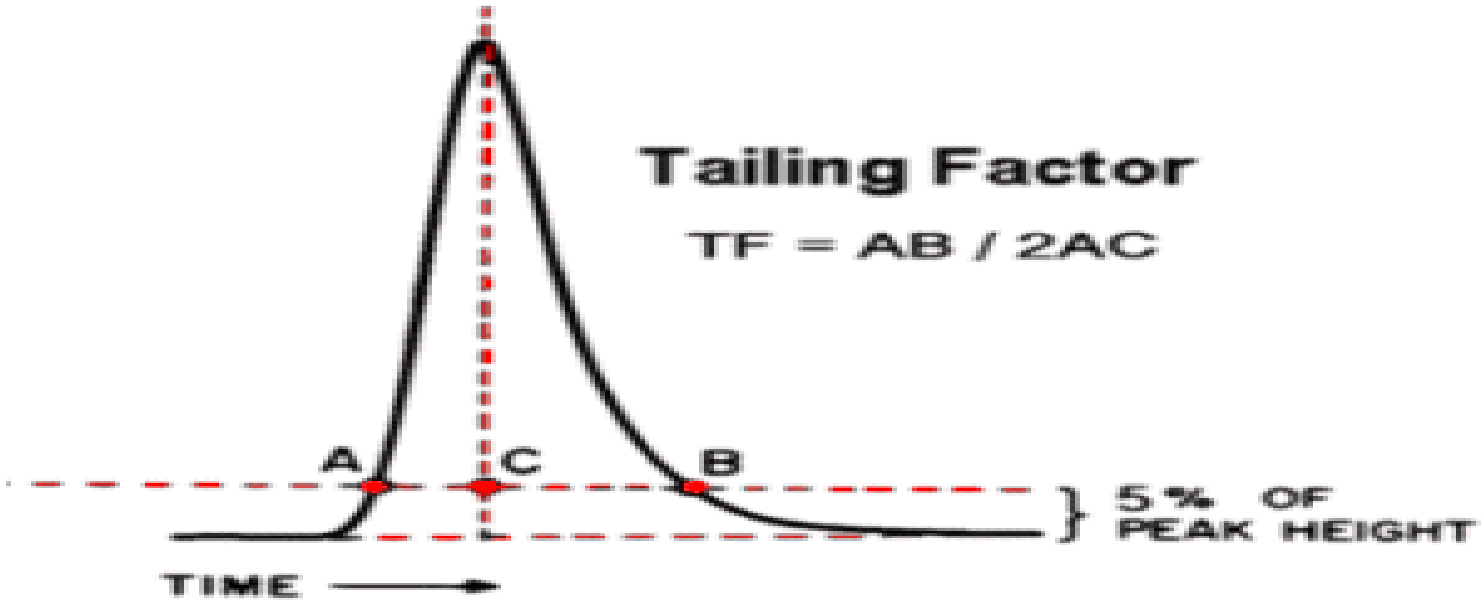
Definition: Asymmetry factor

The **asymmetry factor** is a measure of peak tailing. It is defined as the distance from the center line of the peak to the back slope divided by the distance from the center line of the peak to the front slope, with all measurements made at 10% of the maximum peak height. The asymmetry factor of a peak will typically be similar to the tailing factor for the same peak, but the two values cannot be directly converted



Definition: Tailing factor

The **tailing factor** is a measure of peak tailing. It is defined as the distance from the front slope of the peak to the back slope divided by twice the distance from the center line of the peak to the front slope, with all measurements made at 5% of the maximum peak height. The tailing factor of a peak will typically be similar to the asymmetry factor for the same peak, but the two values cannot be directly converted.

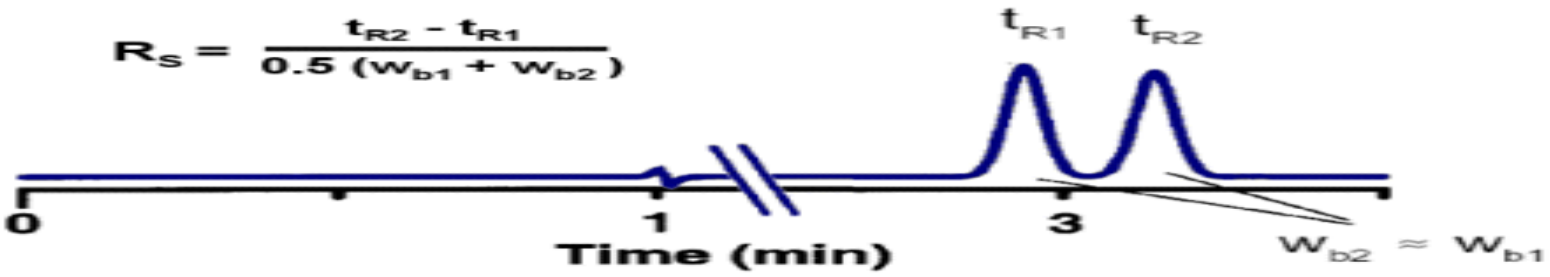


Definition: Resolution

The **resolution** between two chromatographic peaks is a measure of how well they are separated. Resolution is defined as the center-to-center separation between two peaks divided by the average baseline width of those peaks:

Rs = (tR2 - tR1)/[0.5(wb2 + wb1)]

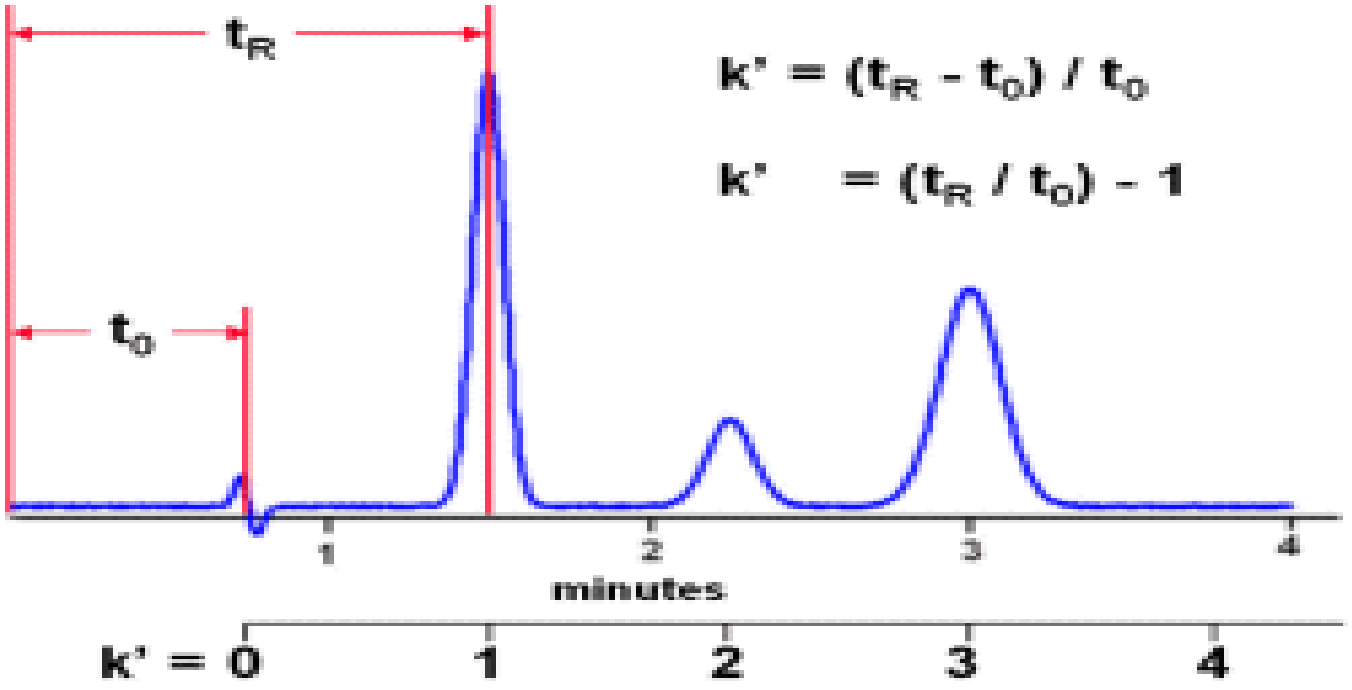
where **tR1** and **tR2** are the retention times of the first and second peaks of interest, and **wb1** and **wb2** are the baseline widths of those peaks



For ideally-shaped (Gaussian) peaks, a resolution of 1.0 corresponds to approximately 5% overlap between equal-size peaks. A resolution of 1.5 corresponds to 1% overlap

Definition: Retention time

The retention time, symbolized by **tR**, is the elapsed time from injection until the highest-concentration part of the peak has eluted from the column



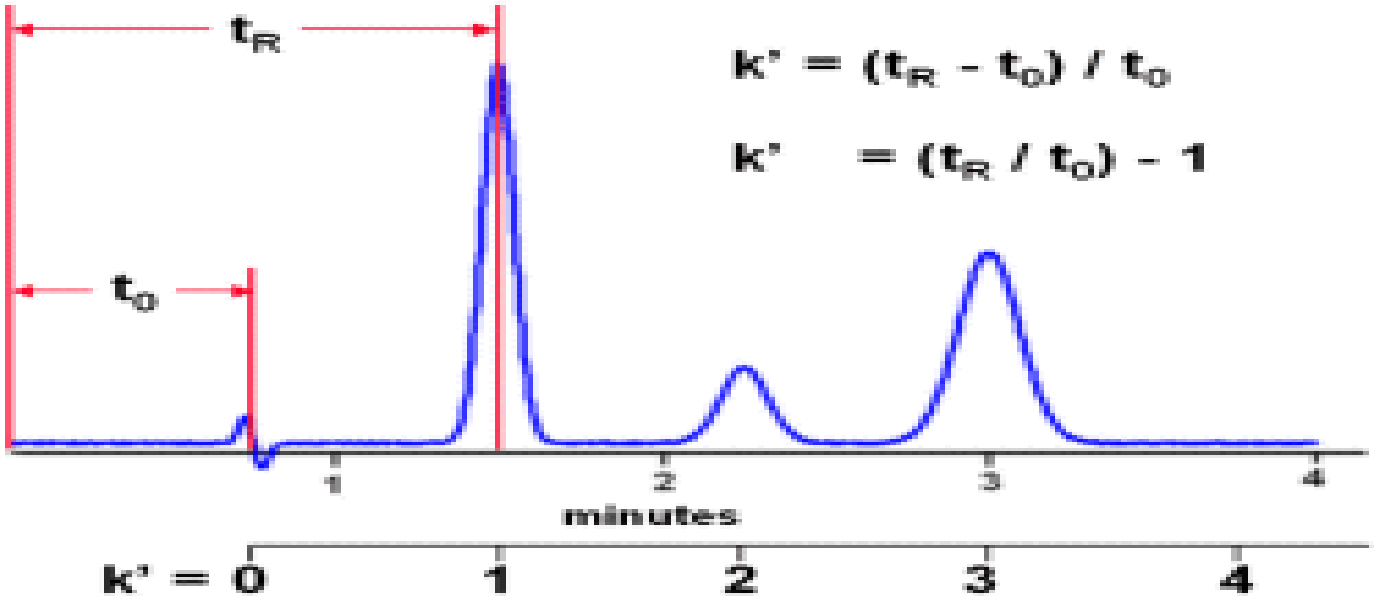
k’ Definition: Capacity factor عامل القدرة

The **capacity factor** (also called "capacity ratio") is symbolized by **k'** (USP terminology) or **k** (IUPAC/ASTM terminology). It is a measure of the retention of a peak that is independent of column geometry or mobile phase flow rate.

The capacity factor is calculated as:

k' = (tR - t0)/t0

where **tR** is the retention time of the peak, and **t0** is the dead time of the column.



Note that the capacity factor cannot be calculated in this manner from a gradient chromatogram. The "apparent" capacity factor obtained when this is done is meaningless chromatographically

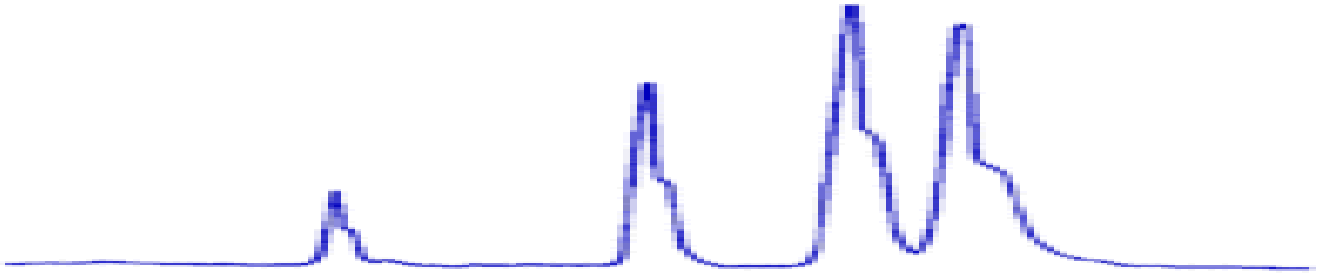
Definition: Separation factor •

The **separation factor**, symbolized by **a or α (Selectivity factor)**, is a measure of the selectivity of a chromatographic system. It is calculated from the k' values of two adjacent peaks:

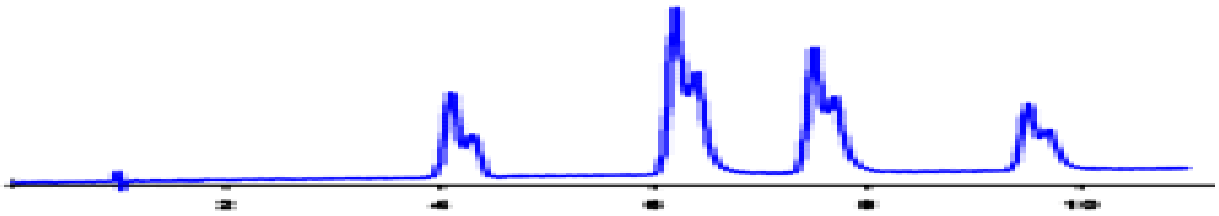
a = k'2/k'1

Definition: Shoulder •

While there is no strict numerical way to describe a "shoulder," most chromatographers recognize the problem when they see it. The peak (or peaks) are asymmetrical, but with a sharply defined tail that appears as though it may be an additional component partially resolved from the main peak (in fact, a partially resolved interference is one possible cause for a peak shoulder).تحلل الجزيئات



In extreme cases, the shoulder may be pronounced enough to make the peaks appear as a doublet or "split" peaks



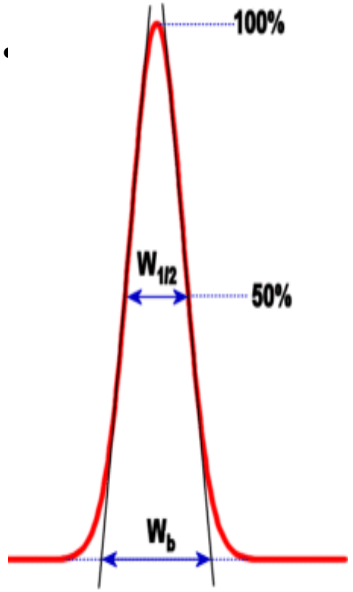
Definition: Peak width

A number of different measurements of **peak width** are used in HPLC. The most common are:

Baseline width (wb). This is measured by drawing tangents to the front and back slopes of the peak. The baseline width is the distance between the intersections of these tangents with the baseline.

The **width-at-half-height (w0.5)** is the distance from the front slope of the peak to the back slope of the peak measured at 50% of the maximum peak height.

If the peak has the ideal "Gaussian" shape, then **wb = 1.7 w0.5**, but for most "real" peaks, baseline width is somewhat larger than this.



Definition: System suitability

System suitability is defined by ICH as "the checking of a system, before or during analysis of unknowns, to ensure system performance."

System suitability criteria may include such factors as plate count, tailing, retention, and/or resolution. System suitability criteria should also include a determination of reproducibility (%RSD) when a system suitability "sample" (a mixture of main components and expected by-products/interferences) is run.

Definition: Plate number •

The **plate number** of a chromatographic peak is a measure of the amount of peak broadening that has occurred. It is approximately independent of the retention of that peak.

The plate number can be defined as:

$$\mathbf{N = [tR/s]^2}$$

where **tR** is the retention time of the peak and **s** is the square root of the variance of the peak.

More commonly, the plate number is calculated from either the baseline width:

$$\mathbf{N = 16(tR/wb)^2}$$

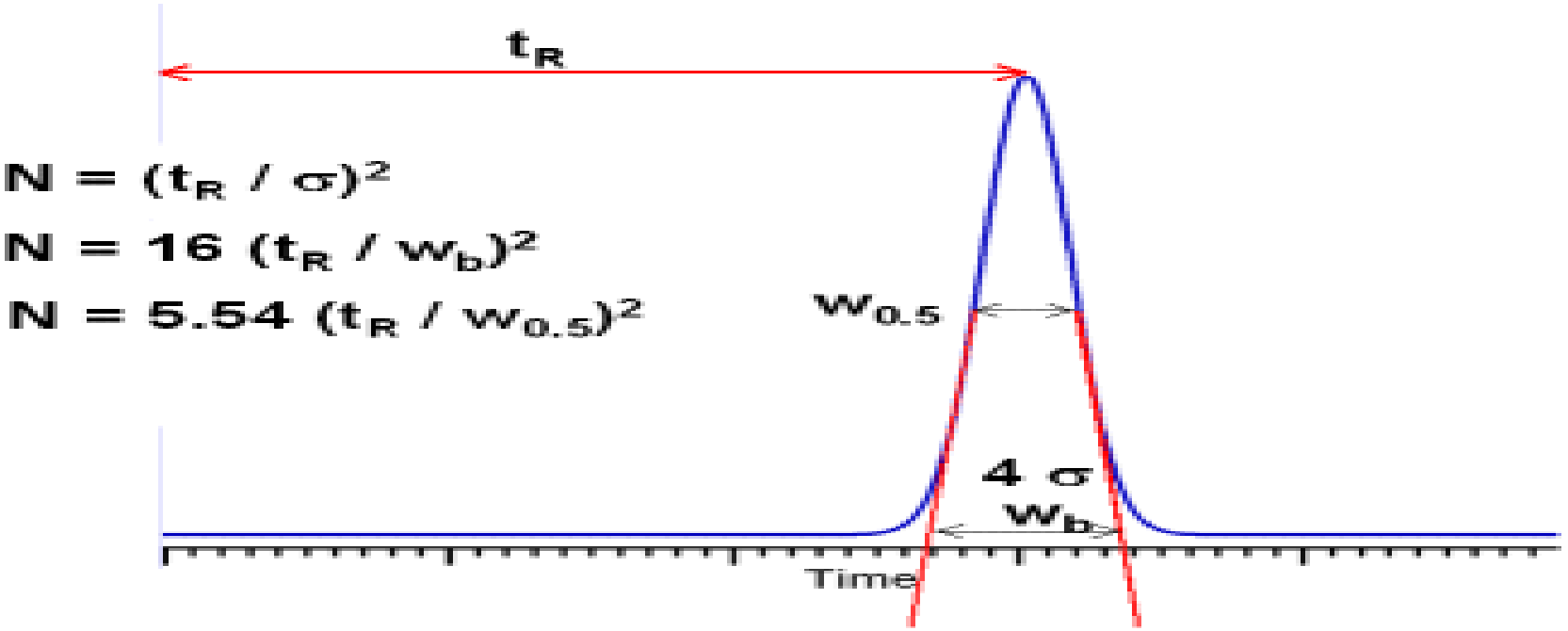
or from the width at half height:

$$\mathbf{N = 5.54(tR/w0.5)^2}$$

If a peak is "ideally" shaped (Gaussian, perfectly symmetrical), all three measurements will give the same plate number. In practice, the baseline width method gives lower (worse) plate numbers than does the width-at-half-height method.

-

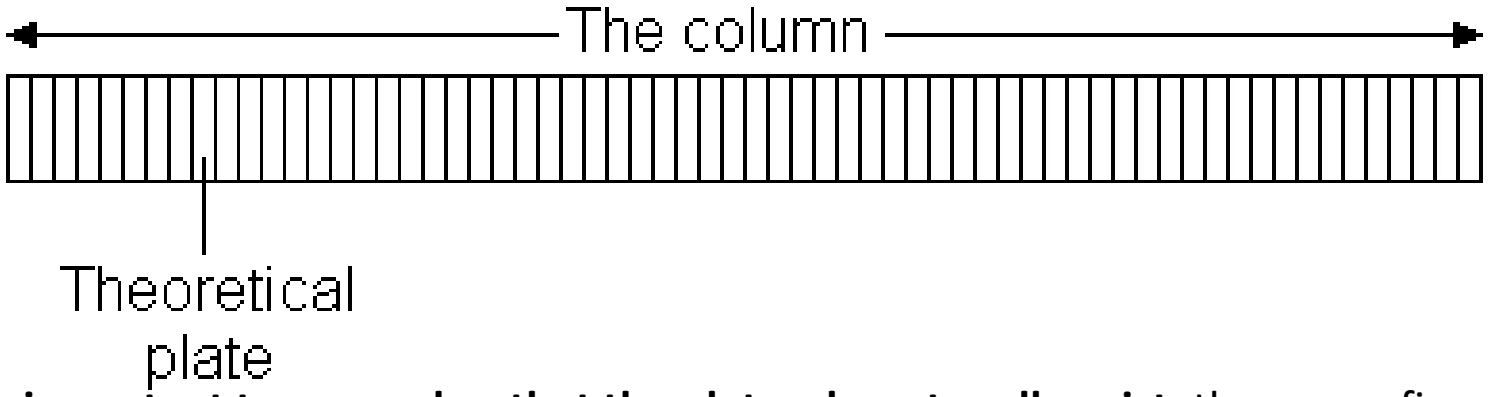
Note that the plate number cannot be calculated in this manner from a gradient chromatogram. The "apparent" plate number obtained when this is done is meaningless chromatographically.



The Theoretical Plate Model of Chromatography (N) or the System efficiency

نموذج نظرية اللوحات او كفائة النظام

The plate model supposes that the chromatographic column is contains a large number of separate layers, called *theoretical plates*. Separate equilibrations of the sample between the stationary and mobile phase occur in these "plates". The analyte moves down the column by transfer of equilibrated mobile phase from one plate to the next.



It is important to remember that the plates do not really exist; they are a figment of the imagination that helps us understand the processes at work in the column.They also serve as a way of measuring column efficiency, either by stating the number of theoretical plates in a column, *N* (the more plates the better), or by stating the plate height; the *Height Equivalent to a Theoretical Plate* (the smaller the better).

If the length of the column is *L*, then the HETP is

HETP = L / N

The number of theoretical plates that a real column possesses can be found by examining a chromatographic peak after elution;

$$N = \frac{5.55\ t_R^{\,2}}{w_{1/2}^{\,2}}$$

where $w_{1/2}$ is the peak width at half-height.

As can be seen from this equation, columns behave as if they have different numbers of plates for different solutes in a mixture

Resolution R عامل القدرة

Although the selectivity factor, a, describes the separation of band centres, it does not take into account peak widths. Another measure of how well species have been separated is provided by measurement of the *resolution*. The resolution of two species, A and B, is defined as

$$R = \frac{2[(t_R)_B - (t_R)_A]}{W_A + W_B}$$

Baseline resolution is achieved when $R = 1.5$

It is useful to relate the resolution to the number of plates in the column, the selectivity factor and the retention factors of the two solutes;

$$R = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{1 + k'_B}{k'_B}\right)$$

To obtain high resolution, the three terms must be maximised. An increase in *N*, the number of theoretical plates, by lengthening the column leads to an increase in retention time and increased band broadening - which may not be desirable. Instead, to increase the number of plates, the height equivalent to a theoretical plate can be reduced by reducing the size of the stationary phase particles.

It is often found that by controlling the capacity factor, k', separations can be greatly improved. This can be achieved by changing the temperature (in Gas Chromatography) or the composition of the mobile phase (in Liquid Chromatography).

The selectivity factor, a, can also be manipulated to improve separations. When a is close to unity, optimising k' and increasing N is not sufficient to give good separation in a reasonable time. In these cases, k' is optimised first, and then a is increased by one of the following procedures:

1-Changing mobile phase composition

2-Changing column temperature

3-Changing composition of stationary phase

4-Using special chemical effects (such as incorporating a species which complexes with one of the solutes into the stationary phase)

The Rate Theory of Chromatography نظريات معدل الفصل الكروماتوجرافي

A more realistic description of the processes at work inside a column takes account of the time taken for the solute to equilibrate between the stationary and mobile phase (unlike the plate model, which assumes that equilibration is infinitely fast). The resulting band shape of a chromatographic peak is therefore affected by the rate of elution. It is also affected by the different paths available to solute molecules as they travel between particles of stationary phase. If we consider the various mechanisms which contribute to band broadening, we arrive at the Van Deemter equation for plate height; (HETP = $A + B / u + C u$) where u is the average velocity of the mobile phase. *A, B,* and *C* are factors which contribute to band broadening.

***A* - Eddy diffusion**

The mobile phase moves through the column which is packed with stationary phase. Solute molecules will take different paths through the stationary phase at random. This will cause broadening of the solute band, because different paths are of different lengths.

***B* - Longitudinal diffusion**

The concentration of analyte is less at the edges of the band than at the center. Analyte diffuses out from the center to the edges. This causes band broadening. If the velocity of the mobile phase is high then the analyte spends less time on the column, which decreases the effects of longitudinal diffusion.

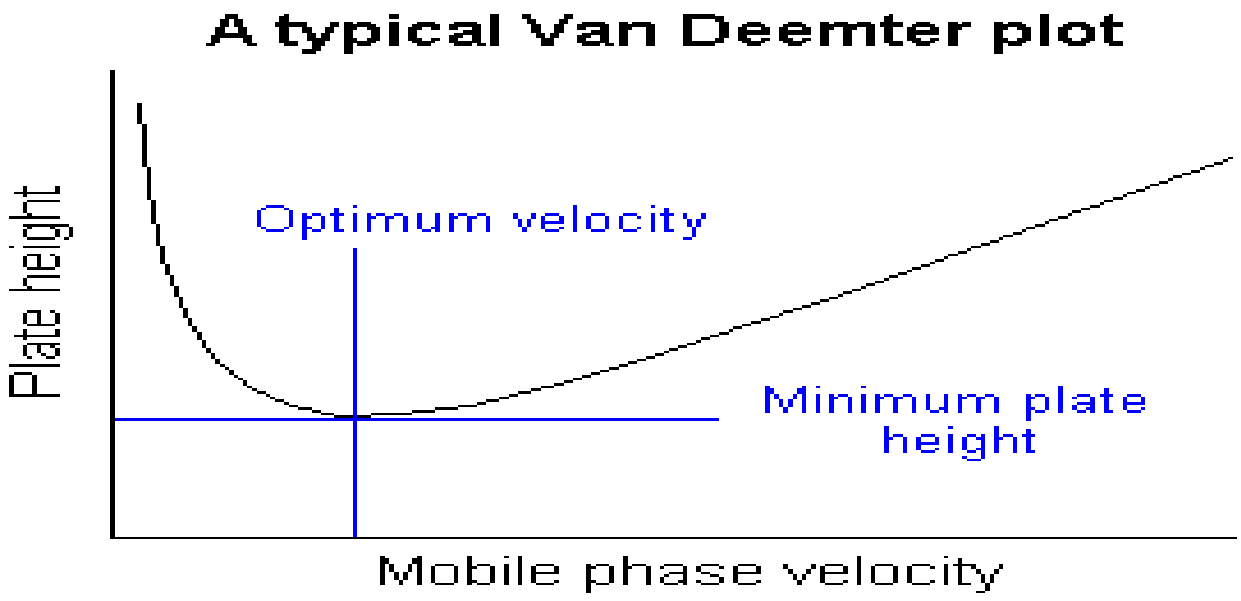
***C* - Resistance to mass transfer**

The analyte takes a certain amount of time to equilibrate between the stationary and mobile phase. If the velocity of the mobile phase is high, and the analyte has a strong affinity for the stationary phase, then the analyte in the mobile phase will move ahead of the analyte in the stationary phase. The band of analyte is broadened. The higher the velocity of mobile phase, the worse the broadening becomes.

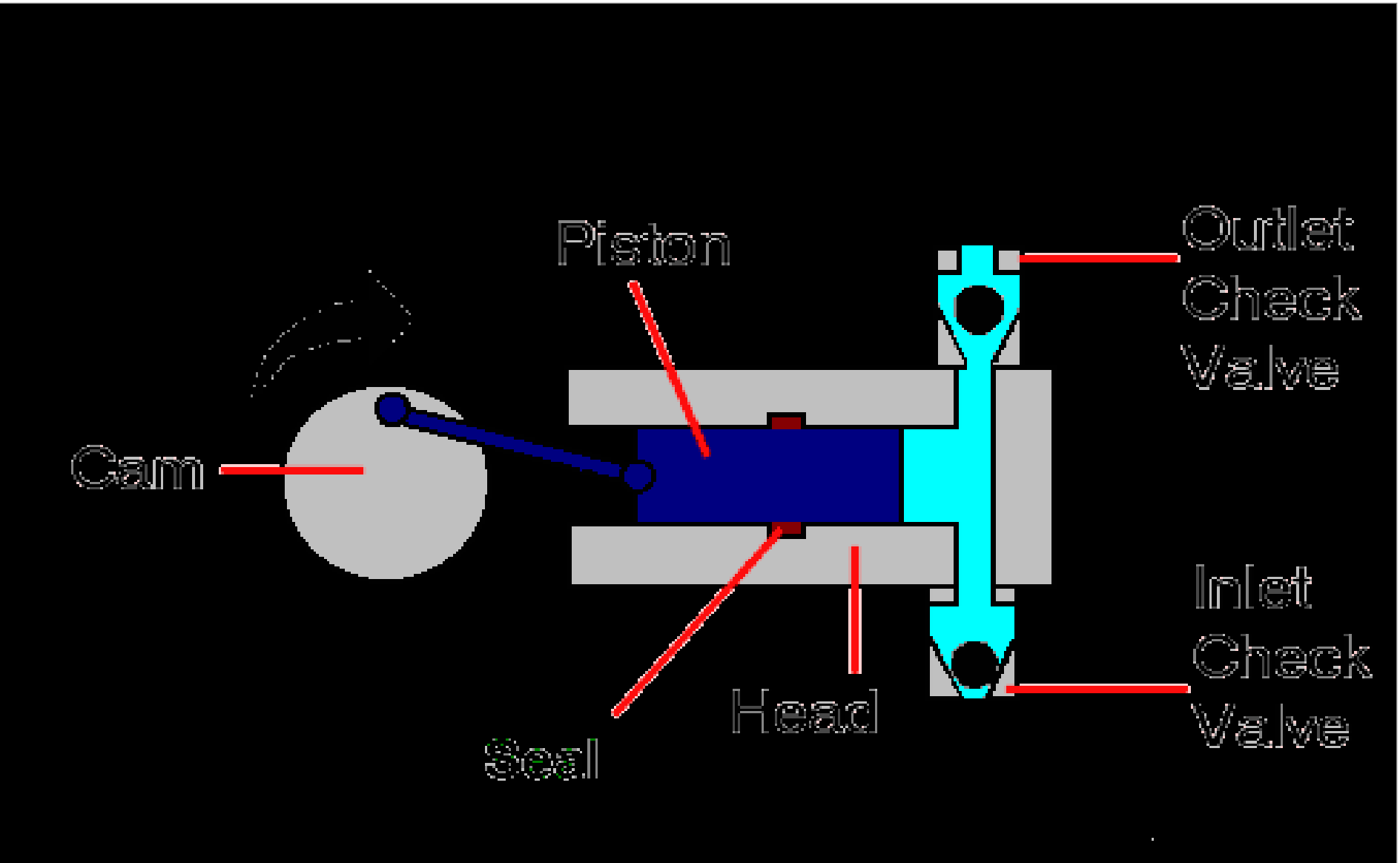
Van Deemter plots

A plot of plate height vs. average linear velocity of mobile phase.

Such plots are of considerable use in determining the optimum mobile phase flow rate.



INLET CHECK VALVE: the check valve on an HPLC pump that allows mobile phase to flow from the reservoir to the pump, but not in the reverse direction



Single- and Variable-Wavelength Detectors

Table 2 UV Cutoff[a] Wavelengths λ_{min} of Mobile-Phase Solvents of Ultrapure Quality

Solvent	λ_{min} (nm)
Acetone	330
Acetonitrile	190
Chloroform	245
Cyclohexane	200
Dichloromethane	235
Diethyl ether	215
Dimethylformamide	270
1,4-Dioxane	215
Ethanol	210
Ethyl acetate	260
n-Hexane	190
Methanol	205
2-Propanol	210
Tetrahydrofuran	220

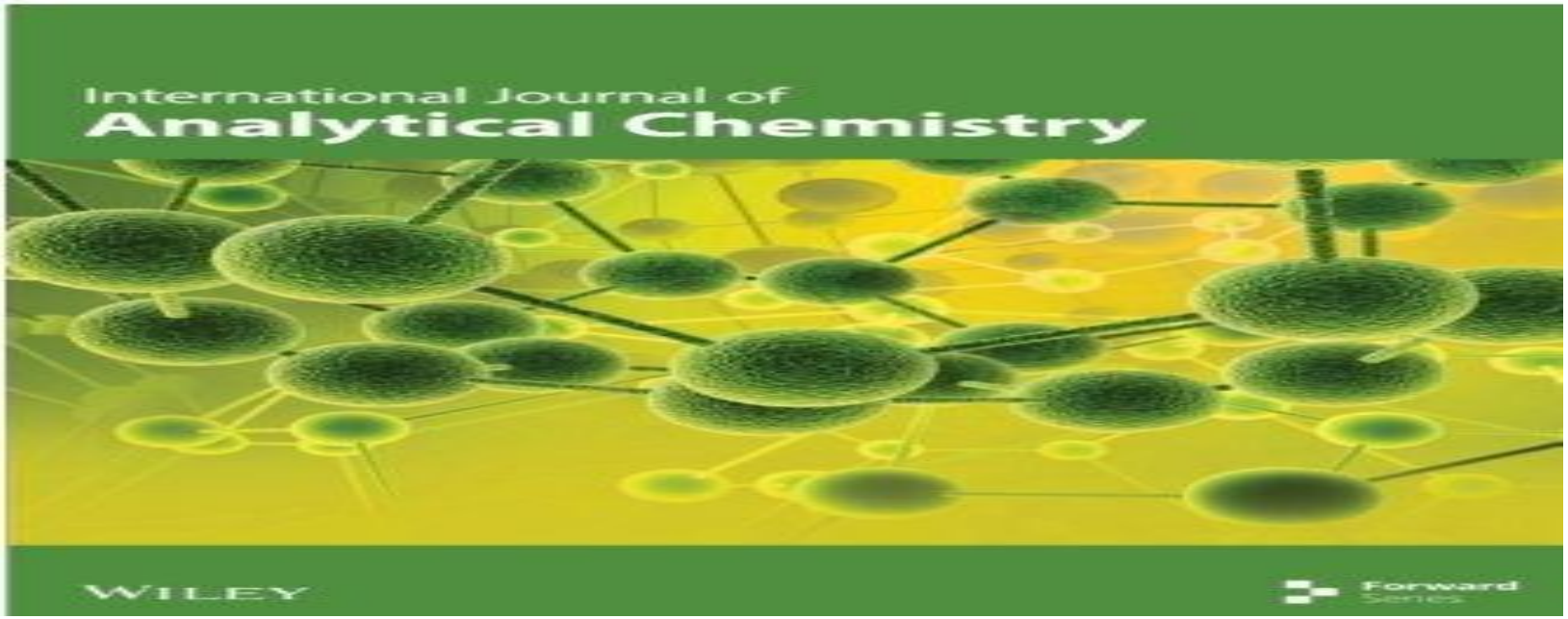
[a]The *cutoff* is defined as the wavelength at which the absorbance is 1 AU in a 10-mm cuvette.

الشق العملي الجزء الثاني

اما الجزء الثاني فكيف تطور**Method Validation**

لطريقة فحص على جهاز HPLCحسب بحثي المنشور 2026في مجلة Wiley

https://www.researchgate.net/publication/400769184_A_High-Performance_Liquid_Chromatography_Method_for_Simultaneous_Analysis_of_Naringin_Ascorbic_Acid_and_Caffeine_Design_Development_and_Validatio



Method Validation

:

"The process by which it is established, through me in

The Pharmaceutical Automated Analysis Laboratory

Institution: An-Najah National University

Department: Department of Pharmacy

studies, that the performance characteristics of the method meet the requirements for its intended purpose, The aim of this step is to ensure that by laboratory tests this method is complete valid according to Validation guidelines and book. according to Validation ICH Q2(R1) guideline. and USP 32 (1225) guideline , CRC Handbook of ANALYTICAL VALIDATION) (book),ISO/IEC 17025:2017 guideline This Validation guidelines and book give a written plan for how validation steps will be worked and give the acceptance criteria.

The aim of Validation

The aim of this research is to validate non-standard methods and build a specific, selective, sensitive, precise (intraday& interday precise), repeatable, accurate, linear, and robust system , validate RP-HPLC routine analytical analysis for quantification and synchronized determination of Ascorbic Acid , caffeine, and naringin in different concentrations in a range from 7.815 µg/ml to 125 µg/ml according to Validation ICH Q2(R1) guideline. {8} and USP 32 (1225) guideline {17}, CRC Handbook of ANALYTICAL VALIDATION) (book){14}, ISO/IEC 17025:2017 guideline {15},) This Validation guidelines and book give a written plan for how validation steps will be worked and give the acceptance criteria.

Method & Chromatographic Assessment:

Apparatus:

The HPLC technique was used to analyze Mix from Ascorbic Acid , Caffeine, and Naringin. The reverse phase chromatography analytical method was performed using Binary HPLC Pump Waters 1525with a six-port manual injector and a Waters 2998 diode array detector, and Breeze 2 software was utilized for instrument control, data collection, and data processing. And with non-Polar Stationary Phase C18 silica based bonded Phase Column (250 mm long X 4.6 mm i.d packed with 5 µm) (column Symmetry Shield™ RP-18) was used.

Wave Length Selection:

The aim of this Step To determine the best wavelength (λ) for analyzing of mix of Ascorbic Acid , Caffeine, and Naringin (part of System suitability)(see System suitability) use a diode Waters 2998 array detector (DAD) with different wave lengths (λ) 220 nm, (λ) 254 nm, (λ) 243 nm, and (λ) 288 nm by (HPLC) Waters 1525, und injection 20 µl mix of 31.25 µg/ml from Ascorbic Acid , Caffeine, and Naringin by flow rate 1 ml /min with Mobile phase Isocratic elution (75% phosphate buffer (pH 3.5) and 25% acetonitrile and read the % Area at different wavelengths (λ) 220 nm, (λ) 254 nm, (λ) 243 nm, and (λ) 288 nm.

245 nm Wave Length Selection:

Ascorbic Acid	Caffeine	Naringin
75.4% Area	**21.34% Area**	**3.23% Area**

{Table 1} % Area of mix of 31.25 μg/ml from Ascorbic Acid , Caffeine, and Naringin at wavelength (λ) 254 nm

Result :

Ascorbic Acid has very high % Area, but Caffeine has low % Area and Naringin have very low % Area. This makes (λ) 254 nm not good for detecting all three compounds synchronized.

243 nm Wave Length Selection:

Ascorbic Acid	Caffeine	Naringin
82.81% Area	**13.48% Area**	**3.66% Area**

{**Table 2**} % Area of mix of 31.25 µg/ml from Ascorbic Acid , Caffeine, and Naringin at wavelength (λ) 243 nm

result :

Ascorbic Acid has very high % Area, but Caffeine has low % Area and Naringin have very low % Area. This makes (λ) 243 nm not good for detecting all three compounds synchronized.

288 nm Wave Length Selection:

Ascorbic Acid	Caffeine	Naringin
10.7% Area	40.24% Area	48.98% Area

{Table 3 } % Area of mix of 31.25 µg/ml from Ascorbic Acid , Caffeine, and Naringin at wavelength (λ) 288 nm

Result :

Ascorbic Acid has very low % Area, but Caffeine high % Area and Naringin have very high % Area. This makes (λ) 288 nm not good for detecting all three compounds synchronized.

220 nm Wave Length Selection:

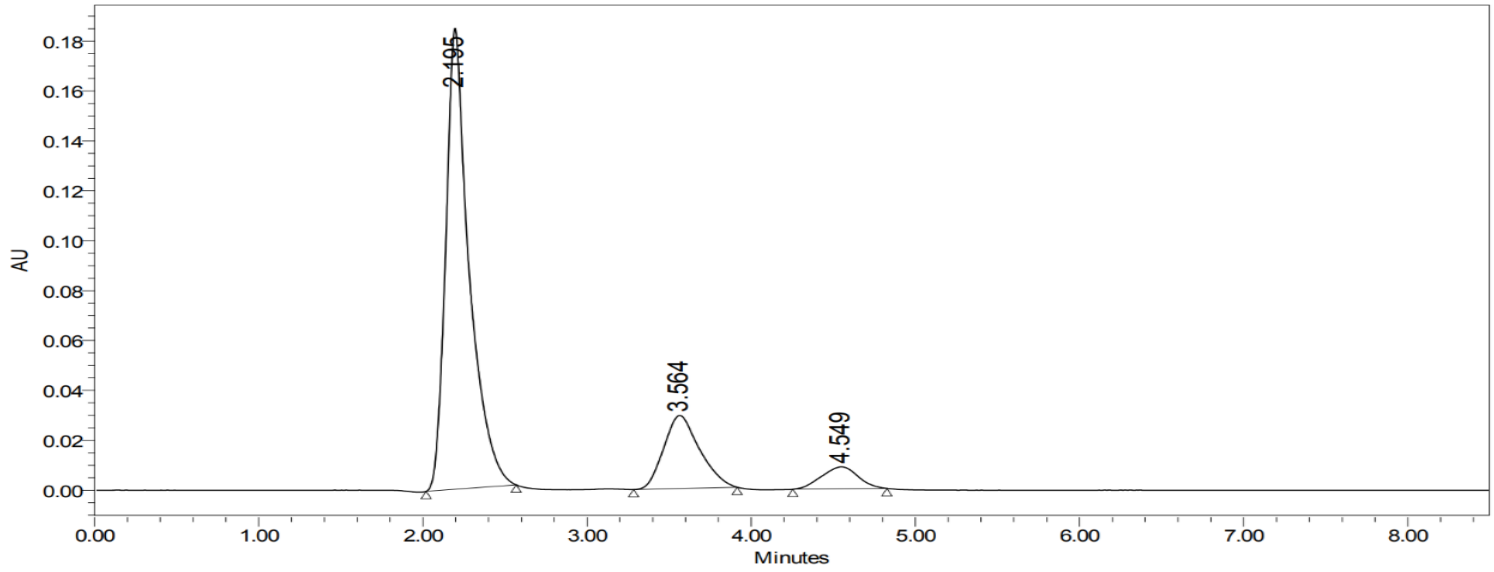
Ascorbic Acid	Caffeine	Naringin
21.85 % Area	40.91 % Area	36.77 % Area

% Area of mix of 31.25 µg/ml from Ascorbic Acid , Caffeine, and Naringin at wavelength (λ) 220 nm

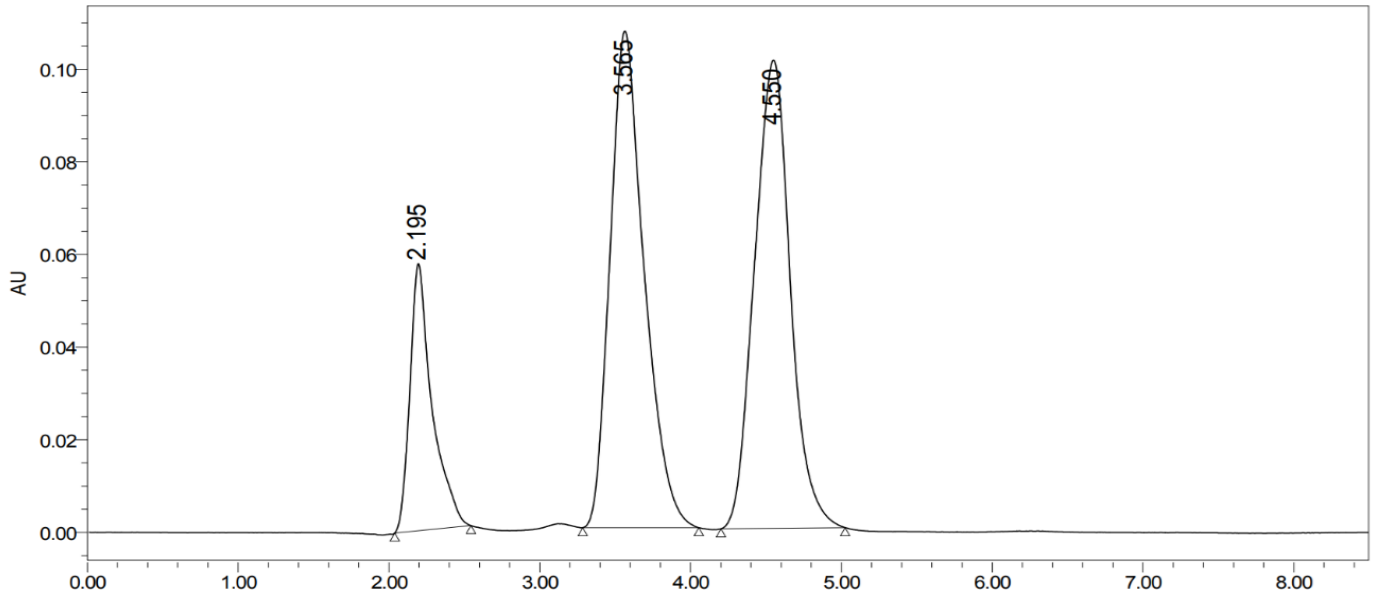
Result :

All three compounds have good % Area, this makes (λ) 220 nm best detecting wavelength for all three compounds synchronized.

Chromatogram of Ascorbic Acid (R.T 2.195 min) 31.25 µg/ml, Caffeine (R.T 3.564 min) 31.25 µg/ml and Naringin (4.789 min) 31.25 µg/ml at flow Rat of Mobile Phase 1 ml /min in 243 nm Mobile Phase: Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile.



Chromatogram of Ascorbic Acid (R.T 2.195 min) 31.25 µg/ml, Caffeine (R.T 3.565 min)) 31.25 µg/ml and Naringin (4.550 min)) 31.25 µg/ml at flow Rat of Mobile Phase 1 ml /min in 220nm
Mobile Phase: Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile



Mobile Phase Selection:

The aim of this Step to Select mobile phase that give best Separation in Short Run time (R.T), more than one mobile phase was tested, which give separation, two succeeded to separating Ascorbic Acid, Caffeine and Naringin, the comparison between them was in the Run time.

1) Mobile phase Polar index 8.2 (40% (methanol Polar index 5.1) with 60% water (Polar index 10.2))

Polar index of Mobile Phase =

(% of Solvent 1 X polar index Solvent 1) + (% of Solvent 2 X polar index Solvent 2)

= (0.40 X 5.1) + (0.60 X 10.2)

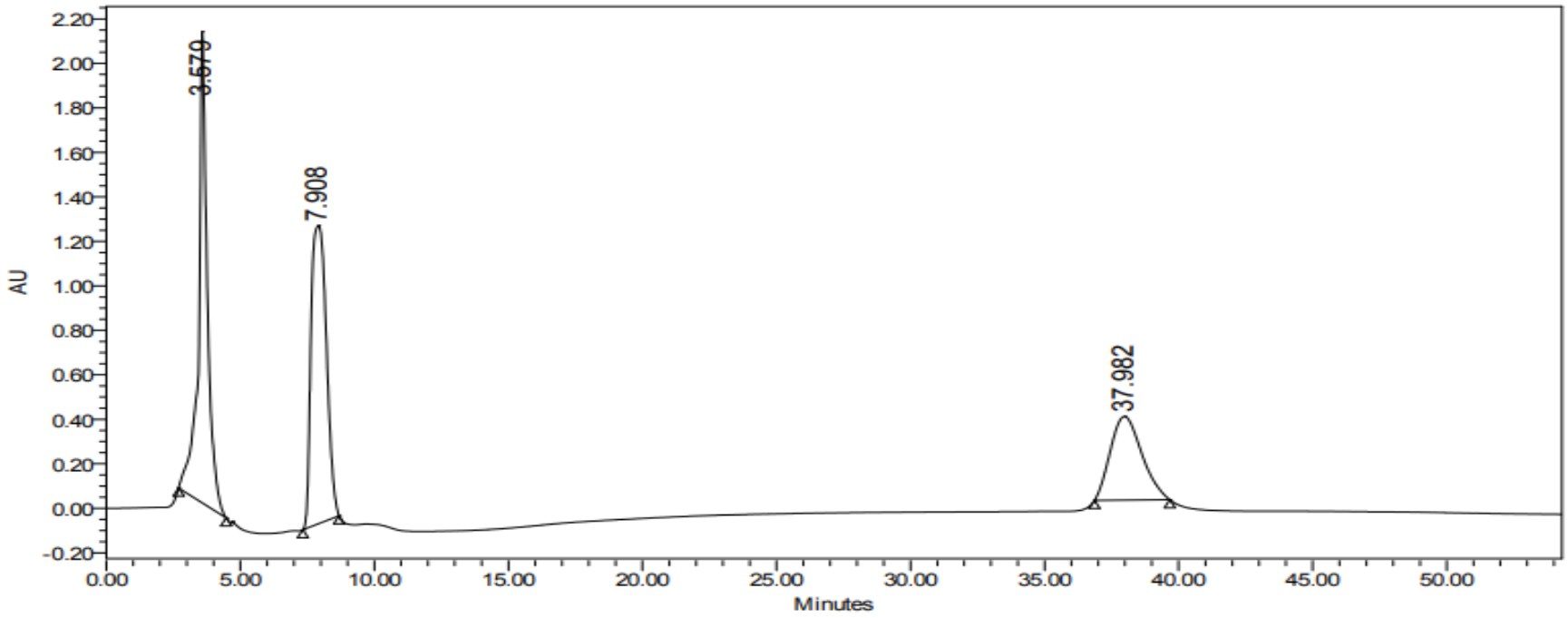
=8.16 ≈

R.T Ascorbic Acid	R.T Caffeine	R.T Naringin
3.118 min	6.968 min	33.485 min

the Analysis time for the 3 Components = 40 minutes
as following :

Mobile phase 40% (methanol with 60% water

Mobile phase (40% Methanol and 60% Water) give good Separation but Long Run time.



Chromatogram of Ascorbic Acid (R.T 3.670min), Caffeine (R.T 7.908 min) and Naringin (37.982 min) at flow Rat of Mobile Phase 1 ml /min Mobile Phase Isocratic elution 40% Water 60% Methanol at 220 nm.

2) Mobile phase (Polar index) Isocratic elution with 75% phosphate buffer (pH 3.5) Polar index 10.2 and 25% acetonitrile Polar index 5.8) Polar index of Mobile Phase =

(% of Solvent 1 X polar index Solvent 1) + (% of Solvent 2 X polar index Solvent 2)

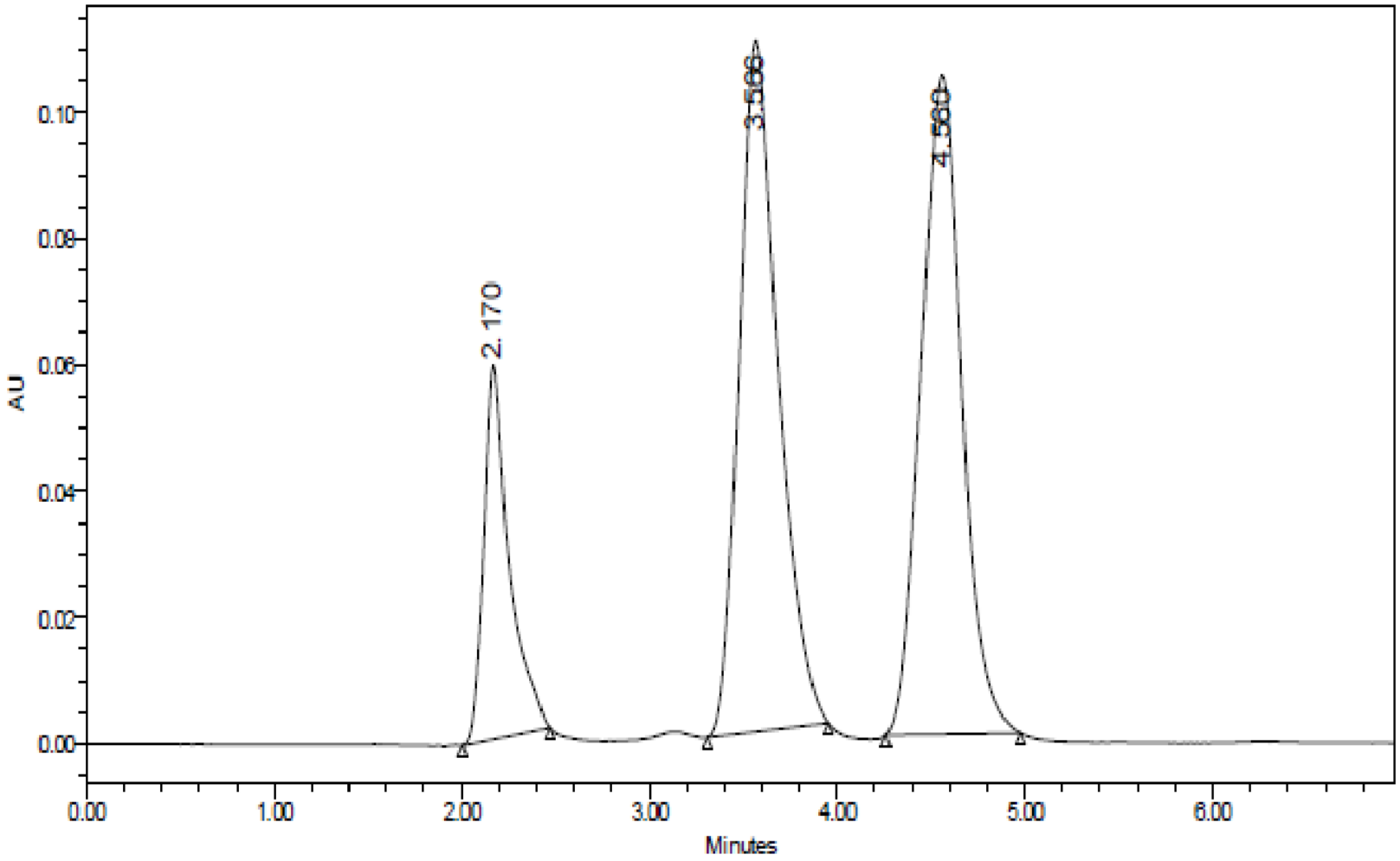
= (0.75 X 10.2) + (0.25 X 5.8)

= 9.1

the Analysis time for the Three Components = 5.5 minute as following :

Ascorbic Acid	Caffeine	Naringin
2.170 min	3.566 min	4.560 min

Mobile Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile :
The method gives good Separation with Short Run time.



Chromatogram of Ascorbic Acid (R.T 2.170 min), Caffeine (R.T 3.566 min) and Naringin (4.560 min) at flow Rat of Mobile Phase 1 ml /min 220nm Mobile Phase: Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile.

HPLC condition :

Mobile phase Isocratic elution with 75% phosphate buffer (pH 3.5) {https://www.drugfuture.com/pharmacopoeia/ep7/data/40102e.pdf} and 25% acetonitrile with Stationery Phase non-Polar Stationary Phase C18 silica based bonded Phase Column (250 mm long X 4.6 mm i.d packed with 5 µm) (column Symmetry Shield™ RP-18) Detectors at wavelength (λ) 220 nm , 20 µl Injection are the Sample and Standard ,Mobile phase Flow Rate =1 ml/min Ambient Temperature

.

Standard preparation:

prepare the Stock Solution of Standard 1mg /ml :

Weight 100mg Ascorbic Acid , 100mg Caffeine and 100 mg Naringin add to 100 volumetric flasks add 50 ml Dilute solution (7.5 % Phosphate buffer solution pH 3.5{To prepare the Linear Standard Solution (125,62.5, 31.25, 15.625 ,7.813 µg/ml)

From stock Standard Solution (1 mg/ml) take 12.5 ml to 100 ml volumetric flask fill with the dilute, to prepare 125 µg/ml

Tack 25 ml from 125 µg/ml to 50 ml Volumetric flask fill with dilute to the mark=62.5 µg/ml

Tack 25 ml from 62.5 µg/ml to 50 ml Volumetric flask fill with dilute to the mark=31.25 µg/ml

Tack 25 ml from 31.25 µg/ml to 50 ml Volumetric flask fill with dilute to the mark=15.625 µg/ml

Tack 25 ml from15.625 µg/ml to 50 ml Volumetric flask fill with dilute to the mark=7.813 µg/ml

Method Validation

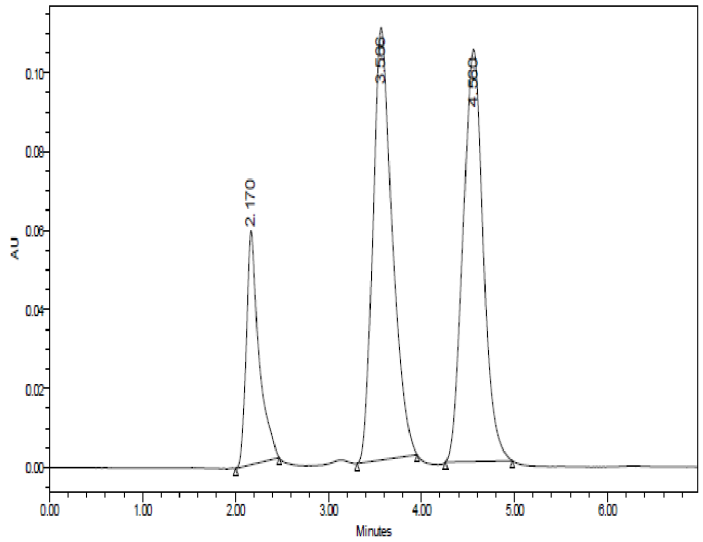
studies, that the performance characteristics of the method meet the requirements for its intended purpose, The aim of this step is to ensure that by laboratory tests this method is complete valid according to Validation guidelines and book. according to Validation ICH Q2(R1) guideline. and USP 32 (1225) guideline , CRC Handbook of ANALYTICAL VALIDATION) (book),ISO/IEC 17025:2017 guideline This Validation guidelines and book give a written plan for how validation steps will be worked and give the acceptance criteria.

Specificity

"Specificity is the ability to measure accurately and specifically the analyte of interest in the presence of other components that may be expected to be present in the matrix. "

The method doesn't show any interference from other components in the Analysis of Ascorbic Acid , Caffeine , Naringin the method is Specific to analysis all three component or one component alone .

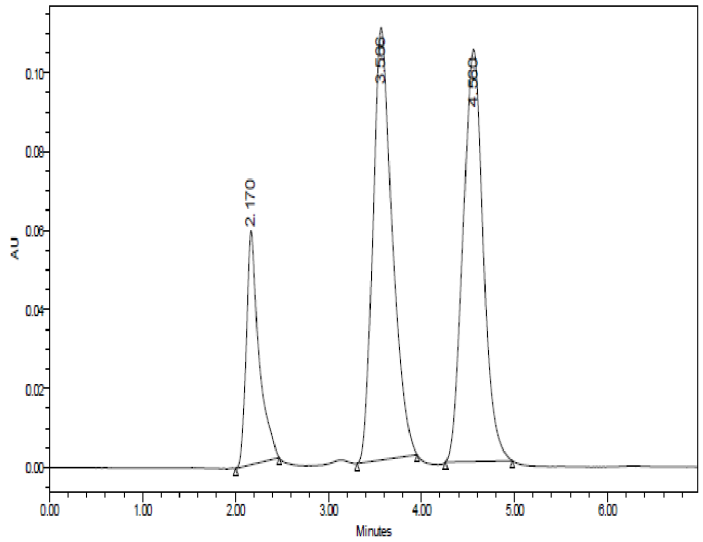
chromatogram of Ascorbic Acid (R.T 2.170 min), Caffeine (R.T 3.566 min) and Naringin (4.560 min) at flow Rat of Mobile Phase 1 ml /min 220nm Mobile Phase: Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile.



Selectivity

“ Selectivity is a relative term to describe to which extent particular analytes in mixtures or matrices can be measured without interferences from other components of similar behaviour “

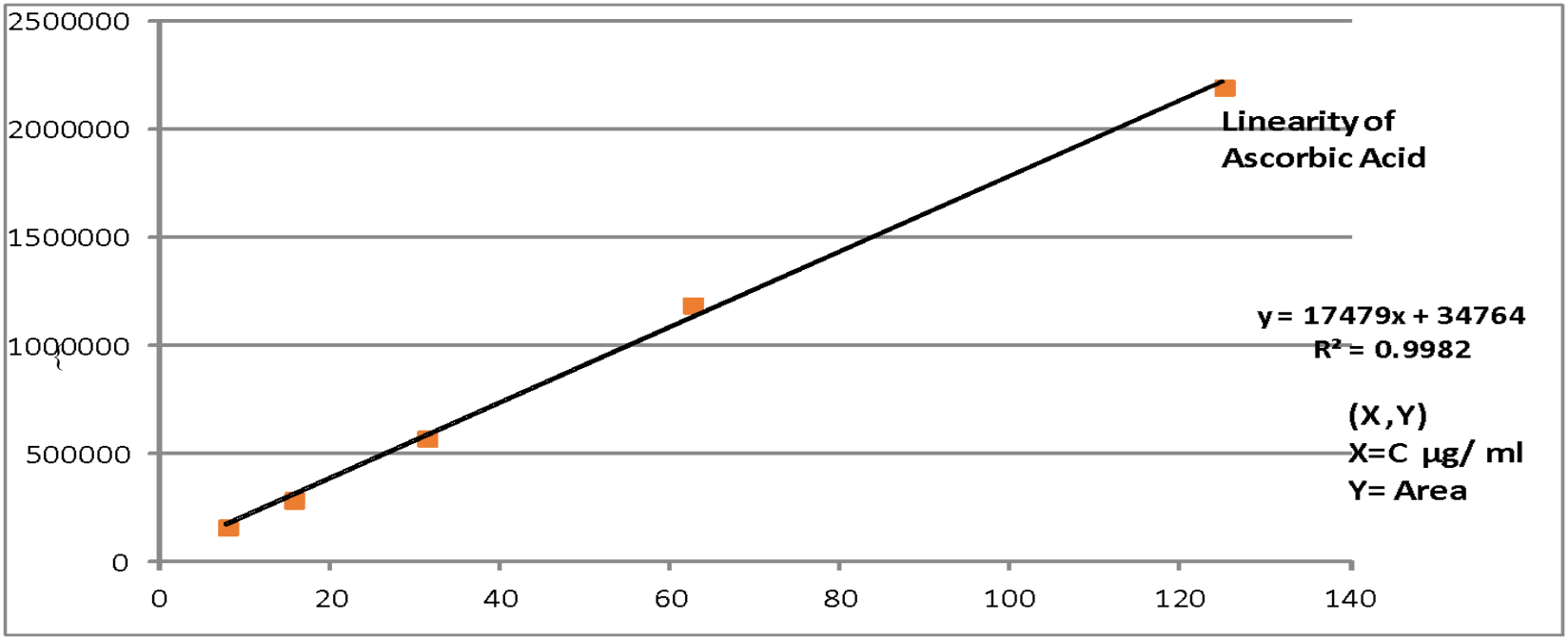
The method gives good separation in the chromatogram of HPLC between the three Peak Area of the three components without any overlap, that main that the method is good Selectivity.



Linearity:

Linearity: (5-point Linearity)

" The ability of the method to elicit test results that are directly, or by a well-defined mathematical transformation, proportional to analyte concentration within a given range."



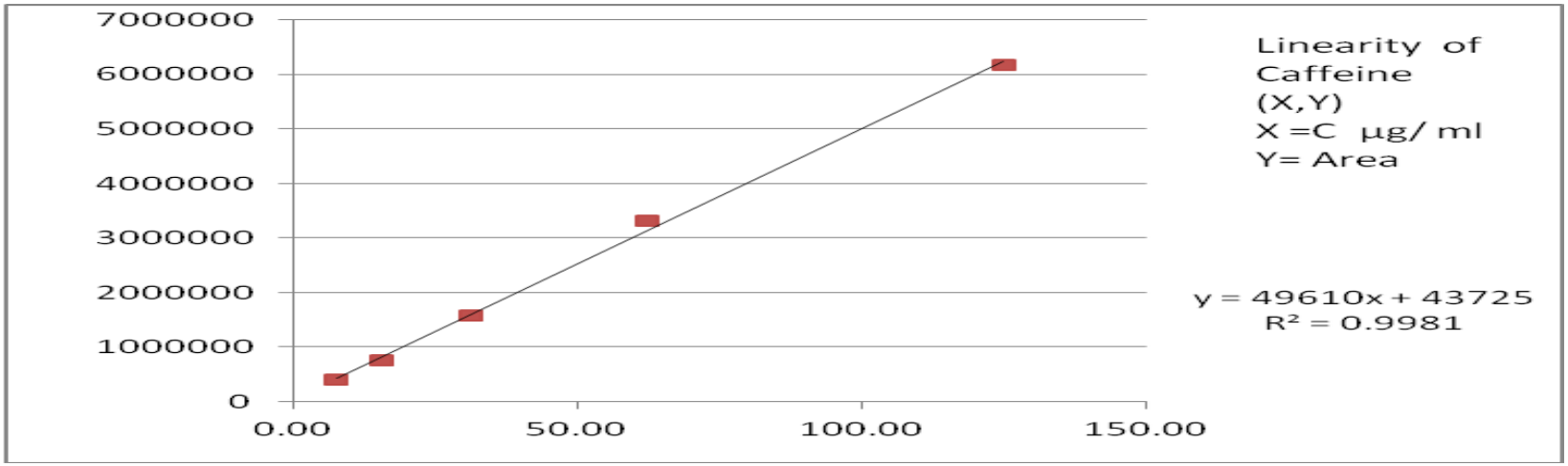
Linearity of Ascorbic Acid

Concentration µg/ ml	Retention Time (min)	Area	S/N
0	0	0	0
7.813	2.119	140981	14.47592155
15.626	2.129	287638	29.53465448
31.25	2.128	574130	58.95163775
62.5	2.127	1188901	122.0762912
125	2.134	2193490	225.2274361

Average = 2.1297	noise	=9739	slope:	=17478.5
SD = 0.003785939	LOD mg/ml	=1.59	R^2	=0.9982
% RSD = 0.17777143	LOQ µg/ml=	=5.29	intercept	= 34764.03

Linearity of Ascorbic Acid

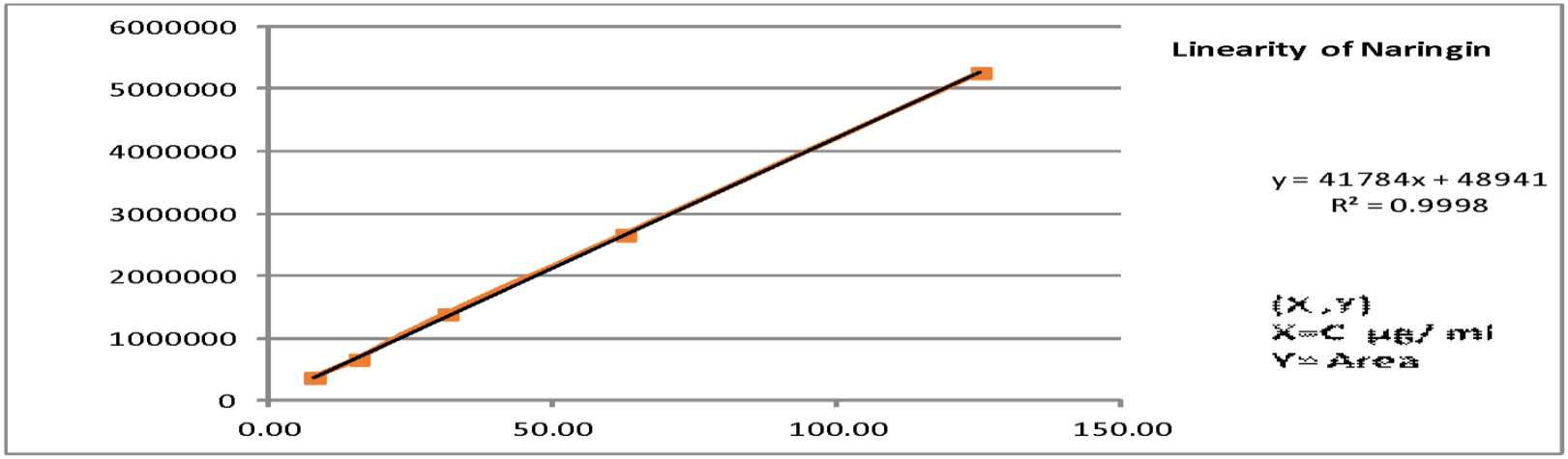
Linearity of Caffeine



Concentration µg/ ml	Retention Time (min)	Area	S/N
0	0	0	0
7.813	3.541	377422	38.75
15.62	3.543	758603	77.89
31.25	3.557	1427232	146.55
62.5	3.567	3323350	341.24
125	3.558	6168455	633.37

Average	= 3.560666667	LOD (µg/ml)	=0.601592		
Std Dev	= 0.005507571	LOQ (µg/ml)	=2.132406	slope:	= 49610.23805
RSD (%)	= 0.154678072			R^2	= 0.9972
		Noise	=9739	Intercept	= 43724.84371

Linearity of Naringin



Concentrations µg/ml	**Retention Time (min)**	**Area**	**S/N**		
0	**0**	**0**	**0**		
7.815	**4.54**	**373198**	**39.40**		
15.62	**4.55**	**667071**	**68.49**		
31.25	**4.525**	**1394905**	**143.22**		
62.5	**4.536**	**2664789**	**288.65**		
125	**4.539**	**5164112**	**519.98**		
Average	**4.538**	**LOD (µg/ml)**	**0.68**	**slope:**	**41784**
STD DEV	**0.008972179**	**LOQ (µg/ml)**	**2.28**	**R2**	**0.9998**
RSD	**0.197712191**	**Noise**	**9739**	**y-intercept**	**48941**

The Linearity concentration range for Ascorbic Acid ,Caffeine,and Naringin is form 7.813 µg/ ml to 125 µg/ ml .Linearity equation= (Area(Y) =(slope(m) X Concentration µg/ ml (X)) + intercept (C) Linearity equation Ascorbic Acid : Y=17479 X + 34764 , Correlation Coff1cint (R^2) = 0.9982 Linearity equation Caffeine : Y=49610 X + 34764, Correlation Coefficient (R^2) = 0.9981 Linearity equation Naringin : Y=41784 X + 48941, Correlation Coff1cint (R^2) =0.9998 Result:All test Result are acceptance with the Validation guidelines and book ((R^2) $\geq$ 0.995) that main that the method give a strong relationship (linear Relationship) between concentration and Area (response) for the Ascorbic Acid ,Caffein,Narginin Retiontin time (R.T) of Ascorbic Acid ,Caffeine,Naringin with five different Concentration .

Precision :

Precision the dgree of agreement among individual test result when an analytical method is used repeatedly to multiple samplings of homogeneous sample The aim of this step to ensure that the method is strong in the real routine analysis Intraday Precision in same days Intraday Precision in different days (Intermediate Precision).

Intraday Precision Ascorbic Acid in diffrant time but in the same day		
concentration 31.25µg/ml	RT (min)	Area
injection(0 Hour)	2.182	581205
injection (3 Hour)	2.183	579969
injection (6 Hour)	2.177	579777
Average	2.180666667	580317
STD DEV	0.00321455	774.99935
RSD	0.147411354	0.1335476

Intraday Precision in different time but in the same day:

The aim of this step to ensure that the methods are stabile and repeatable in same day but long time (three time in 6 Hour).

Intraday Precision Ascorbic Acid in diffrant time but in the same day

RSD % of Retantion time (R.T) Intraday Precision Ascorbic Acid =0.147411354%

RSD % of Area Peak Intraday Precision Ascorbic Acid =0.1335 %

Result :

The RSD % of Area Peak and Retantion time Ascorbic Acid are less the 2 % Acceptabile Value for Limits of Intraday Precision test acording to validation guidelines and book. the result of Intraday Precision test for Ascorbic Acid are Acceptabile for methode Intraday Precision validation limits

Intraday Precision Caffeine in different time but in the same day		
concentration 31.25µg/ml	**RT (min)**	**Area**
injection (0 Hour)	**3.559**	**1562235**
injection (3 Hour)	**3.566**	**1559061**
injection (6Hour)	**3.565**	**1549926**
Average	**3.56333**	**1557074**
STD DEV	**0.00379**	**6390.5396**
RSD	**0.10625**	**0.41042**

Intraday Precision Caffeine in different time but in the same day
RSD % of Retantion time (R.T) Intraday Precision Caffeine =0.10625%
RSD % of Area Peak Caffeine Interday Precision =0.41042%
Result :
The RSD % of Retantion time (R.T) and Area Peak Caffeine are less the 2 % Acceptabile Value for Limits of Intraday Precision test acording to validation guidelines and book, the result of Intraday Precision test for Caffeine are Acceptabile for method Intraday Precision validation limits (≤ 2%)

Intraday Precision Naringin in different time but in the same day		
concentration 31.25µg/ml	RT (min)	Area
injection (0 time)	4.535	1376438
injection (3 Hour)	4.538	1372855
injection 2 (9 Hour)	4.547	1371804
Average	4.54	1373699
STD DEV	0.006244998	2429.555721
RSD	0.137555022	0.176862305

Intraday Precision Naringin in different time but in the same day
RSD % of Retantion time (R.T) Intraday Precision Naringin =0.137555022%
RSD % of Area Peak Naringin Interday Precision =0.176862305%
Result :
The RSD % of Retantion time (R.T) and Area Peak Intraday Precision Naringin in different time but in the same day are less the 2 % Acceptabile Value for Limits of Intraday Precision test acording to Validation , guidelines and book,. the result of Intraday Precision Naringin in different time but in the same day test for Naringin are Acceptabile for methode Intraday Precision validation limits (≤ 2 %)

Intraday Precision Repeatability of Ascorbic acid		
concentration 62.5 µg/ml Ascorbic acid	RT (min)	Area
injection 1	2.204	1122702
injection 2	2.198	1130436
injection 3	2.233	1134106
injection 4	2.198	1130436
injection 5	2.233	1135106
injection 6	2.204	1123502
injection 7	2.231	1119738
Average	2.214428571	1128003.714
STD DEV	0.016939669	6002.888463
RSD %	0.76496795	0.532169211

Intraday Precision Repeatability (in the same time) :

the aim of this step operating the replicate test in short time under the same condition
Intraday Precision Repeatability of Ascorbic acid
RSD % of Retention time Ascorbic acid Intraday Precision Repeatability = 0.76496795%
RSD % of Area Peak Ascorbic acid Intraday Precision Repeatability =0.532169211%
Result :

The RSD % of Retantion time and Area Peak Intraday Precision Repeatability test for Ascorbic acid are less than 2 % the Acceptabile Value for Limits of Intraday Precision Repeatability test acording to Validiation guidelines and book. Intraday Precision Repeatability test for Ascorbic Acid are Acceptabile for methode Intraday Precision Repeatability validation limits (≤ 2%).

Intraday Precision Repeatability of Caffeine		
concentration 62.5 µg/ml	RT (min)	Area
injection 1	3.589	3113985
injection 2	3.565	3109183
injection 3	3.597	3097812
injection 4	3.647	3120816
injection 6	3.62	3112343
injection 6	3.586	3114900
injection 7	3.609	3144312
Average	3.601857143	3116193
STD DEV	0.026510555	14261.96085
RSD	0.736024609	0.457672578

Intraday Precision Repeatability of Caffeine
RSD % of Retention time Caffeine acid Intraday Precision Repeatability = 0.736024609%
RSD % of Area Peak Caffeine Intraday Precision Repeatability =0.457672578%
Result :
The RSD % of Retantion time and Area Peak Intraday Precision Repeatability test for Caffeine are less than 2 % the Acceptabile Value for Limits of Intraday Precision Repeatability test acording to validation guidelines and book. Intraday Precision Repeatability test for Caffeine are Acceptabile for methode Intraday Precision Repeatability validation limits (≤ 2%).

Intraday Precision Repeatability of Naringin		
concentration 62.5 (µg/ml) Naringin	RT (min)	Area
injection 1	4.551	2663771
injection 2	4.545	2669128
injection 3	4.565	2649601
injection 4	4.543	2672021
injection 5	4.559	2661998
injection 6	4.551	2672219
injection 7	4.553	2669335
Average	4.552428571	2665439
STD DEV	0.007634508	7995.244524
RSD %	0.167701876	0.299959764

Intraday Precision Repeatability of Naringin

RSD % of Retention time Naringin Intraday Precision Repeatability = 0.736024609%
RSD % of Area Peak Naringin Intraday Precision Repeatability =0.457672578%
Result :
The RSD % of Retantion time and Area Peak Intraday Precision Repeatability test for Naringin are less than 2 % the Acceptabile Value for Limits of Intraday Precision Repeatability test acording to validation guidelines and book. Intraday Precision Repeatability test for Caffeine are Acceptabile for methode Intraday Precision Repeatability validation limits (≤ 2%).

Interday Precision				
concentration Ascorbic Acid 31.25µg/ml	Area 1day	Area 2 day	Area 3 day	Average 1,2,3 Day
injection 1	582711	579845	576907	583417.25
injection 2	583401	579806	577845	579041.25
injection 3	584557	578457	577866	578018.75
injection 4	583000	578057	579457	
AVERAGE	583417.25	579041.25	578018.8	580159.0833
STD DEV	810.798115	920.31747	1057.997	2867.597267
RSD	0.13897397	0.1589382	0.183039	0.494277751

concentration Caffeine 31.25µg/ml	Area 1day	Area 2 day	Area 3 day	Average 1,2,3 days
injection 1	1560354	1550354	1548354	1563409.5
injection 2	1563926	1554926	1550926	1552909.5
injection 3	1563968	1553968	1548720	1549484
injection 4	1565390	1552390	1549937	
AVERAGE	1563409.5	1552909.5	1549484	1555267.667
STD DEV	2147.64639	1998.929297	1175.467	7256.073324
RSD	0.13736941	0.128721558	0.075862	0.466548201

concentration Naringin 31.25µg/ml	Area 1day	Area 2 day	Area 3day	Average 1,2,3 Day
injection 1	1378986	1378986	1365908	1377759.75
injection 2	1379904	1372904	1366465	1375652
injection 3	1376543	1375438	1365422	1366768.75
injection 4	1375606	1375280	1369280	
AVERAGE	1377759.75	1375652	1366769	1373393.5
STD DEV	2018.301	2506.740247	1727.548	5833.192688
RSD	0.14649151	0.182221975	0.126397	0.424728433

Interday Precision : Intermediate Precision in 3 days

The aim of this step to ensure that the method is strong in the real routine analysis by different days .

interday Precision ASCORBIC ACID RSD % = 0.49428
,Interday Precision Caffeine RSD % =0.44454
Interday Precision Naringin RSD %=0.424728433
Result :
The RSD % Interday Precision of Area Peak for ASCORBIC ACID ,Caffeine, Naringin are less than 3 % the Acceptable Value for Limits of Interday Precision according to validation book and guidelines Interday Precision ASCORBIC ACID RSD % ,Interday Precision Caffeine RSD % and Interday Precision Naringin RSD % are Acceptable for method Intraday Precision validation limits (≤ 3 %).

Accuracy:

"*The closeness of test results to the true value*"

Ascorbic acid Accuracy	concentration µg/ml	Spike (µg/ml)	TOTAL Conc. µg/ml	Area	Recovery	% Recovery
	31.25	0	31.25	579051	31.14033	99.64905
	31.25	0	31.25	587415	31.61886	101.1803
	31.25	0	31.25	578895	31.1314	99.62049
	31.25	0	31.25	578157	31.08918	99.48538
AVREGE	31.25	0	31.25	580879.5	31.24494	99.98382
	31.25	3.906	35.16	659993	35.77127	101.7385
	31.25	3.906	35.16	660000	35.77167	101.7397
	31.25	3.906	35.16	649990	35.19897	100.1108
	31.25	3.906	35.16	652082	35.31866	100.4512
AVREGE	31.25	3.906	35.16	655516.25	35.51514	101.0101
	31.25	8.35	39.6	722140	39.32689	99.31033
	31.25	8.35	39.6	720422	39.2286	99.06211
	31.25	8.35	39.6	721472	39.28867	99.21382
	31.25	8.35	39.6	720634	39.24073	99.09274
AVREGE	31.25	8.35	39.6	721167	39.27122	99.16975

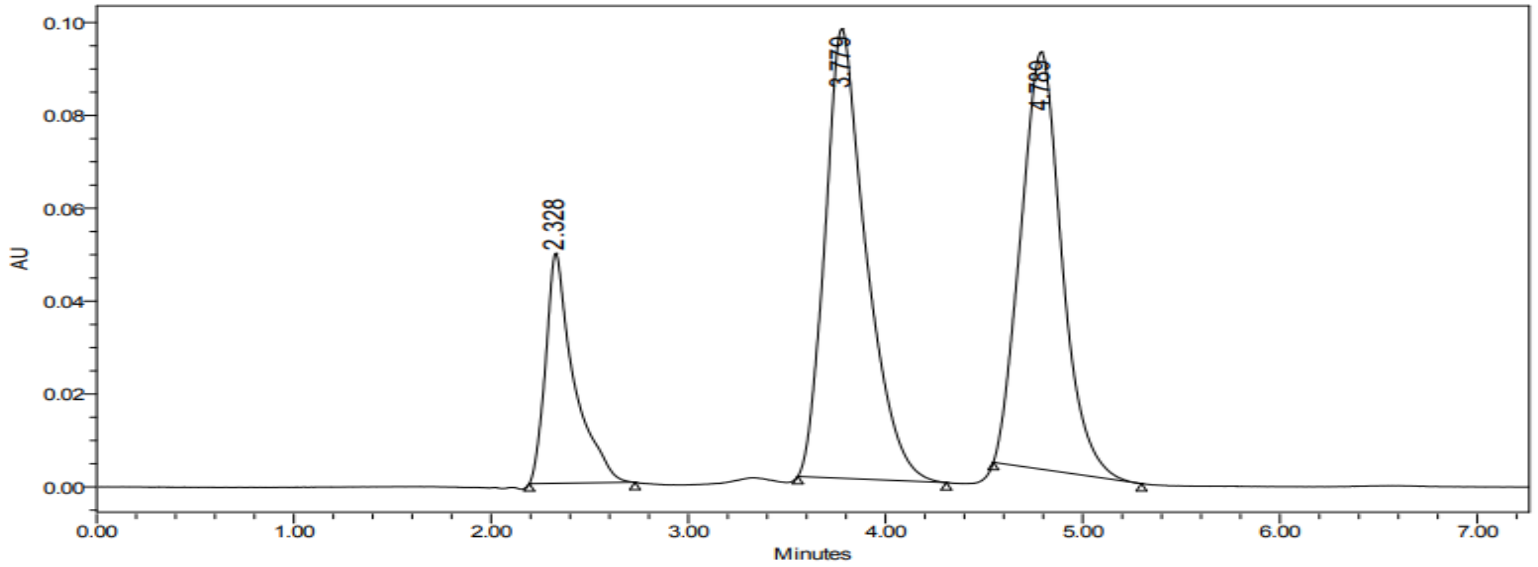
Caffeine Accuracy	conc µg/ml	Spike (µg/ml)	TOTAL Conc. µg/ml	Area	Recovery	% Recovery
	31.25	0	31.25	1514354	31.40642	100.5005
	31.25	0	31.25	1513926	31.39779	100.4729
	31.25	0	31.25	1513968	31.39864	100.4756
	31.25	0	31.25	1516301	31.44567	100.6261
AVREGE	31.25	0	31.25	1514637.25	31.41213	100.5188
	31.25	3.906	35.16	1710457	35.3593	100.5668
	31.25	3.906	35.16	1704359	35.23638	100.2172
	31.25	3.906	35.16	1710219	35.3545	100.5532
	31.25	3.906	35.16	1708360	35.31703	100.4466
AVREGE	31.25	3.906	35.16	1708348.75	35.3168	100.446
	31.25	8.35	39.6	1929157	39.76767	100.4234
	31.25	8.35	39.6	1920345	39.59004	99.97485
	31.25	8.35	39.6	1919903	39.58113	99.95235
	31.25	8.35	39.6	1913210	39.44622	99.61167
AVREGE	31.25	8.35	39.6	1920653.75	39.59626	99.99057

Naringin Accuracy	conc µg/ml	Spike (µg/ml)	Total Conc. µg/ml	Area	Recovery	% Recovery
	31.25	0	31.25	1370211	31.35693	100.3422
	31.25	0	31.25	1369992	31.35166	100.3253
	31.25	0	31.25	1387314	31.76801	101.6576
	31.25	0	31.25	1376101	31.4985	100.7952
AVREGE	31.25	0	31.25	1375904.5	31.49377	100.7801
	31.25	3.906	35.16	1538321	35.39755	100.6756
	31.25	3.906	35.16	1530071	35.19925	100.1116
	31.25	3.906	35.16	1542968	35.50924	100.9933
	31.25	3.906	35.16	1540987	35.46163	100.8579
AVREGE	31.25	3.906	35.16	1538086.75	35.39192	100.6596
	31.25	8.35	39.6	1715404	39.65384	100.136
	31.25	8.35	39.6	1712147	39.57556	99.93827
	31.25	8.35	39.6	1710132	39.52712	99.81597
	31.25	8.35	39.6	1712789	39.59099	99.97724
AVREGE	31.25	8.35	39.6	1712618	39.58688	99.96686

Recovery % of Ascorbic acid =(100.8+100.8+99.9) / 3 = 100.5 %
Recovery % of Caffeine =(100.5+100.5+99.99) / 3= 100.33 %
Recovery % of Naringin =(100.8+100.7+99.96) / 3= 100.5 %
Result :
The Recovery Present of ASCORBIC ACID 100.5 %,Caffeine 100.33 %, Naringin 100.5 %
are in Acceptable Range for Limits of Recovery (98-102%) according to validation guidelines and book,

Robustness :

The capacity of a method to remain unaffected by small, deliberate variations in method parameters ; a measure of the reliability of a method, often used to set the system suitability parameters of a method. {14}



Chromatogram of Ascorbic acid (R.T 2.328 min) ,Caffeine (R.T 3.779 min) and Naringin (4.789 min) at flow Rat of Mobile Phase 0.95 ml /min 220nm Mobile Phase: Isocratic elution with 75% phosphate buffer (pH 3.5) and 25% acetonitrile.

Robustness Ascorbic acid flow Rat 0.95 ml / min 31.25µg/ml 220nm	RT (min)	Robustness Ascorbic acid flow Rat 1.05 ml / min 31.25µg/ml 220 nm	RT (min)
injection 1	2.354	injection 1	2.13
injection 2	2.328	injection 2	2.12
injection 3	2.323	injection 3	2.106
Average	2.335	Average	2.118666667
STD DEV	0.016643317	STD DEV	0.012055428
RSD %	0.712775888	RSD %	0.569010111
Robustness ASCORBIC ACID		Robustness Ascorbic acid	220 nm
pH 3.55 220 nm	RT (min)	pH 3.50	RT (min)
injection 1	2.83	injection 1	2.177
injection 2	2.84	injection 2	2.179
injection 3	2.82	injection 3	2.186
Average	2.83	Average	2.180666667
STD DEV	0.01	STD DEV	0.004725816
RSD %	0.35335689	RSD %	0.21671426

Robustness Ascorbic acid

Robustness (R.T) of Ascorbic acid flow Rat 0.95 ml / min RSD % = 0.712775888 %
Robustness (R.T) of Ascorbic acid flow Rat 1.05 ml / min RSD % = 0.569010111 %
Robustness (R.T) of Ascorbic acid pH 3.5 RSD % = 0.21671426 %
Robustness (R.T) of Ascorbic acid pH 3.55 RSD % = 0.35335689 %
Result : The test result of Robustness change the flow Rat ± 0.05 ml / min and pH (± 0.05) found that the RSD % for retention time for Ascorbic acid in both change are less than 3 % the Acceptable Value for Limits of Robustness validation guidelines and book , the Robustness RSD % result are in the acceptable range ≤ 3 % the method is Robust for Ascorbic acid .

Robustness caffeine		Robustness Caffeine	
flow Rat 0.95 ml / min 31.25µg/ml 220nm	**RT (min)**	**flow Rat 1.05 ml / min 31.25µg/ml 220nm**	**RT (min)**
injection 1	3.82	injection 1	3.472
injection 2	3.779	injection 2	3.446
injection 3	3.771	injection 3	3.437
Average	3.79	Average	3.451666667
STD DEV	0.026286879	STD DEV	0.018175075
RSD %	0.693585194	RSD %	0.526559377
Robustness Caffeine		**Robustness caffeine**	**220 nm**
pH 3.55 220 nm	**RT (min)**	**pH 3.50**	**RT (min)**
injection 1	4.83	injection 1	3.591
injection 2	4.87	injection 2	3.609
injection 3	4.82	injection 3	3.577
Average	4.84	Average	3.592333333
STD DEV	0.026457513	STD DEV	0.016041613
RSD %	0.546642833	RSD %	0.446551338

Robustness (R.T) of Caffeine flow Rat 0.95 ml / min RSD % = 0.693585194 %
Robustness (R.T) of Caffeine flow Rat 1.05 ml / min RSD % = 0.526559377 %
Robustness of (R.T) Caffeine pH 3.5 RSD % = 0.446551338 %
Robustness of (R.T) Caffeine pH 3.55 RSD % = 0.546642833 %
Result : The test result of Robustness change the flow Rat ± 0.05 ml / min and pH (± 0.05) found that the RSD % for retention time for Caffeine in both change are less than 3 % the Acceptable Value for Limits of Robustness validation guidelines and book , the Robustness RSD % result are in the acceptable range ≤ 3 % the method is Robust for Caffeine .

Robustness o caffeine

Robustness Naringin		Robustness Naringin	
flow 0.95 ml / min (31.25 (µg/ml)	**RT (min)**	**flow 1.05 ml / min 31.25 (µg/ml)**	**RT (min)**
injection 1	**4.791**	**injection 1**	**4.515**
injection 2	**4.789**	**injection 2**	**4.507**
injection 3	**4.781**	**injection 3**	**4.515**
Average	**4.787**	**Average**	**4.512333333**
STD DEV	**0.005292**	**STD DEV**	**0.004618802**
RSD	**0.110539**	**RSD**	**0.102359507**
Robustness Naringin		**Robustness Naringin**	**220 nm**
pH 3.55 220 nm	**RT (min)**	**pH 3.50**	**RT (min)**
injection 1	6.523	injection 1	4.514
injection 2	6.518	injection 2	4.524
injection 3	6.536	injection 3	4.528
Average	6.525667	Average	4.522
STD DEV	0.009292	STD DEV	0.007211103
RSD	0.142385	RSD	0.159467106

Robustness (R.T) of Naringin flow Rat 0.95 ml / min RSD % = 0.110539 %
Robustness (R.T) of Naringin flow Rat 1.05 ml / min RSD % = 0.102359507 %
Robustness (R.T) of Naringin pH 3.5 RSD % = 0.159467106 %
Robustness (R.T) of Naringin pH 3.55 RSD % = 0.142385 %
Result: The test result of Robustness changes the flow Rat ± 0.05 ml / min and pH (± 0.05) found that the RSD % for retention time for Naringin in both changes are less than 3 % the Acceptable Value for Limits of Robustness validation guidelines and book, the Robustness RSD % result is in the acceptable range ≤ 3 % the method is Robust for Naringin.

System Suitability :

This Step to ensure that the validation method had a suitable system during validation test by calculated Tailing factor ,Theoretical plate count (N) for Peaks of HPLC chromatography Peaks of Ascorbic acid, caffeine, naringin and the calculation of the Resolution(Rs) between Peak of Ascorbic acid , Caffeine and between Peak of Caffeine, naringin

Start time ,End time, retention time of HPLC chromatography Peaks of Ascorbic acid, caffeine, naringin

Peak Ascorbic acid	min
Retanion time (R.T) Ascorbic acid (min)	2.153
time end Ascorbic acid (min)	2.2895
time start Ascorbic acid (min)	2.10131
Peak Caffeine	
RT Caffeine (min)	3.568
time end Caffeine (min)	3.7529
time start Caffeine (min)	3.4548
Peak 3 Naringin	
RT Naringin (min)	4.53
time end Naringin(min)	4.7301
time start Naringin (min)	4.4306

Tailing factor of Ascorbic acid	**1.820371**
Tailing factor of caffeine	**1.316696**
Tailing factor of Naringin	**1.506539**

Tailing factor for Peaks of HPLC chromatography Peaks of Ascorbic acid, caffeine, naringin ,
Result : The calculation result of Tailing factor for Peaks of HPLC chromatography Peaks of Ascorbic acid 1.820371, caffeine1.316696 , naringin 1.506539 found that the are less than 2 % the Acceptable Value for Limits of Tailing factor for Peaks according to guidelines and book , the Tailing factor for Peaks result are in the acceptable range ≤ 2 %.

Theoretical plate count (N) Ascorbic acid	2094.185
Theoretical plate count (N) Caffeine	2292.164
Theoretical plate count (N) Naringen	3660.351

Table {25 } Theoretical plate count (N) for Peaks of HPLC chromatography Peaks of Ascorbic acid, caffeine, naringin
Result : The calculation result of Theoretical plate count (N) for Peaks of HPLC chromatography Peaks of Ascorbic acid 2094.185, caffeine2292.164 , naringin 3660.351 found that the are more than 2000 the Acceptable Value for Limits of Theoretical plate count (N) for Peaks according to guidelines and book , Theoretical plate count (N) for Peaks result are in the acceptable range ≥ 2000 .

Resolution(Rs) between Peak of Ascorbic acid and Peak of Caffeine	5.819572683
Resolution(Rs) between Peak of Caffeine and Peak of Naringin	3.219544846

Table {26 } Resolution(Rs) between Peak of Ascorbic acid , Peak of Caffeine and Peak of Naringin

Result : The calculation result of Resolution(Rs) between Peak of Ascorbic acid and Peak of Caffeine 5.819572683 ,and the Resolution(Rs) between Peak of Caffeine and Peak of Naringin 3.219544846 found that are more than 2.5 the Acceptable Value for Limits of Resolution(Rs) between Peak according to validation guidelines and book , Theoretical plate count (N) for Peaks result are in the acceptable range >2.5 .
all calculation result of the Resolution(Rs), Theoretical plate count (N), Tailing factor for the Ascorbic acid, caffeine, naringin , found that the methods had a Suitable System

ANOVA :(Analysis of Variance) is a statistical test used to analyze the difference between the means of more than two groups' two-way ANOVA is used to estimate how the mean of a quantitative variable changes according to the levels of two categorical variables. Use a two-way ANOVA when you want to know how two independent variables, in combination, affect a dependent variable.

The aim of this step to use Data Anlysis Anova Tow Factor without Replication with use Microsoft Excel to Extract information that helps to analyses data that good to the methods validation.

Retention time of Ascorbic acid , Caffeine, Naringin with five different Concentration

Concentration	Ascorbic acid R. T	caffeine R. T	Naringin R. T
7.815 µg/mL	2.119	3.541	4.521
15.63µg/mL	2.129	3.543	4.525
31.25µg/mL	2.128	3.557	4.525
62.5µg/mL	2.127	3.567	4.526
125µg/mL	2.134	3.558	4.529

Retention Times (R.T.) and ANOVA Analysis :
The aim this analysis is to determine that There are not much difference in Retantion time in the five diffrant Concentration from the 7.815 to 125 µg/ml (row)and the three type of coponant in columns (Ascorbic acid ,caffeine,and Naringin)
Hyptheses (H 0): There is not so much difference in R.T on five diffrant Concentration (row), There is not so much difference in R.T on thee componant (Columns)
With Anova: Two-Factor Without Replication .
Hyptheses (H 1):There are one of five cocnentration from (rows) at least have effect at the Retantion Time (R.T).
There are one from three coponant (row) at least have effect at the Retantion Time (R.T).
1) ANOVA Row Analysis (Concentration V. Retantion Time)
P-Value = (0.084087)
α = 0.05
P-Value (0.084087) > α = 0.05
that mein we can't Reject (H 0) Retantion Time (R.T) not Change if changed the concentration (row) (Retanion time Stability) (System Subility)
ANOVA (Column) analysis (Componat V. Retantion Time)
P-Value =0.000095233
α = 0.05
P-Value(0.000095233) < α (0.05) that mein we Reject (H0)
Retantion Time (R.T) Changed if changed the three comonant (Column) (Selectivity/ Specificity)
That meine that Retention Time (R.T) are Concentration independent but Retention Time (R.T) depent of the copmonant .
The retantion time not change if changed the concentration (System Sutability)
And the retantion time changed if changed the Componant (Selictivity)

Peak Area (response) and ANOVA Analysis :
The aim this analysis is to determine that There are not much difference in the Peak Area (response) in the five diffrant Concentration from the 7.815 to 125 µg/ml (row)and the three type of coponant in column's (Ascorbic acid ,caffeine,and Naringin)
Hyptheses (H 0): There is not much difference in Peak Area (response) on five diffrant Concentration (row), There is not much difference in Peak Area (response) on three componant (Columns)
With Anova Two-Factor Without Replication .
Hyptheses (H1):There are one of 5 cocnentration from (rows) at least have effect at Peak Area (response)
There are one from 3 coponant (row) at least have effect Peak Area (response).
1) Anova Row Analysis (Concentration V. Peak Area (response))
P-Value = (0.001092)

α = 0.05
P-Value (0.001092) < α = 0.05
that mein we Reject (H 0)
Peak Area (response) Change if changed the concentration (row) (Sensitivity) (Linearity)
2) Anova Column analysis (Componat V. Peak Area (response))
P-Value =0.034917
α = 0.05
P-Value(0.034917) < α (0.05) that mein we Reject (H 0)
Peak Area (response) Changed if changed the three comonant (Column) (Selectivity / Specifity)
That mein that Peak Area (response) are Concentration dependent and depent of the copmonant .
The Peak Area (response) change if changed the concentration that response for the (Sensitivity) (Linearity)
And the Peak Area (response) changed if changed the Componant (Selictivity)

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SOP FOR HPLC

- **Standard Operating Procedure (SOP) for High-Performance Liquid Chromatography (HPLC) System Configuration:**
- Binary HPLC Pump: Waters 1525
- Manual Injector: Six-port injector
- Detector: Waters 2998 Diode Array Detector (DAD)
- **English Version**
- **Objective**
- To provide a detailed and standardized procedure for the operation of the Waters 1525 Binary HPLC system equipped with a six-port manual injector and a Waters 2998 diode array detector for the separation and analysis of chemical compounds.
- **Requirements**
- Equipment and Materials:
  - Waters 1525 Binary HPLC Pump
  - Six-port manual injector
  - Waters 2998 DAD
  - Analytical column appropriate for the analysis
  - Mobile phase (HPLC-grade solvents)
  - Sample solutions
  - Standard calibration solutions
  - Personal Protective Equipment (PPE)
- Tools:
  - Syringes (10 µL – 100 µL)
  - Filters (0.45 µm)
  - Ultrasonic bath (if required for degassing)

- **Procedure**
- **1. Pre-Operation Setup**
- **Inspect the System:**
  - Ensure all system components (pump, injector, column, and detector) are clean and properly connected.
  - Verify that the mobile phase is freshly prepared and degassed.
- **Install the Analytical Column:**
  - Attach the appropriate column for the analysis to the system.
  - Equilibrate the column with the mobile phase for at least 15 minutes.
- **Prepare the Samples:**
  - Filter the sample and standard solutions using 0.45 µm filters.
  - Load the samples into vials and seal them properly.
- **2. System Initialization**
- Power on the HPLC system and detector.
- Launch the chromatography software and initialize the system.
- Input the method parameters, including:
  - Mobile phase composition and gradient program (if applicable).
  - Flow rate (e.g., 1.0 mL/min).
  - Detector wavelength (e.g., 254 nm).
  - Injection volume (e.g., 20 µL).

- **3. Sample Injection**
- Prime the manual injector by flushing it with the mobile phase.
- Load the sample into the injector loop using a syringe.
- Turn the injector valve to the “Inject” position to introduce the sample into the system.
- **4. Data Acquisition and Analysis**
- Start the data acquisition in the software.
- Monitor the chromatogram in real time and ensure proper peak separation.
- Save the data after the run is complete.
- Perform preliminary data analysis to calculate retention times and peak areas.
- **5. Post-Operation**
- Flush the system with the mobile phase and then with HPLC-grade water.
- Clean the column and store it appropriately.
- Turn off the HPLC system and software.
- **Safety Precautions**
- Always wear PPE (gloves, goggles, and lab coat).
- Handle solvents and chemicals in a well-ventilated area.
- Dispose of waste solvents in compliance with local regulations.
-

بعض المواقع الممكن الاستفادة منها في الجزء العملي

Operating an HPLC: Part 1

https://www.youtube.com/watch?v=51DSl8BHUjg

Operating an HPLC: Part 2

https://www.youtube.com/watch?v=YaabRXcm4M4&t=37s

HPLC in ARABIC

https://www.youtube.com/watch?v=nYRAGK4NcJI

بعض المواقع الممكن الاستفادة منها

مواقع ممتازه على YouTube تبسط المادة وتجعلها مادة ممتعة :

1) How HPLC Columns Work
https://www.youtube.com/watch?v=mPO7Tv2mIJU

2) HPLC - High Performance Liquid Chromatography - for beginners simple animation – HD
https://youtu.be/IUwRWn9pEdg?list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy

3) HPLC - What is the function of the mobile phase? – Animated
https://www.youtube.com/watch?v=pmHtGDdagJU&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy&index=2

HPLC - The Stationary Phase – Animated
https://www.youtube.com/watch?v=MYSBOxbnuAw&index=3&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy
HPLC - Normal Phase vs Reverse Phase HPLC – Animated
https://www.youtube.com/watch?v=MLoitPJQH3g&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy&index=4
HPLC - Isocratic vs Gradient Elution – Animated
https://www.youtube.com/watch?v=tAcfJPveWwM&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy&index=5
HPLC - UV / VIS detection of analytes Animated
https://www.youtube.com/watch?v=sfxEj_MxBcs&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy&index=6
HPLC - How to read Chromotogram Easy Explained - Simple Animation HD
https://www.youtube.com/watch?v=qXmSb6Xwr5k&index=7&list=PLetrcQ_JUQSmK4yEiMDyMoiaaZYEeQPDy
Principles of Chromatography
https://www.youtube.com/watch?v=_CWtWeEN7is
High Performance Liquid Chromatography HPLC
https://www.youtube.com/watch?v=kz_egMtdnL4





High-Performance Liquid Chromatography (HPLC): From Theory to Practice – A Comprehensive Guide to Analytical Method Development & Validation HPLC From Zero to Hero

كروماتوغرافيا السائل عالية الأداء HPLC من النظرية إلى التطبيق – دليل شامل لتطوير والتحقق من صحة طرق التحليل.

HPLC من الصفر إلى الاحتراف.

DOI: 10.13140/RG.2.2.12233.25440

Mohammad Asaad Mousa

Edition 10/2026

https://doi.org/10.13140/RG.2.2.12233.25440

ResearchGate https://www.researchgate.net/profile/Mohammad-Mousa-6?ev=hdr_xprf

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